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Parametric Gaussian Noise Mixer (PGNM)

Unified Synthesis of Camera Noise and Film Grain

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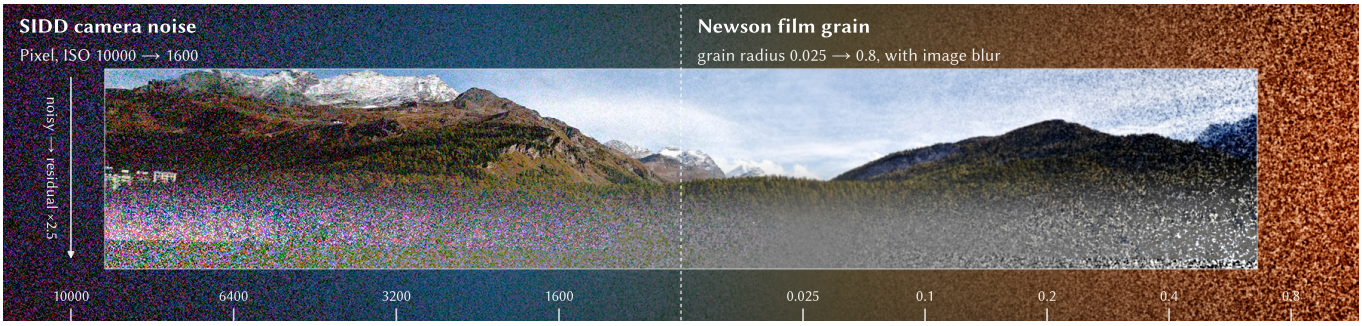

Figure 1 One model, a continuum of noise and grain. From left to right, PGNM interpolates between presets fitted on SIDD camera noise [1] and on Newson film grain [33]; ticks mark the fitted presets. Film presets also apply the renderer's image blur, fitted per radius (Sec. 4.2). Within the photo, the noisy image fades into its residual from top to bottom. Please zoom in.

ABSTRACT

Noise synthesis is commonly approached through physical models of image formation or neural networks trained to generate noisy images. We explore a middle ground: an explicit statistical model fitted directly from paired clean and noisy images. A normalized convolution correlates a Gaussian field across pixels and color channels, and a lookup maps each Gaussian value to a residual given the clean intensity. Because the field stays standard Gaussian, the two components are fitted separately, without reproducing the physical process or training a neural generator. With one formulation for camera noise from SIDD and simulated film grain from Newson, the model has the lowest average distribution and correlation errors on SIDD among the evaluated methods and remains competitive on film grain. A denoiser trained on its noise comes closest to one trained on real noise, and a 1080p frame renders in 10 ms on GPU.

1 Introduction

Photographic noise can be part of an image's intended appearance. Film grain contributes a texture that filmmakers may choose to preserve. For *Oppenheimer*, FotoKem's [14] account of the finishing process explicitly identifies grain structure as part of the distinctive photochemical texture. Reproducing this appearance requires modeling noise as well as removing it.

Noise modeling also supports efficient video delivery. Grain's spatial and temporal randomness makes it expensive to encode. AV1's [34] film-grain synthesis framework encodes a denoised image together with estimated grain parameters and synthesizes grain at the decoder before display. This approach can reduce bandwidth while preserving the intended texture.

In visual-effects compositing, photographed and computer-generated elements must share a consistent photographic appearance. Foundry's Nuke [15] documentation identifies grain matching as a step toward making all elements appear to have been photographed on the same film stock. A noise model therefore provides a means of reproducing the texture of live-action footage on digitally generated or modified content.

Accurate noise synthesis is also useful for denoising data augmentation. Brooks et al. [7] show the importance of modeling sensor noise and image processing when synthesizing raw-image denoising data. Noise Flow [2] further demonstrates training a denoiser

with samples from a learned noise model. These results motivate generating noise representative of the target acquisition process, even when noise removal is the eventual task.

Camera noise and film grain are mostly modeled separately. PGNM renders both by passing a convolved Gaussian field through an intensity-conditioned lookup, each fitted directly from aligned clean/noisy pairs. We find that:

- **Dependence and distribution can be fitted separately.** The normalized convolution keeps each latent value standard Gaussian, so the lookup alone sets the residual distribution. The convolution is then estimated from measured correlations and the lookup from residual quantiles.
- **Direct fits hold up against learned generators.** The learned baselines contain every PGNM ingredient, but with many free parameters and indirect supervision, individual properties are harder to inspect and control. PGNM has the lowest average errors on SIDD, and a denoiser trained on its noise comes closest to one trained on real pairs. On film grain, it matches a physical approximation for fine grain and leads in conditional W_1 for coarser grain.
- **Intensity-only conditioning makes synthesis cheap.** The Gaussian field does not depend on the image, so it is filtered once and replayed (10 ms per 1080p frame on GPU). On one film clip, a prototype that adds this grain after decoding suggests potential bitrate savings over coding the grain.

2 Related Work

Physical and statistical models. Camera models commonly combine photon-counting and readout noise; Foi et al. [13] use a Poisson distribution for photon counts and a Gaussian distribution for readout fluctuations. Wei et al. [39] extend this description to electronic effects in extreme low light. Unprocessing [7] and CycleISP [46] model the processing between raw sensor data and color images; Zou and Fu [51] learn to estimate physical noise parameters. For film grain, Newson et al. [33] render randomly placed grains through Monte Carlo simulation. Zhang et al. [48] derive an analytic approximation when individual grains are unresolved. Tian and Pappas [38] extend efficient simulation to a wider range of magnification and blur. In codecs, AV1 grain synthesis [34] combines autoregression over preceding neighbors with intensity-dependent scaling.

Neural camera-noise models. Among normalizing flows, Noise Flow [2] models raw noise, whereas sRGB-Flow [26] models noise in processed images. Noise2NoiseFlow [30] jointly learns noise modeling and denoising from noisy pairs; NAFlow [23] adapts sampling using a reference pair. InvDN [29] instead uses an invertible network to separate image content and noise.

Among generative adversarial networks (GANs), DANet [45] jointly learns denoising and generation; C2N [20] uses unpaired clean and noisy images. CA-NoiseGAN [10] conditions generation on camera information, while PNGAN [8] aligns generated and real images through pixel-level adversarial training. NeCA [16] models the correlation between neighboring noise values. GMDCN [37] supplements adversarial training with a Gram-matrix style loss.

SeNM-VAE [50] uses a variational autoencoder with paired and unpaired data. Among diffusion models, RNSD [42] uses camera conditioning, GuidNoise [22] uses one guidance pair, and PNG [24] uses learned noise prompts. YeTI [25] learns from two noisy observations and generates another noisy realization from a noisy input.

Learned film-grain models. Deep-FG [4] uses a GAN conditioned on grain parameters; GrainNet [27] uses a lightweight generator with a texture-aware loss. Style-FG [6] transfers grain style, while 3R-INN [5] combines grain removal, reconstruction, and rescaling through an invertible network. Yang et al. [44] separate scene content from degradation to enable transfer, including film grain.

Shared observations. Read together, these models repeatedly measure or encode two observations, which PGNM adopts rather than introduces. First, residual strength depends on clean intensity. Foi et al. [13] model it through random photon counts, Colom and Buades [11] estimate it as a noise curve from weakly textured regions, Noise Flow [2] builds it into an invertible transformation, and NeCA-W [16] and AV1 [34] vary strength with intensity. In processed images, however, a strength curve captures only part of this observation: residual distributions also change in bias, asymmetry, and clipping, and deviate from common parametric forms (Fig. 3). We therefore keep the full intensity-conditioned distribution as a two-dimensional lookup. Second, residuals are spatially correlated; for film grain, grain size appears as the distance over which nearby residual values remain correlated. A per-pixel transformation, such as Noise Flow's intensity-dependent one or our lookup, reshapes each residual distribution but cannot correlate values that start independent. Correlation therefore needs its own component: NeCA models neighboring correlation, NM-FlowGAN [18] adds a GAN for spatial correlation to per-pixel normalizing flows, and AV1 adds autoregression. PGNM realizes this division as a classical Gaussian copula, a correlated Gaussian field transformed pixel by pixel to target marginals [9, 43], akin to texture synthesis [17, 19].

3 Method

PGNM separates intensity-conditioned residual distributions from spatial and color dependence. A full RGB convolution shapes a Gaussian latent field, which determines how residuals relate across pixels and channels; an empirical lookup then maps each latent value, given the clean intensity, to a signed residual (Fig. 2). The two components are fitted separately from aligned clean/noisy RGB pairs and remain fixed during synthesis.

Let $X \in [0,1]^{h\times w\times 3}$ be a clean image, $\mathbf{p} = (p, q)$ a pixel location, and $c \in C = \{R, G, B\}$ a channel. For fitting pairs $\{(X^{(n)}, Y^{(n)})\}_{n=1}^{N}$, the observed residual is

$$r_c^{(n)}(\mathbf{p}) = y_c^{(n)}(\mathbf{p}) - x_c^{(n)}(\mathbf{p}). \tag{1}$$

We describe each component together with its fitting procedure; numerical details are provided in the supplementary material.

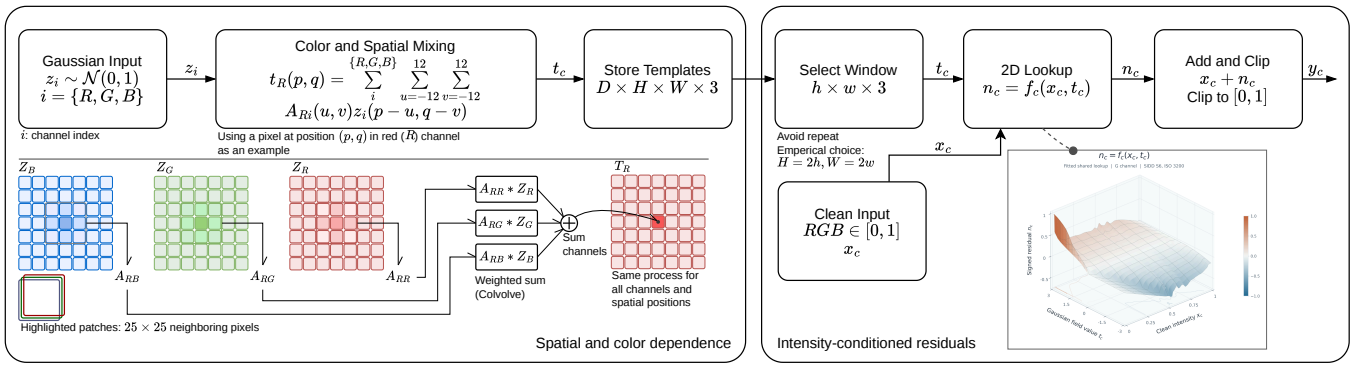

Figure 2 Inference pipeline. A full RGB convolution introduces spatial and color dependence into Gaussian latent templates. A shifted template window and the clean intensity index a residual lookup. We add its signed output to the clean image and clip to $[0, 1]$.

3.1 Spatial and color dependence

Synthesis. Starting from independent Gaussian values across locations and input channels,

$$z_i(p,q) \sim \mathcal{N}(0,1), \qquad i \in C, \tag{2}$$

we construct the latent field with a full $25 \times 25 \times 3 \times 3$ convolution tensor:

$$t_c(p,q) = \sum_{i \in C} \sum_{u=-12}^{12} \sum_{v=-12}^{12} A_{ci}(u,v) z_i(p-u, q-v). \tag{3}$$

Its nine filters A_{ci} couple input channel i to output channel c, jointly introducing spatial and color dependence. The per-output energy constraint

$$\sum_{i \in C} \sum_{u=-12}^{12} \sum_{v=-12}^{12} A_{ci}(u,v)^2 = 1 \qquad \text{for each } c \tag{4}$$

makes each latent value standard Gaussian whatever the filter shapes; for film grain, the spatial extent of the filters reflects grain size. For efficiency, we cache four independently constructed $2h \times 2w \times 3$ latent templates and reuse them by selecting random cyclic windows (details in the supplementary material).

Correlation fitting. We fit the filters so that the latent field reproduces residual correlations, measured where image content and clipping affect residuals least. Following noise estimation from weakly textured regions [11, 28], we define a clean-image flat mask $\mathcal{M}_{\text{flat}}^{(n)}$ by requiring the maximum RGB local standard deviation in a reflected 7×7 window to be at most a threshold. Guarding against clipping and normalizing residuals both need a strength estimate, so we fit a pilot strength curve $\sigma_c^{\text{pilot}}(x)$ on these flat pixels by matching observed second moments to a zero-mean Gaussian residual after image clipping. It serves only the correlation fit; the lookup (Sec. 3.2) is fitted separately on all pixels.

Correlation estimation additionally excludes clean intensities near the clipping boundaries. At each pixel, write $s_c = \sigma_c^{\text{pilot}}(x_c)$, omitting the image index and location; one shared RGB mask is

$$\begin{aligned} \mathcal{M}_{\text{clip}}^{(n)}(\mathbf{p}) &= \prod_{c \in C} \mathbf{1}\{2.5 s_c < x_c < 1 - 2.5 s_c\}, \\ \mathcal{M}_{\text{corr}}^{(n)}(\mathbf{p}) &= \mathcal{M}_{\text{flat}}^{(n)}(\mathbf{p}) \mathcal{M}_{\text{clip}}^{(n)}(\mathbf{p}). \end{aligned} \tag{5}$$

This clean-based guard reduces clipping contamination without selecting on the observed noisy value. If the combined mask is empty across the entire fitting set, which occurs only for the coarsest simulated film grain, we instead set $\mathcal{M}_{\text{corr}}^{(n)} = \mathcal{M}_{\text{flat}}^{(n)}$ for all images. Dividing by the pilot strength places residuals on the unit scale of the latent field, so their correlations estimate those of the field; we also center within each image:

$$u_c^{(n)}(\mathbf{p}) = \frac{r_c^{(n)}(\mathbf{p})}{s_c^{(n)}(\mathbf{p})} - \mu_c^{(n)}, \tag{6}$$

where $\mu_c^{(n)}$ is the mean of the normalized residuals on $\mathcal{M}_{\text{corr}}^{(n)}$.

For displacement $\mathbf{d}$, let $\mathcal{P}_{\mathbf{d}}$ contain all $(n, \mathbf{p})$ whose two endpoints remain within image n and pass its correlation mask. Pooling valid products and counts across images gives all nine channel-pair covariances and correlations:

$$\widehat{C}_{ab}(\mathbf{d}) = \frac{1}{|\mathcal{P}_{\mathbf{d}}|} \sum_{(n,\mathbf{p}) \in \mathcal{P}_{\mathbf{d}}} u_a^{(n)}(\mathbf{p}) u_b^{(n)}(\mathbf{p}+\mathbf{d}), \tag{7}$$

$$\widehat{R}_{ab}(\mathbf{d}) = \frac{\widehat{C}_{ab}(\mathbf{d})}{\sqrt{\widehat{C}_{aa}(\mathbf{0}) \widehat{C}_{bb}(\mathbf{0})}}. \tag{8}$$

We use lags in $[-16, 16]^2$, covering both same-pixel and shifted color dependence. By Eqs. 2 and 3, the latent field's correlation is that of its filters, with $\mathbf{k} = (u, v)$:

$$R_{ab}(\mathbf{d}) = \sum_{i \in C} \sum_{\mathbf{k}} A_{ai}(\mathbf{k}) A_{bi}(\mathbf{k}+\mathbf{d}), \tag{9}$$

so Eq. 4 is exactly $R_{cc}(\mathbf{0}) = 1$, and fitting means choosing filters with $R_{ab} = \widehat{R}_{ab}$. Eq. 9 is quadratic in the filters and couples all displacements, but the Fourier transform turns it into one 3×3 product per frequency: the filter spectrum times its conjugate transpose. We

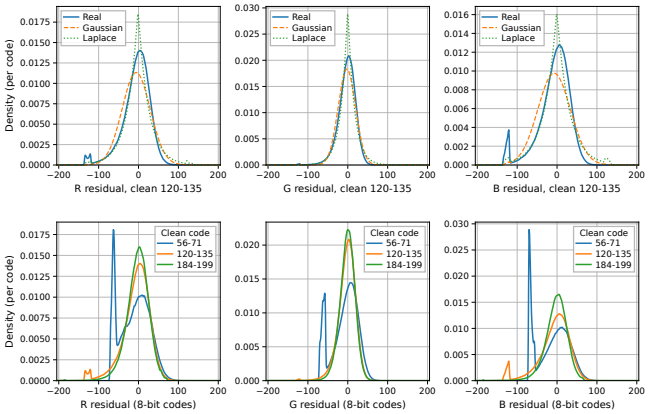

Figure 3 **Intensity-conditioned residual distributions** from SIDD [1]. Top: empirical densities versus common parametric distributions. Bottom: variation with clean intensity.

therefore factor the spectrum of the symmetrized and tapered $\widehat{R}_{ab}$: at each frequency, we project it onto positive semidefinite matrices and take its matrix square root (*spectral factorization*; supplementary material). We convert the spectral factor to spatial kernels, crop to 25×25, and normalize by Eq. 4. Projection and finite support make the fit approximate; the subsequent nonlinear lookup can also alter residual correlations.

3.2 Intensity-conditioned residuals

We represent the intensity-conditioned residual distributions (Fig. 3) with a two-dimensional lookup $f_c(x, t)$ indexed by clean intensity x and latent Gaussian value t.

Fitting. We construct residual histograms by channel and clean intensity using *all training pixels*, including textured and clipped observations. Pooling nearby intensity bins stabilizes the empirical cumulative distributions $\widehat{F}_{c,x}$. At grid points (x_ℓ, t_m), we store their quantiles:

$$f_c(x_\ell, t_m) = \widehat{F}^{-1}_{c,x_\ell}(\Phi(t_m)), \tag{10}$$

where Φ is the standard Gaussian cumulative distribution function. In this *conditional quantile mapping*, the Gaussian value selects a percentile and the clean intensity a residual distribution (Fig. 4). Because each latent value is standard Gaussian (Sec. 3.1), $\Phi(t_c)$ is uniform on $[0, 1]$, so the lookup reproduces $\widehat{F}_{c,x}$ up to interpolation and clipping (Eq. 11), whatever dependence the convolution introduces.

Synthesis. We evaluate $n_c(\mathbf{p}) = f_c(x_c(\mathbf{p}), t_c(\mathbf{p}))$ by bilinear interpolation, with linear extrapolation along the Gaussian axis beyond its stored range (Fig. 4). The final image is

$$\widehat{y}_c(\mathbf{p}) = \mathrm{clip}\big(x_c(\mathbf{p}) + n_c(\mathbf{p}), 0, 1\big). \tag{11}$$

4 Evaluation protocol

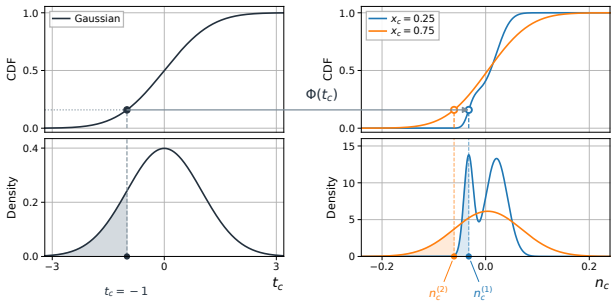

Figure 4 **Quantile mapping.** The Gaussian value t_c selects a percentile and clean intensity x_c selects a residual distribution. The lookup returns the corresponding signed residual n_c.

4.1 Noise synthesis fidelity

We evaluate residual-value distributions and their dependence using four complementary errors; lower values are better for all four. Real and generated images undergo the same clipping and round-half-up conversion to 8 bits before evaluation. Let $r_c = y_c - x_c$ and $\widehat{r}_c = \widehat{y}_c - x_c$ denote their signed residuals in normalized image units.

KL divergence D_{KL}**.** We follow the residual-histogram estimator released with sRGB-Flow [26]. Within each 32×32 tile, we pool all RGB residuals into 130 bins of width four code values over $[-260, 260]$, adding one count per bin before normalization. For real and generated bin probabilities p_j and q_j, the divergence is

$$D_{\mathrm{KL}}(p\|q) = \sum_{j=1}^{130} p_j \log \frac{p_j}{q_j}. \tag{12}$$

Conditional W_1**.** We compare residual distributions at similar clean intensities with the 1-Wasserstein (earth mover's) distance, which in one dimension averages the absolute difference between matching percentiles. Let F_{cb} and G_{cb} be the empirical cumulative distributions of real and generated residuals in channel c and clean-intensity bin b. We measure

$$E_{W_1} = 255 \sum_{c,b} w_{cb} \int_0^1 \left|F^{-1}_{cb}(a) - G^{-1}_{cb}(a)\right| da, \tag{13}$$

where w_{cb} weights bins by their pixel counts and channels equally, with $\sum_{c,b} w_{cb} = 1$, and 255 gives code units. It responds to conditional bias, strength, and distribution shape.

Spatial and color correlation. We measure *spatial correlation error* within each channel across nearby locations and *color correlation error* between channels at neighboring locations. Both compare normalized correlations of real and generated residuals using root mean square error. Normalization, lag supports, and estimator details are provided in the supplementary material.

4.2 Datasets

Smartphone Image Denoising Dataset (SIDD). We evaluate noise synthesis on the official sRGB validation split of the SIDD [1], released for the NTIRE 2019 real image denoising challenge [3].

The release contains 1,280 paired 256 × 256 blocks from 40 source images, with 32 blocks per image. PGNM fits one preset per camera–ISO class on SIDD-Medium sRGB pairs; we exclude the iPhone 7 ISO 1250 source because its class is absent from SIDD-Medium. The resulting evaluation set contains 1,248 blocks from 39 source images, spanning five cameras, 16 camera–ISO classes, and eight physical scenes. Source IDs, acquisition-condition coverage, and the fitting crops are listed in the supplementary material.

Simulated film grain. We use the Boolean film-grain renderer of Newson et al. [33], which places random grains of one radius over the image and blurs the result optically, so grain size is an explicit parameter. We render eight radii from 0.025 to 0.8, covering the fine grain evaluated by GrainNet [27] and coarser grain as a stress case. Fitting uses the grayscale training pairs released with GrainNet, about 100 per radius. Because the renderer blurs the image as well as adding grain, the clean image is not the expected grainy image. For each radius we therefore fit a 7×7 linear filter B on the training pairs and measure residuals against $\mu = B * x$. PGNM and Zhang23 [48] add their grain to μ, while GrainNet and an independent Newson render keep their native pipelines. Evaluation uses GrainNet's 60 natural test images, each rendered by Newson with five independent seeds. On this grayscale data, we report conditional W_1 and spatial correlation error, and add DISTS [12] and LPIPS [49] to compare the overall appearance of the delivered images. Results cover all radii and, separately, radii up to 0.2, where Zhang23's small-grain approximation applies.

4.3 Denoising

Denoising gives an independent view of realism: a denoiser trained on synthetic pairs works on real images only as far as the synthesized noise resembles real noise. We therefore train the same denoiser on real pairs and on pairs from each generator, and compare each generator with this real-data reference. Training follows the protocol released with NM-FlowGAN [18]: a DnCNN [47] trained for 30 epochs on 96×96 patches from 246 SIDD-Medium images, which cover exactly its 15 supported camera–ISO classes. PGNM is fitted per class on the same images; neural generators use the checkpoints of Tab. 1 at native strength. Noise is drawn anew for every patch and epoch and quantized to 8 bits, and all noise sources share initialization, patch order, and augmentation. NAFlow-NAS denotes NAFlow's published setting, which samples noise guided by each patch's real noisy observation. We train one seed per noise source and report PSNR and SSIM at the final epoch on all 1,280 SIDD validation blocks (supplementary material).

5 Results and analysis

Camera noise. PGNM has the lowest mean error on all four measures over all 39 SIDD sources (Tab. 1) and in most of the 16 classes (supplementary), although NM-FlowGAN's spatial error is slightly lower on the 19 sources it covers. The learned baselines already contain every ingredient of PGNM, most with far more free parameters fitted by gradient descent. Each network, however, fits its own training objective, and how its modules interact under that objective is hard to inspect; capacity in the wrong place, or supervision that constrains a property only indirectly, can leave that property off target. Their deviations indeed differ in kind: one baseline makes neighboring pixels too similar, another too independent, and the closest one, GMDCN [37], matches noise strength better than PGNM but the distribution shape and the dependence between channels at neighboring pixels less well (supplementary). PGNM instead gives each property a form just flexible enough for what the observations show and estimates it directly from them. Visually, it is on par with the baselines (Fig. 5).

Method	$D_{KL}\downarrow$	$W_1\downarrow$	Correlation↓ Spatial	Color
All 39 sources (16 camera–ISO classes)				
PGNM (ours)	**0.022**	**1.319**	**0.009**	**0.014**
NAFlow [23]	0.035	2.105	0.019	0.047
GMDCN* [37]	0.031	1.553	0.013	0.032
DANet [45]	0.143	2.589	0.016	0.045
19 sources in sRGB-Flow and NM-FlowGAN				
PGNM (ours)	**0.022**	**1.187**	0.011	**0.014**
NAFlow [23]	0.033	1.746	0.018	0.046
GMDCN* [37]	0.032	1.434	0.013	0.033
DANet [45]	0.150	2.334	0.017	0.042
sRGB-Flow [26]	0.051	4.197	0.023	0.045
NM-FlowGAN [18]	0.037	1.998	**0.010**	0.036

Table 1 Camera-noise synthesis on SIDD [1]. Mean over five generator runs, cameras weighted equally (supplementary); the best value in each block is in bold. *Our implementation of the published model.

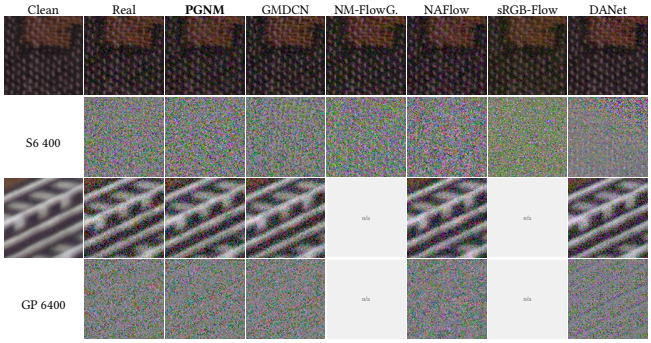

Figure 5 Visual comparisons on SIDD [1]: random (S6 400) and most textured (GP 6400) windows. PGNM is visually on par with the baselines and shows fewer repetitive patterns in the residual (S6 400). Second row of each pair: residual relative to the clean image, amplified by one gain per row (gray is zero). n/a: class not covered by the checkpoint. More in the supplementary material.

PGNM is content-aware through each pixel's clean intensity, which its lookup measures directly, but not tied to the surrounding structure, so on a new image its noise changes only as the intensities do. We applied all SIDD fits and checkpoints, unchanged, to a textured Kodak image (supplementary). NAFlow's [23] noise grows well above its SIDD level, and DANet [45], which has no camera or ISO input, gives about the same level for every class; the others, PGNM included, change less. GMDCN [37] and NM-FlowGAN [18] keep the edge-following part of the residual, which PGNM does not

Training noise	PSNR↑	SSIM↑
Real noise (reference)	37.94	0.896
GMDCN	36.15	0.872
NM-FlowGAN	35.15	0.875
NAFlow	36.72	0.878
NAFlow-NAS	36.88	0.883
PGNM (ours)	**37.22**	**0.889**

Table 2 Real-noise denoising on SIDD Validation (40 images, 1,280 blocks). One DnCNN per training noise (Sec. 4.3), training seed 1. NAFlow-NAS: noise-aware sampling, which also reads the real noisy training patch.

	Residual↓		Appearance↓	
Method	W_1	Spatial	DISTS	LPIPS
All eight radii (0.025–0.8)				
PGNM (ours)	**0.626**	**0.0082**	0.0731	0.0809
Zhang23* [48]	1.402	0.0203	0.1014	0.1397
GrainNet [27]	3.072	0.0202	**0.0690**	**0.0799**
Ind. Newson render	0.424	0.0068	0.0653	0.0785
Radii up to 0.2				
PGNM (ours)	**0.208**	0.0079	**0.0442**	**0.0338**
Zhang23* [48]	0.232	**0.0073**	**0.0442**	**0.0338**
GrainNet [27]	3.313	0.0275	0.0473	0.0357
Ind. Newson render	0.143	0.0066	0.0430	0.0333

Table 3 Simulated film grain [33]. Mean over the 25 output/reference pairs of each image and radius, then over the 60 test images and the radii of each block; the best fitted method is in bold. W_1: conditional W_1 in 8-bit code units; Spatial: spatial correlation error. The independent Newson render compares two realizations of the simulator and is not bolded. *Our implementation of the published model, with the shared blur B applied to the clean image.

model (Sec. 7).

Denoising. As in Tab. 1, PGNM ranks first among synthetic sources: a DnCNN trained on its noise is 0.72 dB below one trained on real pairs (Tab. 2). On all 40 images, the 95% scene-bootstrap interval of its lead over each baseline excludes zero; on the 21 images outside the training classes, its lead over NAFlow-NAS is a tie (supplementary).

Simulated film grain. Because the Boolean renderer is a physical model, we can compare PGNM with Zhang23's analytic approximation of it (Tab. 3). For small grain, Zhang23 is on par with PGNM, and up to $r = 0.6$ it is slightly closer in spatial correlation, as its isotropic correlation model follows the renderer's isotropic grain; as grains become resolved, the approximation degrades, and at the coarsest radius it no longer reproduces the renderer (Fig. 6). Without a physical model, PGNM has the lowest conditional W_1 from a radius of 0.075 upward. GrainNet matches the overall appearance best at coarse radii, where its spatial correlation error is also slightly lower than PGNM's (supplementary). Its output, however, is offset from the expected image by up to 7 codes at every radius, and its finest grain is too strong. The residual errors register this offset,

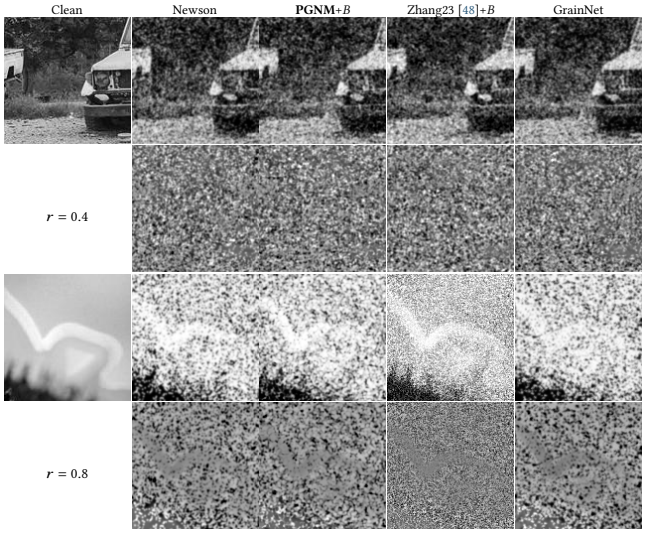

Figure 6 Uncurated film-grain comparison. PGNM and Zhang23 [48] add grain to the shared training-fitted blur $\mu = B * x$. Second row of each pair: residual relative to μ, amplified for visibility.

while the appearance distances largely tolerate it.

Model size and speed. A PGNM model stores the $25 \times 25 \times 3 \times 3$ convolution tensor (5,625 values) and a $3 \times 256 \times 25$ residual lookup (19,200 values; Tab. 4); on grayscale film, only 7,025 of its values per radius are distinct. One model is 5.7–1,250× smaller than GMDCN, NAFlow, DANet, and NM-FlowGAN, but larger than sRGB-Flow and GrainNet, and because PGNM fits one model per condition, its total over the 16 SIDD classes (397,200) exceeds GMDCN. Drawing and filtering a new noise field for every frame, as the baselines do, PGNM is 5–19× faster than the neural baselines on GPU and 3.6–57× faster on CPU; on CPU, Zhang23 remains cheaper. Because the filter does not depend on the image, all reported results read a random window from four templates filtered once per model and resolution (Sec. 3.1), as codec grain synthesis reuses a grain template [34]. Counting the time to build it, this bank is faster than per-frame filtering after about 30 frames on GPU and 24 on CPU.

Grain synthesis for video delivery. Netflix's *Nocturne* [32] has a clean master and a film-grain version. In a prototype outside any codec, we code 100 frames of the grain version, or code the master and add PGNM grain after decoding (Tab. 5). Coding the grain costs 9–14× the bits at equal quantizer (Q 15–31). Even at their highest lossy rates, both encoders stay farther from the grain version in DISTS than PGNM at 0.94 Mbps. They have lower spatial error, and VVC slightly lower LPIPS. Matched DISTS saves at least 90% (85% with nlmeans denoising). At the same rate, PGNM is also closer than AV1's grain synthesis in DISTS, LPIPS and W_1, with nearly equal spatial error. Sent once, its model costs less than AV1's parameters beyond about 20 s.

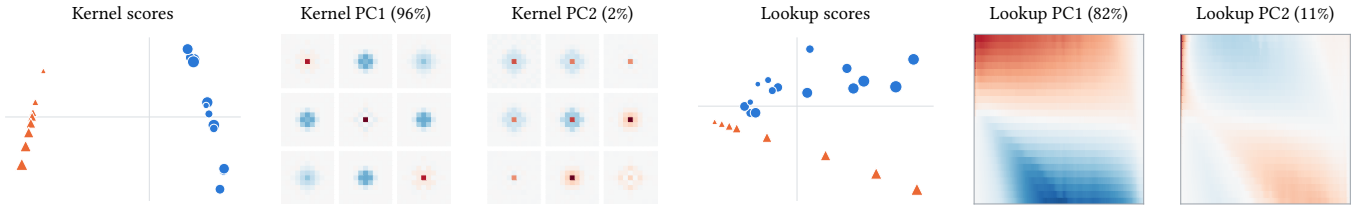

Figure 7 Joint PCA of the 24 fitted presets: convolution tensors (left) and residual lookups (right). Scores in PC1 (horizontal) and PC2: SIDD classes as blue circles and Newson radii as orange triangles, larger for higher ISO or radius. Kernel PC1 and lookup PC2 separate grain from camera noise; kernel PC2 groups SIDD by camera, and lookup PC1 orders presets by strength. Components: central 11 × 11 taps (rows output, columns input) and the green lookup over clean intensity (horizontal) and latent value (−3 to 3); red positive, blue negative.

	Parameters		Time per frame	
Method	Per model	Models	GPU ms	CPU s
SIDD (16 camera–ISO classes)				
sRGB-Flow† [26]	6,160	1	112.1	3.79
GMDCN [37]	140,688	1	395.1	35.39
NAFlow [23]	1,078,091	1	105.0	10.97
DANet [45]	9,153,859	1	163.7	16.04
NM-FlowGAN† [18]	31,057,359	1	220.0	59.47
PGNM (ours)	24,825	16		
new field per frame			20.4	1.04
bank playback			10.2	0.051
bank construction			307	23.45
Boolean film grain (8 radii)				
Zhang23 [48]	51	8	–	0.59
GrainNet [27]	9,857	1	118.2	8.62
PGNM (ours)	24,874	8		

Table 4 Model size and speed. Parameters: learned values read at inference (neural baselines: the noise generator; PGNM's fixed knot grid not counted; film rows include the shared 7 × 7 blur). Models: separately fitted models. Time: one 1920 × 1080 RGB frame, 8-bit clean to noisy image in host memory (median, batch 1, fp32), on one RTX A6000 (GPU) or two i9-14900K efficiency-core threads (CPU); setup details in the supplementary material. †Trained on NM-FlowGAN's 15 camera–ISO classes, 7 of the 16 evaluated. –: no GPU implementation.

Encoder	Grain	Mbps	DISTS↓	LPIPS↓	W_1 ↓	Spatial↓
AV1, Q 15	coded	73.0	0.089	0.102	1.30	0.056
VVC, QP 17	coded	94.5	0.091	0.085	1.21	0.046
AV1, Q 31	AV1 [34]	0.92	0.098	0.118	1.79	0.064
VVC, 2-pass	VVC [31]	0.99	0.201	0.250	1.09	0.076
AV1, Q 31	**PGNM**	1.63	0.055	0.080	0.73	0.037
AV1, Q 55	**PGNM**	0.94	0.069	0.086	0.94	0.065

Table 5 Video delivery (*Nocturne*, 100 frames, 1080p60; VVC: VVenC [40]). Coded: grain encoded, temporal filter on or off, the closer one. Otherwise synthesized after decoding (VVC: estimated from the grain version); PGNM's rate includes its 161 KB model.

6 Ablation

We ablate one fitted component at a time on SIDD, and the kernel support also on film grain (Tab. 6). Each variant is refitted on the same pairs and rendered from the same random streams, so differences come from that component alone; the full variant reproduces the main-table PGNM scores. The ablation was specified after the main evaluation, before its outputs existed, and did not change the method.

Fitting masks. The lookup does not use the fitting masks, so removing them changes only the correlation fit. Without the flat mask, textured pixels enter that fit and weaken the fitted cross-channel dependence between neighboring pixels, which raises color correlation error and, slightly, KL. The flat mask alone recovers full PGNM; the clipping guard alone does not (supplementary material).

Kernel support. Spatial support is essential, but the fitted dependence is compact. On SIDD, the errors level off by 5 × 5. On Newson film grain, 1 × 1 and 3 × 3 supports lose the grain's spatial structure and appearance, increasingly so for coarser grain, whereas 5 × 5 and 11 × 11 match 25 × 25 up to the coarsest radius, with slightly lower spatial correlation error, as on SIDD. Distribution measures improve slightly as the support shrinks (KL on SIDD, conditional W_1 on film grain): spatially independent noise makes each image's empirical distribution fluctuate less, and these measures do not see dependence. We nevertheless fixed a generous 25 × 25 support in advance for every dataset, leaving room for longer-range dependence (correlations are estimated up to 16 pixels) instead of tuning the support on evaluation data. The margin costs little: all timings in Tab. 4 use 25 × 25, and with template playback the support affects only the one-time filtering. In deployment, a smaller support such as 11 × 11, with five times fewer kernel values, is a free option. The fitted presets are also highly structured: across all 24 presets, two principal components explain 98% of the kernel and 93% of the lookup variation (Fig. 7).

Conditional distribution. Replacing the lookup by a Gaussian with the same mean and variance at each clean intensity more than doubles KL and conditional W_1, while the correlation errors stay tied: a Gaussian cannot reproduce the skewed, heavy-tailed shape of SIDD residuals at a fixed intensity (Fig. 3), which the lookup preserves.

7 Limitations and Conclusion

Limitations. PGNM represents only what its two components express: a stationary latent filter and a lookup that sees the clean image through each pixel's intensity alone. Higher-capacity generators can learn residual structure that follows image content, varies

SIDD	$D_{KL}\downarrow$	$W_1\downarrow$	Spatial↓	Color↓
Full PGNM	0.0219	1.32	0.0094	0.0144
No fitting masks	+1.2%	+0.0%	−1.7%	+46%
Kernel 1 × 1	−5.4%	+0.0%	+586%	+770%
Kernel 5 × 5	−0.8%	+0.2%	−6.5%	+0.1%
Kernel 11 × 11	−0.9%	+0.1%	−8.5%	+1.2%
Gaussian conditional	+138%	+108%	+4.5%	+2.0%
Film grain	$W_1\downarrow$	Spatial↓	DISTS↓	LPIPS↓
Full PGNM	0.626	0.0082	0.0731	0.0809
Kernel 1 × 1	−11%	+1164%	+193%	+230%
Kernel 5 × 5	−0.2%	−8.6%	+0.2%	−0.2%
Kernel 11 × 11	−0.2%	−5.9%	+0.2%	+0.0%

Table 6 Ablation. Full PGNM's scores, then each variant's change (lower is better): worse, better, or tie. SIDD (39 sources; D_{KL} on 32×32 tiles): tie when the 95% scene-bootstrap interval includes zero. Film grain (mean over eight Newson radii): colored when most of the 60 images change in that direction at every radius. Further variants and intervals are in the supplementary material.

between captures, or has dependence far from Gaussian; PGNM cannot. On SIDD, part of the residual follows image edges: if this stems from references averaged over many captures and thus softer than the noisy images, PGNM avoids fitting a dataset artifact; if it is a real noise pattern, PGNM misses it. A gap to real noise remains: training a denoiser on PGNM noise falls short of training on real pairs (Tab. 2), and the independent Newson render is closer to the reference on every measure (Tab. 3). Each condition needs its own fit from aligned clean/noisy pairs; one-shot fitting is not yet supported.

Conclusion. The idea behind PGNM is to model noise only as far as it can be measured. Each component is fitted to an observable statistic, so a model can be checked component by component, and camera noise and film grain become points in one parameter space (Fig. 7). Two such measurements already account for much of the realism of far larger generators; what they miss marks where learned capacity is needed.

AI use disclosure

This work used Claude (Anthropic) through Claude Code as an assistant for software development (the PGNM implementation, the experiment, evaluation and figure scripts, and the supplementary video) and for drafting and editing the text. The author directed its use, reviewed its output, and takes full responsibility for the content.

References

[1] Abdelrahman Abdelhamed, Stephen Lin, and Michael S. Brown. A high-quality denoising dataset for smartphone cameras. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1692–1700. IEEE, 2018.

[2] Abdelrahman Abdelhamed, Marcus A. Brubaker, and Michael S. Brown. Noise Flow: Noise modeling with conditional normalizing flows. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3165–3173. IEEE, 2019.

[3] Abdelrahman Abdelhamed, Radu Timofte, Michael S. Brown, et al. NTIRE 2019 challenge on real image denoising: Methods and results. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019.

[4] Zoubida Ameur, Wassim Hamidouche, Edouard François, Miloš Radosavljević, Daniel Menard, and Claire-Hélène Demarty. Deep-Based Film Grain Removal and Synthesis. *IEEE Transactions on Image Processing*, 32:5046–5059, 2023.

[5] Zoubida Ameur, Claire-Hélène Demarty, Daniel Menard, and Olivier Le Meur. 3R-INN: How to be climate friendly while consuming/delivering videos? In *European Conference on Computer Vision (ECCV)*, 2024.

[6] Zoubida Ameur, Claire-Hélène Demarty, Olivier Le Meur, and Daniel Ménard. Style-FG: A Style-based Framework for Film Grain Analysis and Synthesis. *ACM Transactions on Multimedia Computing, Communications, and Applications*, 21(9):1–24, 2025.

[7] Tim Brooks, Ben Mildenhall, Tianfan Xue, Jiawen Chen, Dillon Sharlet, and Jonathan T. Barron. Unprocessing images for learned raw denoising. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11028–11037. IEEE, 2019.

[8] Yuanhao Cai, Xiaowan Hu, Haoqian Wang, Yulun Zhang, Hanspeter Pfister, and Donglai Wei. Learning to Generate Realistic Noisy Images via Pixel-level Noise-aware Adversarial Training. In *Advances in Neural Information Processing Systems*, 2021.

[9] Marne C. Cario and Barry L. Nelson. Autoregressive to Anything: Time-Series Input Processes for Simulation. *Operations Research Letters*, 19 (2):51–58, 1996.

[10] Ke-Chi Chang, Ren Wang, Hung-Jin Lin, Yu-Lun Liu, Chia-Ping Chen, Yu-Lin Chang, and Hwann-Tzong Chen. Learning Camera-Aware Noise Models. In *European Conference on Computer Vision (ECCV)*, 2020.

[11] Miguel Colom and Antoni Buades. Analysis and extension of the Ponomarenko et al. method, estimating a noise curve from a single image. *Image Processing On Line*, 3:173–197, 2013.

[12] Keyan Ding, Kede Ma, Shiqi Wang, and Eero P. Simoncelli. Image quality assessment: Unifying structure and texture similarity. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(5):2567–2581, 2022.

[13] Alessandro Foi, Mejdi Trimeche, Vladimir Katkovnik, and Karen Egiazarian. Practical Poissonian-Gaussian Noise Modeling and Fitting for Single-Image Raw-Data. *IEEE Transactions on Image Processing*, 17(10):1737–1754, 2008.

[14] FotoKem. DP Van Hoytema and FotoKem talk 65mm and Oppenheimer, 2023. Published November 22, 2023. Accessed October 13, 2026.

[15] Foundry. *Applying Grain*. Nuke 16.0 documentation. Accessed October 13, 2026.

[16] Zixuan Fu, Lanqing Guo, and Bihan Wen. sRGB Real Noise Synthesizing with Neighboring Correlation-Aware Noise Model. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1683–1691, 2023.

[17] Bruno Galerne, Yann Gousseau, and Jean-Michel Morel. Random Phase Textures: Theory and Synthesis. *IEEE Transactions on Image Processing*, 20(1):257–267, 2011.

[18] Young Joo Han and Ha-Jin Yu. NM-FlowGAN: Modeling sRGB Noise without Paired Images using a Hybrid Approach of Normalizing Flows and GAN, 2023. arXiv:2312.10112.

[19] David J. Heeger and James R. Bergen. Pyramid-Based Texture Analysis/Synthesis. In *Proceedings of the 22nd Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pages 229–238, 1995.

[20] Geonwoon Jang, Wooseok Lee, Sanghyun Son, and Kyoung Mu Lee. C2N: Practical Generative Noise Modeling for Real-World Denoising. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 2350–2359, 2021.

[21] A. Khintchine. Korrelationstheorie der stationären stochastischen Prozesse. *Mathematische Annalen*, 109(1):604–615, 1934.

[22] Changjin Kim, HyeokJun Lee, and YoungJoon Yoo. GuidNoise: Single-Pair Guided Diffusion for Generalized Noise Synthesis. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(7):5639–5647, 2026.

[23] Dongjin Kim, Donggoo Jung, Sungyong Baik, and Tae Hyun Kim. sRGB Real Noise Modeling via Noise-Aware Sampling with Normalizing Flows. In *International Conference on Learning Representations (ICLR)*, 2024.

[24] Jaekyun Ko, Dongjin Kim, Soomin Lee, Guanghui Wang, and Tae Hyun Kim. Diffusion-Based sRGB Real Noise Generation via Prompt-Driven Noise Representation Learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 35956–35966, 2026.

[25] Jaekyun Ko, Byung Wan Lim, Soomin Lee, Dongjin Kim, and Tae Hyun Kim. YeTI: You Only Need Two Noisy Images for Real-World sRGB Noise Generation. In *European Conference on Computer Vision (ECCV)*, 2026.

[26] Shayan Kousha, Ali Maleky, Michael S. Brown, and Marcus A. Brubaker. Modeling sRGB Camera Noise with Normalizing Flows. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17463–17471, 2022.

[27] Gwilherm Lesné, Yann Gousseau, Saïd Ladjal, and Alasdair Newson. Neural film grain rendering. *Computer Graphics Forum*, 44(2):e70076, 2025.

[28] Xinhao Liu, Masayuki Tanaka, and Masatoshi Okutomi. Single-image noise level estimation for blind denoising. *IEEE Transactions on Image Processing*, 22(12):5226–5237, 2013.

[29] Yang Liu, Zhenyue Qin, Saeed Anwar, Pan Ji, Dongwoo Kim, Sabrina Caldwell, and Tom Gedeon. Invertible Denoising Network: A Light Solution for Real Noise Removal. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13365–13374, 2021.

[30] Ali Maleky, Shayan Kousha, Michael S. Brown, and Marcus A. Brubaker. Noise2NoiseFlow: Realistic Camera Noise Modeling without Clean Images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[31] Vignesh V Menon, Adam Wieckowski, Christian Stoffers, Jens Brandenburg, Christian Lehmann, Benjamin Bross, Thomas Schierl, and Detlev Marpe. Enhancing Film Grain Coding in VVC: Improving Encoding Quality and Efficiency, 2024. arXiv:2407.12465; accepted at IBC 2024.

[32] Netflix. Nocturne. Netflix Open Content, Creative Commons Attribution 4.0, 2018. Clean master and film-grain version. Accessed October 21, 2026.

[33] Alasdair Newson, Julie Delon, and Bruno Galerne. A Stochastic Film Grain Model for Resolution-Independent Rendering. *Computer Graphics Forum*, 36(8):684–699, 2017.

[34] Andrey Norkin and Neil Birkbeck. Film grain synthesis for AV1 video codec. In *2018 Data Compression Conference*, pages 3–12. IEEE, 2018.

[35] Athanasios Papoulis and S. Unnikrishna Pillai. *Probability, Random Variables, and Stochastic Processes*. McGraw-Hill, 4th edition, 2002.

[36] Masanobu Shinozuka. Simulation of Multivariate and Multidimensional Random Processes. *The Journal of the Acoustical Society of America*, 49(1B):357–368, 1971.

[37] Mingyang Song, Yang Zhang, Tunç O. Aydın, Elham Amin Mansour, and Christopher Schroers. A Generative Model for Digital Camera Noise Synthesis, 2023. arXiv:2303.09199.

[38] Daizong Tian and Thrasyvoulos N. Pappas. Efficient Digital Film Grain Simulation. *IEEE Transactions on Image Processing*, 35:8851–8862, 2026.

[39] Kaixuan Wei, Ying Fu, Jiaolong Yang, and Hua Huang. A Physics-based Noise Formation Model for Extreme Low-light Raw Denoising. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[40] Adam Wieckowski, Jens Brandenburg, Tobias Hinz, Christian Bartnik, Valeri George, Gabriel Hege, Christian Helmrich, Anastasia Henkel, Christian Lehmann, Christian Stoffers, Ivan Zupancic, Benjamin Bross, and Detlev Marpe. VVenC: An open and optimized VVC encoder implementation. In *2021 IEEE International Conference on Multimedia & Expo Workshops (ICMEW)*, pages 1–2. IEEE, 2021.

[41] Norbert Wiener. Generalized Harmonic Analysis. *Acta Mathematica*, 55:117–258, 1930.

[42] Qi Wu, Mingyan Han, Ting Jiang, Chengzhi Jiang, Jinting Luo, Man Jiang, Haoqiang Fan, and Shuaicheng Liu. Realistic Noise Synthesis with Diffusion Models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(8):8432–8440, 2025.

[43] Fumio Yamazaki and Masanobu Shinozuka. Digital Generation of Non-Gaussian Stochastic Fields. *Journal of Engineering Mechanics*, 114(7): 1183–1197, 1988.

[44] Wenbo Yang, Zhongling Wang, and Zhou Wang. Towards a Universal Image Degradation Model via Content-Degradation Disentanglement. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

[45] Zongsheng Yue, Qian Zhao, Lei Zhang, and Deyu Meng. Dual Adversarial Network: Toward Real-world Noise Removal and Noise Generation. In *European Conference on Computer Vision (ECCV)*, 2020.

[46] Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Ming-Hsuan Yang, and Ling Shao. CycleISP: Real Image Restoration via Improved Data Synthesis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[47] Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 26(7):3142–3155, 2017.

[48] Kaixuan Zhang, Jingxian Wang, Daizong Tian, and Thrasyvoulos N. Pappas. Film grain rendering and parameter estimation. *ACM Transactions on Graphics*, 42(4), 2023.

[49] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595, 2018.

[50] Dihan Zheng, Yihang Zou, Xiaowen Zhang, and Chenglong Bao. SeNM-VAE: Semi-Supervised Noise Modeling with Hierarchical Variational Autoencoder. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 25889–25899, 2024.

[51] Yunhao Zou and Ying Fu. Estimating Fine-Grained Noise Model via Contrastive Learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12682–12691, 2022.

Parametric Gaussian Noise Mixer (PGNM)

Explanation and evidence

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For exact reimplementation, jump to this part for algorithms, numerical conventions, data splits, random seeds, and evaluation procedures.

Reading the scores. Conditional W_1 compares residual distributions at similar clean intensities; spatial and color correlation errors compare dependence; tile KL compares local residual histograms. Film DISTS and LPIPS compare overall image appearance. Lower values are better for these scores. Exact estimators are in Sec. J. Numbers with the prefix S and lettered sections refer to this supplement, plain numbers to the main paper. Image crops are shown without interpolation; please zoom in.

A PGNM in equations

This section restates PGNM as plain formulas, one step per paragraph; Sec. I gives the same steps as exact algorithms. Images are in $[0, 1]$, x is the clean image, y the noisy one, $r = y - x$ the residual, c a color channel and $\mathbf{p}$ a pixel location.

Synthesis. Independent Gaussian values are filtered by a full RGB convolution and then mapped, pixel by pixel, to signed residuals:

$$t_c = \sum_{i \in \{R,G,B\}} A_{ci} * z_i, \qquad z_i(\mathbf{p}) \sim \mathcal{N}(0,1), \tag{S1}$$

$$1 = \sum_i \sum_{u,v} A_{ci}(u,v)^2 \quad \text{for each } c, \tag{S2}$$

$$\hat{y}_c(\mathbf{p}) = \mathrm{clip}\big(x_c(\mathbf{p}) + f_c\big(x_c(\mathbf{p}), t_c(\mathbf{p})\big),\ 0,\ 1\big), \tag{S3}$$

$$f_c(x,t) = F_{c,x}^{-1}\big(\Phi(t)\big). \tag{S4}$$

The energy constraint (S2) makes every $t_c(\mathbf{p})$ standard Gaussian, so $\Phi(t_c)$ is uniform and the residual at clean intensity x follows the fitted distribution $F_{c,x}$ whatever the filters, up to interpolation and clipping. In practice t is read from a random cyclic window of one of four precomputed fields (Alg. S5).

Residual lookup. $F_{c,x}$ is the empirical distribution of the integer residual codes $\mathrm{rint}(255\, r_c)$ of channel c over *all* training pixels whose clean code lies within $\pm\rho$ codes of $\mathrm{rint}(255\, x)$, with $\rho \geq 2$ the smallest radius that collects at least 2,000 samples (or spans all codes). The lookup stores f_c at the 256 clean codes and at 25 Gaussian values $t \in \{-3, -2.75, \ldots, 3\}$, treating each residual code as a unit-width bin with uniform mass (Alg. S4); it is evaluated by bilinear interpolation, with linear extrapolation in t (Alg. S5).

Flat pixels. The filters are fitted only where residuals are not mixed with image structure. With $\kappa(\mathbf{p}) = \max_c \mathrm{sd}_7(x_c)(\mathbf{p})$ the largest local standard deviation over the channels in a 7×7 window,

$$\mathcal{M}_{\text{flat}} = \{\mathbf{p} : \kappa(\mathbf{p}) \leq \tau\}, \tag{S5}$$

$$\tau = \mathrm{clip}\big(2\, q_{2\%}(\kappa),\ 0.003,\ 0.03\big), \tag{S6}$$

where $q_{2\%}$ is the second percentile over the training images.

Clipping-aware pilot strength. To put residuals from different intensities on one scale before measuring their correlation, we need the strength $\sigma_c(x)$ of the noise *before* the image was clipped to $[0, 1]$. Near black or white, clipping pins part of the noise at the boundary and the observed spread underestimates σ (Fig. S1a). Left uncorrected, residuals there would be inflated by the normalization below, and the clipping guard would admit pixels whose noise does clip. We therefore model a flat pixel of clean intensity I as $y = \mathrm{clip}(I + \sigma\varepsilon, 0, 1)$ with $\varepsilon \sim \mathcal{N}(0,1)$ and use the exact expected squared residual of this model. The noise is clipped at 0 when $\varepsilon < a = -I/\sigma$ and at 1 when $\varepsilon > b = (1-I)/\sigma$; with ϕ and Φ the standard Gaussian density and distribution function,

$$\begin{aligned} m(\sigma, I) = \mathbb{E}\big[(y-I)^2\big] &= \int_{-\infty}^{\infty} (y-I)^2\, \phi(\varepsilon)\, d\varepsilon \\ &= \underbrace{\int_{-\infty}^{a} I^2\, \phi(\varepsilon)\, d\varepsilon}_{\text{clipped at 0}} + \underbrace{\int_{a}^{b} \sigma^2\varepsilon^2\, \phi(\varepsilon)\, d\varepsilon}_{\text{not clipped}} \\ &\quad + \underbrace{\int_{b}^{\infty} (1-I)^2\, \phi(\varepsilon)\, d\varepsilon}_{\text{clipped at 1}} \\ &= I^2\, \Phi(a) + \sigma^2\big[\Phi(b) - \Phi(a) + a\phi(a) - b\phi(b)\big] \\ &\quad + (1-I)^2\big(1 - \Phi(b)\big), \end{aligned} \tag{S7}$$

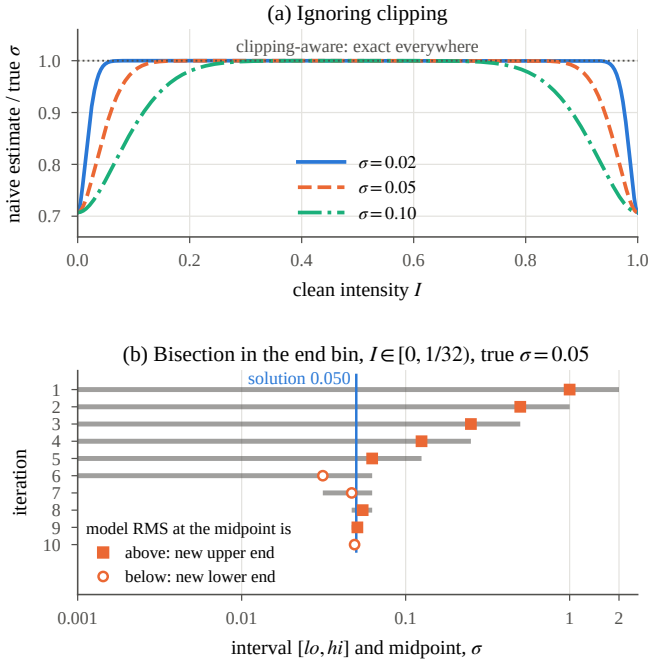

Figure S1 Clipping-aware pilot strength (Eqs. S7–S10), computed with the fitting code on illustrative values. (a) For a Gaussian residual clipped to $[0, 1]$, the naive strength $\sqrt{\text{mean}\, r^2}$ falls below the true σ near black and white, by 29% at the boundary itself and over a wider range for stronger noise; the clipping-aware pilot is exact at every intensity. (b) Solving Eq. S10 in the end bin $k = 0$, with clean intensities spread over $[0, 1/32)$ and a true σ of 0.05 (observed RMS 0.037, the naive estimate). Each midpoint whose model RMS exceeds the observed RMS becomes the new upper end of the interval, any other the new lower end, so the interval halves at every step (60 steps in the fitter).

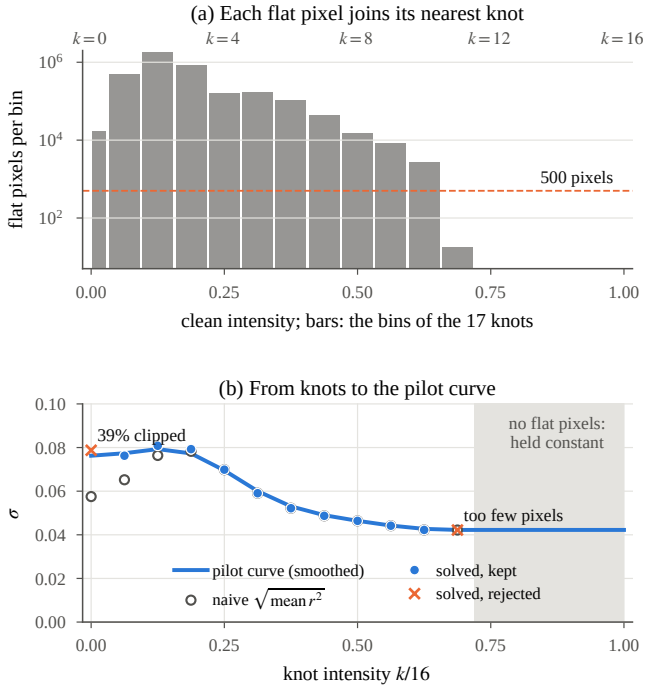

Figure S2 Building the pilot curve on real data: SIDD GP ISO 3200, green channel, with the fitter's own flatness threshold and accumulator on the class's fitting crops. (a) Flat pixels per bin: each pixel joins its nearest of the 17 knots, so a bin spans 16 codes and an end bin 8. (b) At the dark knots $k \le 3$, the naive estimate falls short of the clipping-aware solution. Knot 0 is rejected because 39% of its bin's noise values are predicted to clip and knot 11 because its bin holds 18 pixels; bins 12–16 have no flat pixels. Their values come from the nearest kept knots before one smoothing pass.

where the outer terms are constants times the Gaussian tail probabilities $\Phi(a)$ and $1 - \Phi(b)$. For the middle term, the standard Gaussian density satisfies $\phi'(\varepsilon) = -\varepsilon\,\phi(\varepsilon)$, so integration by parts gives

$$\begin{aligned}\int_a^b \varepsilon^2\phi(\varepsilon)\,d\varepsilon &= \int_a^b \varepsilon\big(-\phi'(\varepsilon)\big)\,d\varepsilon \\ &= \big[-\varepsilon\,\phi(\varepsilon)\big]_a^b + \int_a^b \phi(\varepsilon)\,d\varepsilon \\ &= \Phi(b) - \Phi(a) + a\,\phi(a) - b\,\phi(b).\end{aligned}$$

Far from the boundaries $m \approx \sigma^2$; near them m is clearly smaller, and m always increases with σ.

Fitting σ with Eq. S7 rests on two ideas. A single squared residual says little about σ, so the pilot pools many flat pixels into a few intensity bins, separately per channel. The pixels of a bin keep their own clean intensities, however, and near black and white these decide how much of their noise clips; the model's prediction for a bin therefore averages Eq. S7 over the pixels' own intensities, each pixel counting equally, rather than evaluating it once at the bin's position. What follows spells out the bins, the two bin averages and how the fitter computes them.

The pilot uses 17 knots at $k/16$, $k = 0, \ldots, 16$ (codes 0, 15.9, 31.9, …, 239.1, 255), and each flat pixel joins its nearest knot. The knots' bins are therefore the clean codes $[0, 8)$, $[8, 24)$, $[24, 40)$, …, $[232, 248)$ and $[248, 256)$: 16 codes each, and 8 for the two end knots, which sit at 0 and 255 (Fig. S2a). With $P_{c,k}$ the flat pixels of bin k in channel c, the bin's observation is its mean squared residual,

$$\overline{r^2}_{c,k} = \frac{1}{|P_{c,k}|} \sum_{\mathbf{p} \in P_{c,k}} r_c(\mathbf{p})^2. \tag{S8}$$

The model's prediction for the same pixels is the average of Eq. S7 at their clean intensities. The fitter streams over millions of pixels and cannot keep each one, so it splits every bin into 32 equal subbins of about half a code (the end bins use 16) and keeps, for subbin j, the number $n_{c,k,j}$ of the bin's flat pixels in it and their mean clean intensity $\bar{x}_{c,k,j}$. The prediction is then

$$\begin{aligned}\bar{m}_{c,k}(\sigma) &= \sum_j \frac{n_{c,k,j}}{|P_{c,k}|}\, m\big(\sigma, \bar{x}_{c,k,j}\big), \\ |P_{c,k}| &= \sum_j n_{c,k,j},\end{aligned} \tag{S9}$$

so each subbin counts with its share of the bin's pixels. For 8-bit clean images, each occupied subbin holds a single code and $\bar{x}_{c,k,j}$

is that code, so Eq. S9 equals the per-pixel average exactly. Alg. S2 rounds the shares to multiples of 1/512, at least one per occupied subbin; on all 16 SIDD classes, this changes the kept knot strengths by at most 0.32%. The subbins are finer than 8-bit codes because clean images need not be 8-bit: for the continuous blurred film images μ (Sec. I.4), $\bar{x}_{c,k,j}$ averages nearby intensities, for example to 7.5 codes, where Eq. S7 is evaluated directly. Because m increases with σ, the knot's strength $\sigma_{c,k}$ is the single solution of

$$\bar{m}_{c,k}(\sigma_{c,k}) = \overline{r^2}_{c,k}, \tag{S10}$$

found by bisection over $0 < \sigma \le 2$ with 60 halvings of the interval (Fig. S1b).

Each pixel enters at its own intensity because clipping changes quickly near black and white. In bin 0 (codes 0 to 7), Eq. S7 clips half of the noise values at code 0 but only a few at code 7 for typical noise. Evaluating all of the bin's pixels at the knot value 0 would treat the pixels at code 7 as half clipped and predict too little residual energy for them; to match the observation, the solve would raise σ, inflating the pilot near black, and likewise near white (for equally many pixels at codes 0 and 7 and $\sigma = 0.02$, it returns 0.023 instead of 0.020). Away from the boundaries $m \approx \sigma^2$, and the intensities within a bin barely matter.

A knot's solution is kept only if three conditions hold. First, its bin has at least 500 flat pixels. Second, the solution is below the upper end $\sigma = 2$ of the search, twice the whole intensity range $[0, 1]$; a strength this large means that clipped Gaussian noise does not explain the bin, for instance when misregistration leaves image structure in the residuals. Third, at most 35% of the bin's noise values are predicted to clip. Rejected knots take the value interpolated linearly between kept knots, or the nearest kept value beyond the outermost ones; the 17 values are then smoothed once with weights $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, end values held, and $\sigma_c(x)$ is linear between knots (Fig. S2b; Alg. S2). Without the pooling, with one knot per clean code, the final spatial correlation error is 1.8% higher (Sec. B). The pilot only prepares the correlation fit: the lookup describes the observed, clipped residuals directly and needs no correction.

Correlation. Pixels whose noise is likely to clip are excluded with a guard on the clean image, and residuals are normalized by the pilot and centered within each image:

$$\mathcal{M} = \mathcal{M}_{\text{flat}} \cap \bigcap_c \{\mathbf{p} : 2.5\,\sigma_c(x_c) < x_c < 1 - 2.5\,\sigma_c(x_c)\}, \tag{S11}$$

$$u_c = \frac{r_c}{\sigma_c(x_c)} - \text{mean}_{\mathcal{M}} \frac{r_c}{\sigma_c(x_c)}. \tag{S12}$$

Pooled over all training images, the measured correlation of channel a at $\mathbf{p}$ with channel b at $\mathbf{p} + \mathbf{d}$, $\mathbf{d} \in [-16, 16]^2$, is $\widehat{R}_{ab}(\mathbf{d}) = \widehat{C}_{ab}(\mathbf{d}) / \sqrt{\widehat{C}_{aa}(\mathbf{0})\,\widehat{C}_{bb}(\mathbf{0})}$ with $\widehat{C}_{ab}(\mathbf{d})$ the mean of $u_a(\mathbf{p})\,u_b(\mathbf{p}+\mathbf{d})$ over pairs inside $\mathcal{M}$ (Eqs. 7–8 of the main paper). If $\mathcal{M}$ is empty in the whole training set, $\mathcal{M}_{\text{flat}}$ is used instead.

Spectral factorization. A covariance describes how variables vary together; a linear transformation describes how to construct that dependence from shared inputs. If a zero-mean vector $\mathbf{z}$ has covariance I, then $\mathbf{t} = L\mathbf{z}$ has covariance

$$\text{Cov}(\mathbf{t}) = L\,\text{Cov}(\mathbf{z})L^{\mathsf{T}} = LL^{\mathsf{T}}.$$

Constructing variables with a desired covariance C therefore amounts to finding a factor L such that $C = LL^{\mathsf{T}}$. For a stationary spatial field, covariance depends only on the displacement between locations. Fourier coordinates separate this translation-invariant construction by frequency, replacing a large spatial factorization with one scalar or small channel-matrix factorization at each frequency. This is the general idea behind spectral factorization. Here it turns the measured correlations $\widehat{R}$ into the convolution filters of Eq. S1.

To see how a filter creates dependence, consider independent, zero-mean, unit-variance inputs $z(p)$ and let

$$t(p) = \frac{z(p-1) + z(p)}{\sqrt{2}}.$$

The neighboring outputs $t(p)$ and $t(p{+}1)$ share $z(p)$: a large value of that input raises both outputs. Each output has variance one, while their covariance is 1/2; outputs two or more positions apart share no inputs and have covariance zero. More generally, convolution forms $t(p) = \sum_u a(u)z(p-u)$, and its output covariance is

$$\begin{aligned} R(d) &:= \mathbb{E}[t(p)t(p+d)] \\ &= \sum_{u,v} a(u)a(v)\,\mathbb{E}[z(p-u)z(p+d-v)] \\ &= \sum_u a(u)a(u+d). \end{aligned} \tag{S13}$$

Only terms referring to the same input location survive the expectation. Thus $R(d)$ is the overlap of the filter with a copy shifted by d (Fig. S3b,c). At zero displacement, $R(0) = \sum_u a(u)^2$; with unit filter energy, covariance and normalized correlation coincide. Equation S13 is the forward problem: a filter determines a covariance. Fitting solves the inverse problem: construct a filter from a desired covariance.

In spatial coordinates, this inverse problem couples filter coefficients through their products. To separate it by frequency, define the filter's frequency response and the covariance's power spectrum as

$$\tilde{a}(\omega) = \sum_u a(u)e^{-\mathrm{i}\omega u}, \qquad S(\omega) = \sum_d R(d)e^{-\mathrm{i}\omega d},$$

where $\mathrm{i}^2 = -1$ and $\omega \in [-\pi, \pi]$ is angular frequency. The complex gain $\tilde{a}(\omega)$ describes how the filter scales and shifts a wave of frequency ω; $S(\omega)$ describes output power at that frequency. Substitution of Eq. S13, followed by $v = u + d$, gives

$$\begin{aligned} S(\omega) &= \sum_{u,v} a(u)a(v)e^{-\mathrm{i}\omega(v-u)} \\ &= \left(\sum_u a(u)e^{\mathrm{i}\omega u}\right)\left(\sum_v a(v)e^{-\mathrm{i}\omega v}\right) \\ &= \overline{\tilde{a}(\omega)}\,\tilde{a}(\omega) = |\tilde{a}(\omega)|^2. \end{aligned} \tag{S14}$$

The bar denotes complex conjugation. The Wiener–Khinchin relation identifies the Fourier transform of the covariance as the power spectrum [21, 41]; the calculation above expresses that spectrum in terms of the filter. Filtering multiplies the amplitude at each

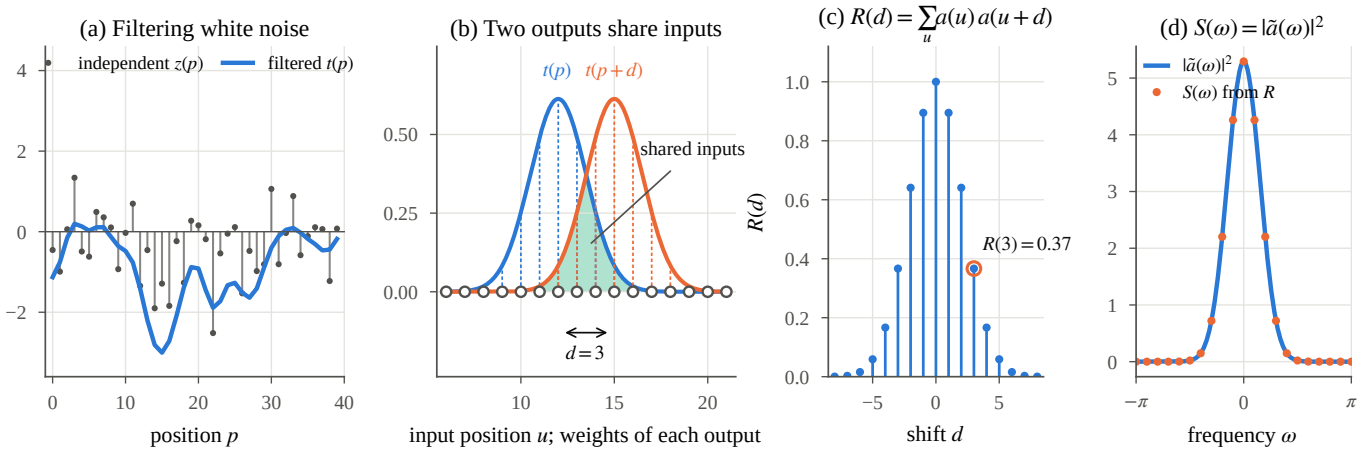

Figure S3 **Why filtering correlates noise**, in one dimension. (a) Independent Gaussian samples $z(p)$ and their output t through a 9-tap Gaussian filter: neighboring outputs become similar. (b) The filter's weights for the outputs at p and $p+d$ (dashed: the weight each input receives); the inputs under both windows (shaded) are shared, which is why $t(p)$ and $t(p+d)$ are correlated. (c) The correlation is the overlap of the filter with its shift (Eq. S13). (d) Its Fourier transform $S(\omega)$ equals $|\tilde{a}(\omega)|^2$ at every frequency (Eq. S14), so a filter with a given correlation follows from $\sqrt{S}$.

frequency by $\tilde{a}(\omega)$ and hence its power by $|\tilde{a}(\omega)|^2$; unit-variance white inputs have unit power spectrum [35].

A valid covariance has a nonnegative spectrum. The desired filter can therefore be constructed one frequency at a time, then transformed back to spatial coordinates:

$$\begin{aligned}\tilde{a}(\omega) &= \sqrt{S(\omega)},\\ a(u) &= \frac{1}{2\pi}\int_{-\pi}^{\pi}\sqrt{S(\omega)}e^{\mathrm{i}\omega u}\,d\omega.\end{aligned} \tag{S15}$$

For example, the two-tap filter above gives $R(0) = 1$, $R(\pm1) = 1/2$, and hence $S(\omega) = 1 + \cos\omega$. This example illustrates why emphasizing low frequencies smooths a signal, as in Fig. S3a.

The covariance determines the magnitude of $\tilde{a}$, but leaves its phase free: multiplying by a unit-magnitude phase factor leaves Eq. S14 unchanged. Phases must also respect conjugate symmetry for the spatial filter to remain real. The nonnegative scalar root chooses zero phase and gives a real, symmetric filter. It is one valid construction, and the covariance alone cannot identify a unique original filter. This choice need not have finite support even when a finite filter could produce the same covariance.

For multiple channels, each output mixes several independent input fields through $t_c(\mathbf{p}) = \sum_i \sum_{\mathbf{k}} A_{ci}(\mathbf{k}) z_i(\mathbf{p}-\mathbf{k})$. In two dimensions, $\mathbf{p}$, $\mathbf{k}$, and $\mathbf{d}$ are spatial vectors and $\boldsymbol{\omega} = (\omega_y, \omega_x)$. At each frequency, the filters form a matrix $\widetilde{A}$, with rows for output channels and columns for input channels. Using the same lag convention as in the measured correlations,

$$\begin{aligned}\widetilde{A}_{ci}(\boldsymbol{\omega}) &= \sum_{\mathbf{k}} A_{ci}(\mathbf{k})e^{-\mathrm{i}\boldsymbol{\omega}\cdot\mathbf{k}},\\ R_{ab}(\mathbf{d}) &:= \mathbb{E}[t_a(\mathbf{p})t_b(\mathbf{p}+\mathbf{d})],\\ S_{ab}(\boldsymbol{\omega}) &= \sum_{\mathbf{d}} R_{ab}(\mathbf{d})e^{-\mathrm{i}\boldsymbol{\omega}\cdot\mathbf{d}}\\ &= \sum_i \overline{\widetilde{A}_{ai}(\boldsymbol{\omega})}\widetilde{A}_{bi}(\boldsymbol{\omega}).\end{aligned} \tag{S16}$$

The last equality follows from Eq. 9 of the main paper just as Eq. S14 follows from Eq. S13. Thus $S = \overline{\widetilde{A}}\,\widetilde{A}^{\mathsf{T}}$ is a Hermitian positive semidefinite matrix at each frequency: $S = S^{\mathsf{H}}$ and all its eigenvalues are nonnegative. This is the matrix counterpart of $S = |\tilde{a}|^2$, so a scalar square root becomes a matrix square root [36]. Write $S = V\Lambda V^{\mathsf{H}}$, where H denotes conjugate transpose. Its Hermitian square root gives a valid factor:

$$W = V\sqrt{\Lambda}V^{\mathsf{H}}, \qquad \widetilde{A} = \overline{W}.$$

The eigenvectors identify channel mixtures with separate powers; taking the square root of each power gives the corresponding filter gain. Transforming back to the original channel coordinates yields a matrix square root. The conjugation follows from our definition of $R_{ab}(\mathbf{d})$: it ensures $\overline{\widetilde{A}}\,\widetilde{A}^{\mathsf{T}} = WW^{\mathsf{H}} = S$ and preserves the specified lag direction. For RGB, the matrix is 3×3 because there are three channels; this size is independent of the filters' spatial support.

The fit applies this construction in three steps (Alg. S3). First, the measured correlations $\widehat{R}$ are symmetrized to satisfy $R_{ab}(\mathbf{d}) = R_{ba}(-\mathbf{d})$ and tapered at distant lags:

$$\begin{aligned}\widehat{R}^{\mathrm{s}}_{ab}(\mathbf{d}) &= \tfrac{1}{2}\left[\widehat{R}_{ab}(\mathbf{d}) + \widehat{R}_{ba}(-\mathbf{d})\right]\eta(\mathbf{d}),\\ \eta(\mathbf{d}) &= \cos^2\left[\frac{\pi}{2}\,\mathrm{clip}\left(\frac{\|\mathbf{d}\|_2 - 8}{8}, 0, 1\right)\right].\end{aligned} \tag{S17}$$

The taper leaves lags within radius 8 unchanged and smoothly reduces them to zero at radius 16. Second, $\widehat{R}^{\mathrm{s}}$ is transformed to $\widehat{S}$ and each frequency matrix is factored:

$$\begin{aligned}\widehat{S} &= V\Lambda V^{\mathsf{H}}, \qquad \Lambda_+ = \max(\Lambda, 0),\\ S_+ &= V\Lambda_+ V^{\mathsf{H}},\\ W &= V\Lambda_+^{1/2}V^{\mathsf{H}}, \qquad \widetilde{A} = \overline{W}.\end{aligned} \tag{S18}$$

The maximum and square root act on the diagonal eigenvalues. Negative eigenvalues, which can arise in an estimated and tapered spectrum, are set to zero to obtain a valid spectral matrix, so

$\widetilde{A}\widetilde{A}^{\mathsf{T}} = WW^{\mathsf{H}} = S_+$. Third, the frequency responses are inverse-transformed, cropped to $\mathcal{K} = \{-12, \dots, 12\}^2$, and normalized per output channel. For $\mathbf{k} \in \mathcal{K}$, the retained coefficients are

$$\Gamma_{ci}(\mathbf{k}) = \mathcal{F}^{-1}[\widetilde{A}_{ci}](\mathbf{k}),$$
$$A_{ci}(\mathbf{k}) = \frac{\Gamma_{ci}(\mathbf{k})}{\left(\sum_j \sum_{\boldsymbol{\ell} \in \mathcal{K}} \Gamma_{cj}(\boldsymbol{\ell})^2\right)^{1/2}}. \tag{S19}$$

Here $\mathcal{F}^{-1}$ is the two-dimensional inverse Fourier transform. The transforms are computed on a 64×64 frequency grid. The result is nine 25×25 filters satisfying the unit-energy constraint of Eq. S2. The covariance construction requires only uncorrelated, zero-mean, unit-variance inputs; choosing independent standard Gaussian inputs also makes the filtered field Gaussian, with standard normal marginals for the lookup of Eq. S4. Tapering, spectral projection, cropping, and renormalization make the covariance fit approximate; the subsequent nonlinear lookup can further change residual correlations.

B Ablation details

Design. Each variant changes one step of Fit (Alg. S1) and is otherwise refitted on the same training pairs, rendered by the unchanged synthesis from the same random streams, and scored like the main results. The variants, and which of them the main paper shows, were fixed before any was scored, except the last one below, and all are reported. A refit with the full recipe reproduces the main evaluation's presets, outputs and scores exactly on SIDD and its presets at every film radius, so every difference is due to the changed step. The variants are:

- *No fitting masks*: the pilot uses all pixels and the correlation fit uses neither the flat mask nor the clipping guard.
- *Flat mask only* / *clipping guard only*: one of the two masks for the correlation fit (the guard-only variant also fits the pilot on all pixels).
- *Kernel $k \times k$*: the factor is cropped to $k \times k$ instead of 25×25 before normalization.
- *Gaussian conditional*: the lookup is replaced by a Gaussian with the mean and standard deviation of the same pooled residual histogram at each clean intensity, stored as a lookup linear in t.
- *Pilot, one knot per code*: the pilot strength is solved at one knot per 8-bit clean code instead of at 17 knots (Sec. A), with the same rules for keeping, filling and smoothing knots. This variant was added while preparing this supplement, after the others had been scored.

The Gaussian lookup exceeds the residual range $|g| \le 1$ for the dark red and blue channels of S6 ISO 800 and ISO 3200, whose residuals there have means of 69 to 124 codes and standard deviations of 60 to 78 codes. The main row therefore clips those table entries to ± 1 (29 entries in the two presets, changing fewer than 10^{-5} of the channel values rendered on their training data), a choice made after some Gaussian scores of the other classes existed but before any was looked at; an extra row compares the unclipped variant with full PGNM on the 33 sources of the other classes.

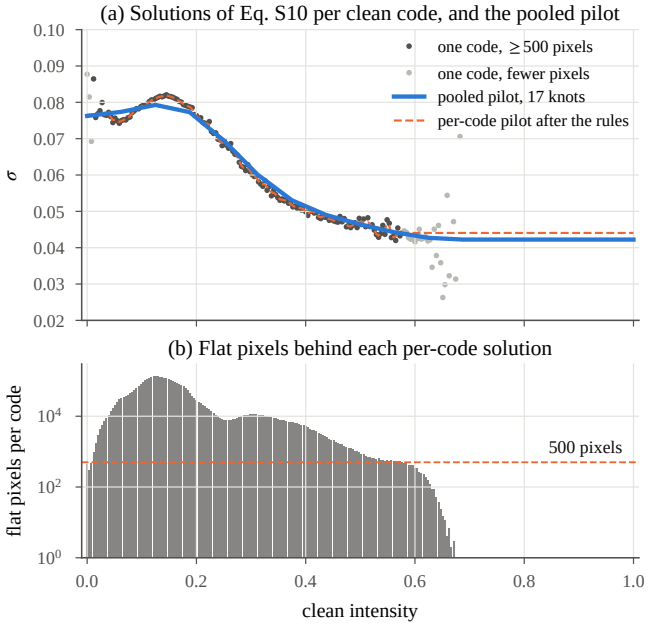

Figure S4 Pooled and per-code pilot for SIDD GP ISO 3200, green channel (the class of Fig. S2). (a) Eq. S10 solved at every clean code with flat pixels (points; light: fewer than 500 pixels), the pooled 17-knot pilot of PGNM, and the per-code pilot after the same rules for keeping, filling and smoothing knots. (b) Flat pixels per code. The per-code strengths follow the pooled pilot where flat pixels are plentiful and scatter where they are few, in the first codes and at the brightest flat intensities.

Results. Tab. S1 lists every SIDD variant with its interval. The flat mask alone recovers full PGNM on every metric, whereas the clipping guard alone leaves the color correlation error 34% higher, close to the 46% of no masks: textured pixels in the correlation fit weaken the fitted cross-channel dependence between neighboring pixels. A 3×3 kernel removes most of the 1×1 loss on SIDD but is still 46% worse in spatial correlation; from 5×5 on, no correlation error is worse than full PGNM's (11×11 is 8.5% better in spatial correlation). Smaller kernels lower the tile KL slightly, by 0.6–0.9% from 3×3 to 11×11 and by 5.4% for 1×1, where both correlation errors grow several-fold. Tab. S2 gives the film-grain kernel sweep by radius: a 1×1 support loses the grain structure at every radius, a 3×3 support raises the spatial correlation error by 76–180% at every radius, and from 5×5 on the changes are small and mostly improvements.

Solving the pilot at one knot per clean code instead of pooling at 17 knots raises the spatial correlation error by 1.8% and the tile KL by 0.15%, with the other measures tied: the per-code strengths agree with the pooled pilot where flat pixels are plentiful and scatter where they are few (Fig. S4).

Variant	KL, 32^2 tiles	KL, 256^2 blocks	W_1	Spatial	Color
Full PGNM	0.0219	0.0108	1.319	0.0094	0.0144
No fitting masks	+1.2% [+0.6, +1.8]	0.0% [0.0, +0.1]	0.0% [0.0, +0.1]	−1.7% [−11.8, +6.8]	+46.3% [+14.9, +71.0]
Flat mask only	+0.1% [−0.6, +0.8]	0.0% [−0.1, 0.0]	0.0% [0.0, +0.1]	−4.3% [−16.3, +7.3]	+3.2% [−8.8, +15.6]
Clipping guard only	+0.6% [+0.4, +0.7]	0.0% [0.0, +0.1]	0.0% [0.0, +0.1]	+1.9% [−1.6, +6.3]	+33.8% [+12.9, +54.0]
Kernel 1 × 1	−5.4% [−6.2, −4.7]	−0.2% [−0.6, +0.1]	0.0% [−0.2, +0.3]	+586% [+570, +603]	+770% [+743, +801]
Kernel 3 × 3	−0.6% [−1.3, 0.0]	+0.1% [−0.3, +0.5]	+0.2% [−0.1, +0.4]	+45.8% [+37.8, +54.2]	+47.3% [−4.5, +88.5]
Kernel 5 × 5	−0.8% [−1.4, −0.2]	0.0% [−0.3, +0.3]	+0.2% [−0.1, +0.4]	−6.5% [−14.3, +0.3]	+0.1% [−2.5, +3.2]
Kernel 11 × 11	−0.9% [−1.4, −0.5]	0.0% [−0.3, +0.4]	+0.1% [−0.1, +0.3]	−8.5% [−14.4, −3.3]	+1.2% [−1.2, +4.1]
Gaussian conditional‡	+138% [+122, +151]	+294% [+263, +321]	+108% [+97, +119]	+4.5% [−1.4, +12.7]	+2.0% [−4.5, +6.6]
Gaussian conditional, 33 sources§	+133% [+112, +147]	+276% [+232, +308]	+102% [+80, +123]	+4.7% [−1.5, +15.1]	+4.3% [−2.9, +9.4]
Pilot, one knot per code¶	+0.1% [+0.1, +0.2]	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]	+1.8% [+0.5, +3.3]	0.0% [−0.9, +0.8]

Table S1 SIDD ablation, all variants (39 sources, 8 scenes, five generator runs). Full PGNM's scores, then each variant's change relative to them with the 95% scene-bootstrap interval (100,000 draws resampling the physical scenes): worse, better, or tie when the interval includes zero. Each variant changes one fitting step and is rendered from the same random streams; the 25 × 25 refit of full PGNM reproduces the main evaluation's presets, outputs and scores exactly. ‡The moment-matched Gaussian lookup exceeds the residual range $|g| \leq 1$ in two classes; there its table is clipped to ±1 (Sec. B). §The same comparison restricted to the classes where the unclipped Gaussian is valid (excludes S6 ISO 3200, S6 ISO 800). ¶Added while preparing this supplement, after the other variants were scored: the pilot strength is solved at one knot per 8-bit clean code instead of 17 knots (Sec. A), with the same rules.

		W_1				Spatial				DISTS				LPIPS			
Radius	Full	k=1	k=3	k=5	k=11	k=1	k=3	k=5	k=11	k=1	k=3	k=5	k=11	k=1	k=3	k=5	k=11
0.025	0.107	−3.3	−1.1	−0.9	−0.1	+792	+76.4	−23.0	−16.3	+160	+2.2	−0.1	+0.5	+58.8	+8.0	+0.4	+0.3
0.05	0.138	−7.0	−2.3	−0.6	0.0	+1247	+103	−7.4	−5.5	+200	+2.2	+0.9	+0.1	+137	+3.2	−0.6	−0.2
0.075	0.168	−8.8	−0.4	+1.1	+0.8	+1298	+97.1	−6.5	−3.5	+222	+2.2	−0.3	+0.3	+161	+2.7	−0.3	+0.1
0.1	0.206	−11.8	−2.2	−0.4	−0.2	+1314	+94.6	−5.4	−2.7	+232	+2.2	+0.1	+0.7	+173	+2.7	−0.2	−0.2
0.2	0.419	−11.6	−2.3	−0.4	0.0	+1350	+101	−2.4	−1.4	+233	+2.3	+0.3	+0.5	+198	+2.0	−0.2	−0.1
0.4	0.852	−12.2	−2.7	−1.0	−0.8	+1269	+114	−6.3	−4.3	+207	+3.3	+0.2	+0.4	+212	+1.2	0.0	−0.1
0.6	1.305	−11.3	−2.3	0.0	−0.7	+1120	+128	−10.6	−6.6	+179	+4.9	+0.4	−0.4	+244	+2.8	+0.1	+0.2
0.8	1.810	−10.2	−2.2	+0.2	+0.2	+1109	+180	−2.3	−3.1	+154	+4.5	+0.1	+0.3	+282	+4.2	−0.5	+0.1
All radii	0.626	−10.7	−2.2	−0.2	−0.2	+1164	+113	−8.6	−5.9	+193	+3.5	+0.2	+0.2	+230	+2.8	−0.2	0.0
$r \leq 0.2$	0.208	−9.7	−1.9	−0.2	+0.1	+1167	+92.8	−10.1	−6.7	+219	+2.3	+0.2	+0.4	+176	+2.6	−0.2	−0.1

Table S2 Film-grain kernel support, all sizes and radii. Change of each metric (%) when the 25 × 25 support of full PGNM is cropped to $k \times k$ in the fit, before normalization; Full: full PGNM's conditional W_1 in 8-bit codes. Per radius, a change is worse or better when more than half of the 60 test images move in that direction, else tie; in the last two rows, when this holds at every radius of the scope (the rule of Tab. 6 of the main paper). Each image averages 25 output–reference pairs. The 25 × 25 refit equals the evaluated preset at every radius.

C SIDD sources, results by camera and ISO, and film grain by radius

Sources. Tab. S3 lists the 40 SIDD Validation source images. The camera–ISO labels follow the order of the SIDD benchmark scene instances: Validation image i is the i-th benchmark instance in sorted order, which we read from the benchmark archive's index and confirmed for all 40 images by matching their content to SIDD-Medium images of the same scene and camera. The benchmark instances are not part of SIDD-Medium; they come from eight of its ten scenes and its five cameras but are separate captures.

Results by camera and ISO. Tabs. S5–S9 break down Tab. 1 of the main paper by camera and camera–ISO class, one table per metric, and add the 256×256 block KL (Tab. S6). Each value averages the five generator runs and the class's source images; camera rows weight the camera's classes equally, and the last row reproduces the main table. A class contains one to five source images, so the tables are descriptive and carry no intervals.

PGNM has the lowest value in 13 of the 16 classes for the tile KL, 10 for the block KL, 10 for conditional W_1, 10 for spatial and 14 for color correlation error; values equal at the printed precision count for every method concerned. GMDCN has all remaining lowest KL values and five of the six remaining lowest conditional W_1, including both G4 classes and S6 ISO 400 on all three of these metrics; NAFlow has the lowest conditional W_1 for S6 ISO 800, and NM-FlowGAN the lowest spatial correlation error in four of the seven classes it covers.

Film grain by radius. Tab. S4 breaks down Tab. 3 of the main paper by grain radius, with the main table's weighting. Zhang23 has the lowest conditional W_1 at $r = 0.025$ and 0.05, and PGNM from $r = 0.075$ on. In spatial correlation error, Zhang23 is lowest up to $r = 0.6$ and GrainNet at $r = 0.8$, where Zhang23's error rises to 0.11; PGNM's is above GrainNet's from $r = 0.4$ on. In DISTS and LPIPS, PGNM and Zhang23 differ by at most 0.0013 up to $r = 0.1$, and GrainNet is lowest from $r = 0.2$ on.

D Further analyses

These analyses were computed after the main evaluation, on its scored outputs or with its fits and checkpoints unchanged; they

Image	SIDD scene instance	Phone	ISO	Exposure (s)	Illuminant (K)	Light	Scene	Evaluated	19-source
0	0009_001_S6_00800_00350_3200_L	S6	800	1/350	3200	L	001	✓	✓
1	0021_001_GP_10000_05000_5500_N	GP	10000	1/5000	5500	N	001	✓	
2	0024_001_N6_03200_01500_5500_N	N6	3200	1/1500	5500	N	001	✓	
3	0026_001_G4_00400_00160_5500_L	G4	400	1/160	5500	L	001	✓	
4	0031_001_IP_01600_02000_3200_N	IP	1600	1/2000	3200	N	001	✓	✓
5	0037_002_GP_06400_03200_3200_L	GP	6400	1/3200	3200	L	002	✓	
6	0041_002_IP_01600_04000_5500_L	IP	1600	1/4000	5500	L	002	✓	✓
7	0046_002_G4_00400_00350_3200_L	G4	400	1/350	3200	L	002	✓	
8	0049_002_N6_00800_00800_5500_L	N6	800	1/800	5500	L	002	✓	✓
9	0053_002_S6_03200_02000_5500_N	S6	3200	1/2000	5500	N	002	✓	✓
10	0056_003_N6_03200_04000_5500_N	N6	3200	1/4000	5500	N	003	✓	
11	0058_003_G4_00400_00500_5500_L	G4	400	1/500	5500	L	003	✓	
12	0061_003_S6_00800_00500_4400_L	S6	800	1/500	4400	L	003	✓	✓
13	0067_003_GP_01600_01500_3200_L	GP	1600	1/1500	3200	L	003	✓	
14	0071_003_IP_02000_04000_5500_L	IP	2000	1/4000	5500	L	003	✓	
15	0074_004_N6_00400_00040_3200_L	N6	400	1/40	3200	L	004	✓	
16	0079_004_G4_00800_00160_3200_N	G4	800	1/160	3200	N	004	✓	✓
17	0082_004_S6_03200_00500_4400_L	S6	3200	1/500	4400	L	004	✓	✓
18	0085_004_GP_06400_02000_4400_N	GP	6400	1/2000	4400	N	004	✓	
19	0093_004_IP_01250_00250_3200_L	IP	1250	1/250	3200	L	004	excluded	
20	0095_005_N6_00400_00200_3200_L	N6	400	1/200	3200	L	005	✓	
21	0100_005_G4_00800_00400_3200_N	G4	800	1/400	3200	N	005	✓	✓
22	0103_005_S6_01600_00800_4400_L	S6	1600	1/800	4400	L	005	✓	✓
23	0109_005_GP_10000_08000_4400_N	GP	10000	1/8000	4400	N	005	✓	
24	0112_005_IP_01000_07500_5500_L	IP	1000	1/7500	5500	L	005	✓	
25	0119_006_N6_00400_00100_3200_L	N6	400	1/100	3200	L	006	✓	
26	0124_006_G4_00800_00350_3200_N	G4	800	1/350	3200	N	006	✓	✓
27	0128_006_S6_03200_01600_4400_L	S6	3200	1/1600	4400	L	006	✓	✓
28	0131_006_GP_01600_01250_4400_N	GP	1600	1/1250	4400	N	006	✓	
29	0141_006_IP_01600_01620_3200_L	IP	1600	1/1620	3200	L	006	✓	✓
30	0143_007_N6_00800_00800_4400_N	N6	800	1/800	4400	N	007	✓	✓
31	0148_007_G4_00400_00400_4400_L	G4	400	1/400	4400	L	007	✓	
32	0153_007_S6_03200_03200_5500_L	S6	3200	1/3200	5500	L	007	✓	✓
33	0158_007_GP_03200_03200_5500_N	GP	3200	1/3200	5500	N	007	✓	
34	0162_007_IP_01600_01600_3200_L	IP	1600	1/1600	3200	L	007	✓	✓
35	0171_008_N6_03200_01600_4400_L	N6	3200	1/1600	4400	L	008	✓	
36	0174_008_G4_00800_00800_4400_N	G4	800	1/800	4400	N	008	✓	✓
37	0176_008_S6_00400_00100_5500_L	S6	400	1/100	5500	L	008	✓	✓
38	0183_008_GP_06400_06400_5500_N	GP	6400	1/6400	5500	N	008	✓	
39	0187_008_IP_01600_01600_3200_L	IP	1600	1/1600	3200	L	008	✓	✓

Table S3 SIDD Validation sources. Validation image i (32 blocks each) is the i-th SIDD benchmark scene instance in sorted order; its fields follow SIDD's naming (exposure 1/shutter; light: low light L or normal brightness N). The iPhone 7 ISO 1250 source is excluded because no SIDD-Medium training data share its camera–ISO class. 19-source: covered by the sRGB-Flow and NM-FlowGAN checkpoints. Evaluated sources by phone (G4 8, GP 8, IP 7, N6 8, S6 8); by illuminant (3200 K 14, 4400 K 11, 5500 K 14); by brightness (low 24, normal 15); by scene (001: 5, 002: 5, 003: 5, 004: 4, 005: 5, 006: 5, 007: 5, 008: 5); 16 camera–ISO classes with 1–5 sources each. Phones: G4: LG G4; GP: Google Pixel; IP: iPhone 7; N6: Nexus 6; S6: Galaxy S6 Edge.

describe the reported results and are not additional comparisons.

D.1 Where the SIDD errors come from

Tab. S10 complements conditional W_1 with three per-bin error measures, the difference of residual means (bias), of standard deviations (strength) and of standardized quantiles (shape), and lists the correlations of residuals standardized per bin. On all 39 sources, PGNM has the lowest bias (0.92 codes) and shape error (0.041σ), whereas GMDCN matches the per-bin strength best (1.26 against 1.84 codes for PGNM); with PGNM's lower color correlation error (Tab. 1 of the main paper), this is the comparison summarized in Sec. 5. That error covers neighboring pixels only; at the same pixel, GMDCN's channel correlation is closer to the real one (0.361 against 0.357, PGNM 0.341). The dependence columns show the two directions of error in neighboring-pixel correlation named there: NAFlow's is too high (0.49 against 0.44 for real noise) and sRGB-Flow's too low (0.32 against 0.45 on the 19-source block); NM-FlowGAN's same-pixel color correlation is the lowest (0.29 against 0.33).

Because strength is PGNM's largest remaining error, we checked how much of the W_1 gap to GMDCN it explains with an oracle that rescales each generated residual to the standard deviation of the

	Radius r								Mean over radii	
Method	0.025	0.05	0.075	0.1	0.2	0.4	0.6	0.8	All	≤ 0.2
Conditional W_1 (8-bit codes)										
PGNM	0.107	0.138	**0.168**	**0.206**	**0.419**	**0.852**	**1.305**	**1.810**	**0.626**	**0.208**
Zhang23*	**0.070**	**0.118**	0.174	0.235	0.564	1.662	3.226	5.167	1.402	0.232
GrainNet	3.652	3.456	3.321	3.224	2.913	2.434	2.473	3.102	3.072	3.313
Ind. Newson render	0.040	0.080	0.121	0.158	0.314	0.618	0.892	1.167	0.424	0.143
Spatial correlation error										
PGNM	0.0104	0.0074	0.0072	0.0072	0.0071	0.0077	0.0090	0.0097	**0.0082**	0.0079
Zhang23*	**0.0090**	**0.0069**	**0.0069**	**0.0068**	**0.0069**	**0.0071**	**0.0074**	0.1115	0.0203	**0.0073**
GrainNet	0.0596	0.0329	0.0209	0.0153	0.0088	0.0076	0.0078	**0.0083**	0.0202	0.0275
Ind. Newson render	0.0062	0.0065	0.0067	0.0067	0.0069	0.0069	0.0073	0.0074	0.0068	0.0066
DISTS										
PGNM	0.0204	0.0339	0.0436	**0.0512**	0.0720	0.0991	0.1214	0.1429	0.0731	**0.0442**
Zhang23*	**0.0192**	**0.0334**	**0.0435**	0.0514	0.0735	0.1064	0.1361	0.3475	0.1014	**0.0442**
GrainNet	0.0273	0.0378	0.0466	0.0531	**0.0718**	**0.0939**	**0.1068**	**0.1148**	**0.0690**	0.0473
Ind. Newson render	0.0187	0.0329	0.0429	0.0504	0.0699	0.0919	0.1047	0.1113	0.0653	0.0430
LPIPS										
PGNM	0.0052	**0.0161**	**0.0279**	**0.0395**	0.0803	0.1387	0.1658	0.1732	0.0809	**0.0338**
Zhang23*	**0.0051**	**0.0161**	**0.0279**	**0.0395**	0.0803	0.1401	0.1690	0.6392	0.1397	**0.0338**
GrainNet	0.0099	0.0189	0.0296	0.0404	**0.0797**	**0.1351**	**0.1601**	**0.1651**	**0.0799**	0.0357
Ind. Newson render	0.0050	0.0159	0.0276	0.0391	0.0792	0.1355	0.1606	0.1649	0.0785	0.0333

Table S4 Simulated film grain by radius (lower is better). Breakdown of Tab. 3 of the main paper: each value averages the 25 output–reference pairs of each test image, then the 60 test images; the last two columns average the radii of each scope with equal weights and reproduce the main table. Lowest fitted method of each column in bold, values equal at the printed precision together; the independent Newson render compares two simulator runs and is not bolded. Descriptive only, without intervals. *Our implementation of the published model, with the shared blur B; its small-grain approximation applies up to $r = 0.2$.

Class	n	**PGNM**	NAFlow	GMDCN*	DANet	sRGB-Flow	NM-FlowG.
G4 400	4	0.026	0.018	**0.015**	0.106	–	–
G4 800	4	0.023	0.027	**0.020**	0.189	0.041	0.033
G4	8	0.024	0.022	**0.018**	0.148	–	–
GP 1600	2	**0.018**	0.036	0.020	0.175	–	–
GP 3200	1	**0.015**	0.049	0.018	0.470	–	–
GP 6400	3	**0.028**	0.035	0.031	0.047	–	–
GP 10000	2	**0.025**	0.044	0.039	0.149	–	–
GP	8	**0.021**	0.041	0.027	0.210	–	–
IP 1000	1	**0.021**	0.028	0.028	0.120	–	–
IP 1600	5	**0.023**	0.037	0.042	0.068	0.042	0.033
IP 2000	1	**0.009**	0.040	0.066	0.078	–	–
IP	7	**0.018**	0.035	0.045	0.089	–	–
N6 400	3	**0.019**	0.034	**0.019**	0.095	–	–
N6 800	2	**0.012**	0.027	0.025	0.216	0.067	0.035
N6 3200	3	**0.018**	0.041	0.036	0.114	–	–
N6	8	**0.017**	0.034	0.026	0.142	–	–
S6 400	1	0.031	0.044	**0.021**	0.136	0.030	0.043
S6 800	2	**0.025**	0.028	0.032	0.164	0.053	0.039
S6 1600	1	**0.029**	0.035	0.044	0.082	0.057	0.042
S6 3200	4	**0.031**	0.056	0.061	0.118	0.080	0.071
S6	8	**0.029**	0.040	0.039	0.125	0.055	0.049
All	39	**0.022**	0.035	0.031	0.143	–	–

Table S5 SIDD by camera and camera–ISO class: D_{KL}, 32×32 tiles (lower is better). Mean over five generator runs and over the n source images of each class; italic rows weight the camera's classes equally, and All weights cameras equally, which reproduces Tab. 1 of the main paper. Lowest value of each row in bold; descriptive only, without intervals. –: class or camera not fully covered by the checkpoint (sRGB-Flow and NM-FlowGAN: 19-source block only). *Our implementation of the published model.

Class	n	**PGNM**	NAFlow	GMDCN*	DANet	sRGB-Flow	NM-FlowG.
G4 400	4	0.0183	0.0074	**0.0049**	0.0455	–	–
G4 800	4	0.0116	0.0121	**0.0071**	0.1073	0.0290	0.0165
G4	8	0.0150	0.0098	**0.0060**	0.0764	–	–
GP 1600	2	0.0080	0.0224	**0.0069**	0.1224	–	–
GP 3200	1	**0.0011**	0.0359	0.0036	0.3664	–	–
GP 6400	3	0.0125	0.0162	**0.0124**	0.0207	–	–
GP 10000	2	**0.0054**	0.0238	0.0203	0.0728	–	–
GP	8	**0.0068**	0.0246	0.0108	0.1456	–	–
IP 1000	1	**0.0134**	0.0197	0.0190	0.0628	–	–
IP 1600	5	**0.0124**	0.0266	0.0294	0.0342	0.0356	0.0218
IP 2000	1	**0.0023**	0.0289	0.0524	0.0420	–	–
IP	7	**0.0094**	0.0251	0.0336	0.0463	–	–
N6 400	3	0.0094	0.0220	**0.0065**	0.0417	–	–
N6 800	2	**0.0022**	0.0144	0.0104	0.1633	0.0562	0.0195
N6 3200	3	**0.0031**	0.0241	0.0152	0.0493	–	–
N6	8	**0.0049**	0.0202	0.0107	0.0847	–	–
S6 400	1	0.0257	0.0364	**0.0134**	0.0750	0.0222	0.0358
S6 800	2	**0.0162**	0.0163	0.0215	0.1081	0.0438	0.0281
S6 1600	1	**0.0161**	0.0198	0.0272	0.0595	0.0425	0.0265
S6 3200	4	**0.0147**	0.0324	0.0301	0.0626	0.0580	0.0415
S6	8	**0.0182**	0.0262	0.0230	0.0763	0.0416	0.0330
All	39	**0.0108**	0.0212	0.0168	0.0859	–	–

Table S6 SIDD by camera and camera–ISO class: D_{KL}, 256×256 blocks (lower is better). Weighting and notation as in Tab. S5.

scored real image (Tab. S11). The oracle reads the evaluation data and is not a method. With one gain per channel, PGNM's conditional W_1 falls from 1.32 to 1.05 codes and is lower than GMDCN's in 15 of the 16 classes, against 11 as scored; per-bin corrections keep this ranking. The tile KL changes little under any correction.

Class	n	**PGNM**	NAFlow	GMDCN*	DANet	sRGB-Flow	NM-FlowG.
G4 400	4	1.459	0.795	**0.705**	1.272	–	–
G4 800	4	1.348	1.406	**1.174**	2.485	3.804	1.636
G4	8	1.403	1.100	**0.940**	1.879	–	–
GP 1600	2	**0.962**	1.864	1.240	2.835	–	–
GP 3200	1	**0.583**	2.844	1.106	6.542	–	–
GP 6400	3	2.738	3.373	**2.600**	3.297	–	–
GP 10000	2	**1.894**	4.137	3.220	4.404	–	–
GP	8	**1.544**	3.055	2.041	4.269	–	–
IP 1000	1	0.992	1.681	**0.899**	1.722	–	–
IP 1600	5	**0.941**	1.684	1.556	1.263	2.442	1.409
IP 2000	1	**0.371**	1.551	1.460	1.212	–	–
IP	7	**0.768**	1.638	1.305	1.399	–	–
N6 400	3	**0.861**	1.350	0.896	1.619	–	–
N6 800	2	**0.592**	1.445	1.058	2.536	3.299	1.732
N6 3200	3	**1.576**	4.047	2.649	2.883	–	–
N6	8	**1.009**	2.281	1.534	2.346	–	–
S6 400	1	1.800	1.865	**1.029**	1.669	3.334	2.355
S6 800	2	1.394	**1.383**	1.774	2.247	5.695	2.297
S6 1600	1	**2.024**	2.201	2.029	5.035	7.605	3.079
S6 3200	4	**2.261**	4.353	2.956	3.259	12.333	5.123
S6	8	**1.870**	2.451	1.947	3.052	7.242	3.214
All	39	**1.319**	2.105	1.553	2.589	–	–

Table S7 SIDD by camera and camera–ISO class: conditional W_1 (8-bit codes) (lower is better). Weighting and notation as in Tab. S5.

Class	n	**PGNM**	NAFlow	GMDCN*	DANet	sRGB-Flow	NM-FlowG.
G4 400	4	**0.0079**	0.0205	0.0108	0.0190	–	–
G4 800	4	**0.0046**	0.0185	0.0125	0.0149	0.0247	0.0084
G4	8	**0.0062**	0.0195	0.0117	0.0170	–	–
GP 1600	2	0.0114	0.0219	**0.0103**	0.0132	–	–
GP 3200	1	**0.0046**	0.0216	0.0134	0.0173	–	–
GP 6400	3	**0.0063**	0.0168	0.0120	0.0120	–	–
GP 10000	2	**0.0091**	0.0185	0.0134	0.0173	–	–
GP	8	**0.0078**	0.0197	0.0123	0.0150	–	–
IP 1000	1	0.0191	0.0259	0.0152	**0.0149**	–	–
IP 1600	5	0.0252	0.0225	0.0189	0.0234	0.0215	**0.0146**
IP 2000	1	**0.0086**	0.0243	0.0237	0.0191	–	–
IP	7	**0.0176**	0.0242	0.0193	0.0192	–	–
N6 400	3	**0.0074**	0.0215	0.0139	0.0195	–	–
N6 800	2	**0.0076**	0.0160	0.0122	0.0149	0.0246	0.0098
N6 3200	3	**0.0095**	0.0140	**0.0095**	0.0106	–	–
N6	8	**0.0082**	0.0172	0.0119	0.0150	–	–
S6 400	1	**0.0079**	0.0192	0.0132	0.0172	0.0202	0.0128
S6 800	2	0.0058	0.0175	0.0102	0.0143	0.0214	**0.0051**
S6 1600	1	0.0081	0.0158	0.0073	0.0117	0.0208	**0.0042**
S6 3200	4	0.0076	0.0126	0.0083	0.0111	0.0233	**0.0066**
S6	8	0.0073	0.0163	0.0097	0.0136	0.0214	**0.0072**
All	39	**0.0094**	0.0194	0.0130	0.0159	–	–

Table S8 SIDD by camera and camera–ISO class: spatial correlation error (lower is better). Weighting and notation as in Tab. S5.

D.2 A clean image from outside SIDD

Tab. S12 gives the numbers behind the Kodak check of Sec. 5, and Fig. S17 shows one class. Each SIDD fit and checkpoint was applied unchanged to *kodim19*, whose most textured windows have more than twice the local contrast of the most textured SIDD window, for five camera–ISO classes, one per camera. Residual strength depends on the clean intensities present, so every method changes somewhat between images. NAFlow's strength rises by 26–78% above its own SIDD level; DANet, which receives no camera or ISO input, produces 7.8–8.3 codes in every class, while its SIDD level varies from 9.0 to 33.6; the other methods change by at most 24%.

The detail slope measures whether a residual follows the clean image's fine structure. Real SIDD residuals have a positive slope in the G4, N6 and S6 classes (+0.53 to +0.68): the noisy captures are slightly sharper than their reference images. GMDCN and NM-FlowGAN reproduce this on SIDD and keep a positive slope on Kodak in these classes (+0.23 to +0.30), which we read as learned behavior rather than a failure; NAFlow's slope is negative on both. PGNM's slope stays within about ±0.02 on Kodak, as expected from a model that uses the clean image only through its intensity.

Class	n	**PGNM**	NAFlow	GMDCN*	DANet	sRGB-Flow	NM-FlowG.
G4 400	4	**0.0110**	0.0479	0.0276	0.0568	–	–
G4 800	4	**0.0135**	0.0565	0.0439	0.0607	0.0414	0.0354
G4	8	**0.0122**	0.0522	0.0358	0.0588	–	–
GP 1600	2	**0.0213**	0.0480	0.0316	0.0535	–	–
GP 3200	1	**0.0062**	0.0545	0.0286	0.0687	–	–
GP 6400	3	**0.0196**	0.0592	0.0241	0.0753	–	–
GP 10000	2	**0.0113**	0.0407	0.0346	0.0546	–	–
GP	8	**0.0146**	0.0506	0.0297	0.0630	–	–
IP 1000	1	0.0287	0.0607	0.0304	**0.0285**	–	–
IP 1600	5	**0.0142**	0.0593	0.0369	0.0456	0.0615	0.0231
IP 2000	1	**0.0151**	0.0706	0.0498	0.0590	–	–
IP	7	**0.0193**	0.0635	0.0390	0.0444	–	–
N6 400	3	**0.0117**	0.0258	0.0356	0.0299	–	–
N6 800	2	**0.0135**	0.0277	0.0277	0.0286	0.0357	0.0597
N6 3200	3	**0.0121**	0.0342	0.0204	0.0237	–	–
N6	8	**0.0124**	0.0293	0.0279	0.0274	–	–
S6 400	1	0.0264	0.0281	0.0516	0.0516	**0.0255**	0.0401
S6 800	2	**0.0109**	0.0445	0.0217	0.0368	0.0398	0.0285
S6 1600	1	**0.0047**	0.0526	0.0148	0.0198	0.0458	0.0151
S6 3200	4	**0.0124**	0.0397	0.0143	0.0252	0.0595	0.0151
S6	8	**0.0136**	0.0412	0.0256	0.0334	0.0427	0.0247
All	39	**0.0144**	0.0474	0.0316	0.0454	–	–

Table S9 SIDD by camera and camera–ISO class: color correlation error (lower is better). Weighting and notation as in Tab. S5.

The same check with the film fits on a gray version of *kodim19* (Tab. S13) leaves PGNM's grain strength within 4% of the Newson reference at every radius (+0.3% to +3.6%), comparable to Zhang23 (−1.1% to +2.2%) and to the independent Newson render (−1.0% to +0.5%); GrainNet's finest grain is 60% stronger.

D.3 Offset and strength of film-grain outputs

Tab. S14 measures, on the scored film outputs, how far each method's mean residual departs from the expected image $\mu = B * x$ and how strong its grain is, and Tab. S15 resolves the mean residual by intensity. Both Newson runs keep every bin mean within ±0.41 codes. PGNM stays within ±0.30 codes up to $r = 0.2$ and within ±0.83 codes beyond, with a pixel-weighted offset within 0.06 codes of the reference at every radius. Zhang23 stays within ±0.16 codes up to $r = 0.2$ and reaches 4.3 codes at $r = 0.8$, where its approximation no longer holds. GrainNet's mean residual depends on intensity at every radius, from up to +7 codes in mid-tones at the finest radius (+2.4 to +3.2 from $r = 0.4$ on) to about −6 codes near white; its pixel-weighted offset is 2.3–3.6 codes against 0.05–0.73 for the reference. Its grain is 44% stronger than the reference at $r = 0.025$ and 7% at $r = 0.05$, and within 3% from $r = 0.075$ on. These offsets enter the residual errors of Tab. 3 of the main paper, while the appearance distances largely tolerate them.

Method	Per-bin residual distribution↓				Dependence of standardized residuals		
	W_1 (codes)	Bias (codes)	Strength (codes)	Shape (σ)	RGB, same pixel	Lag 1, horizontal	Lag 1, vertical
All 39 sources (16 classes)							
Real					0.357	0.440	0.441
PGNM	**1.319**	**0.92**	1.84	**0.041**	0.341	0.429	0.430
NAFlow	2.105	1.85	2.17	0.059	0.370	0.485	0.490
GMDCN*	1.553	1.12	**1.26**	0.059	0.361	0.440	0.443
DANet	2.589	1.09	2.95	0.089	0.350	0.443	0.439
19 sources covered by sRGB-Flow and NM-FlowGAN							
Real					0.329	0.451	0.451
PGNM	**1.187**	**0.76**	1.64	**0.041**	0.308	0.438	0.439
NAFlow	1.746	1.55	1.71	0.057	0.365	0.488	0.492
GMDCN*	1.434	1.09	**1.03**	0.064	0.337	0.442	0.446
DANet	2.334	1.01	2.64	0.103	0.356	0.444	0.438
sRGB-Flow	4.197	4.27	2.35	0.091	0.393	0.324	0.320
NM-FlowGAN	1.998	1.53	2.08	0.083	0.288	0.446	0.444

Table S10 Where the SIDD residual errors come from. Descriptive per-bin error measures beside conditional W_1 on the scored outputs (all pixels, five runs, weighted as in Tab. 1 of the main paper). In each clean-intensity bin: bias, the difference of residual means; strength, the difference of standard deviations (both root mean square over bins, weighted by pixel count, averaged over RGB); shape, the mean absolute difference of 256 quantiles after each residual is standardized by its own bin mean and standard deviation (bins with a standard deviation below 0.5 codes are skipped). Right: correlations of residuals standardized per bin, same pixel across channels and to the next pixel within a channel; the Real rows give the target. Lowest error in bold. *Our implementation of the published model.

Oracle correction	W_1 (codes)		KL, 32^2 tiles		PGNM lower (W_1 / KL)
	PGNM	GMDCN*	PGNM	GMDCN*	
As scored	1.318	1.554	0.0218	0.0312	11 / 12 of 16
One gain per channel	1.045	1.432	0.0241	0.0290	15 / 12 of 16
Strength per bin	0.847	1.346	0.0231	0.0284	15 / 13 of 16
Mean and strength per bin	0.489	0.885	0.0199	0.0237	15 / 13 of 16

Table S11 Oracle strength check (diagnostic only; it reads the scored real image). Each generated residual is rescaled about its own mean so that its standard deviation equals the real one, per channel or per clean-intensity bin; the last row also moves each bin mean to the real one. Integer residuals are dequantized with uniform dither first. First two generator runs, all 39 sources, weighted as in the main table; last column: camera–ISO classes in which PGNM's score is lower than GMDCN's.

Residual standard deviation (codes): SIDD → Kodak (change)							
Class	Real	**PGNM**	GMDCN*	NM-FlowGAN	NAFlow	sRGB-Flow	DANet
G4 800	14.6	13.7 → 12.2 (−11%)	14.1 → 13.4 (−5%)	15.2 → 14.4 (−5%)	14.2 → 21.7 (+52%)	14.8 → 15.8 (+7%)	15.3 → 7.8 (−49%)
GP 6400	31.8	34.6 → 31.0 (−10%)	31.0 → 29.1 (−6%)	–	30.0 → 37.8 (+26%)	–	33.6 → 8.3 (−75%)
IP 1600	9.6	10.2 → 7.7 (−24%)	9.6 → 8.7 (−10%)	9.6 → 8.9 (−7%)	9.6 → 17.1 (+78%)	10.0 → 10.8 (+8%)	9.0 → 8.0 (−11%)
N6 800	14.3	14.5 → 12.6 (−13%)	14.5 → 14.0 (−3%)	14.4 → 13.9 (−3%)	12.9 → 21.1 (+64%)	11.6 → 12.2 (+5%)	13.5 → 7.8 (−42%)
S6 800	17.5	19.0 → 17.2 (−9%)	19.1 → 16.1 (−16%)	19.8 → 20.3 (+2%)	16.7 → 25.7 (+54%)	19.9 → 20.8 (+4%)	18.4 → 8.3 (−55%)
Detail slope: SIDD → Kodak							
Class	Real	**PGNM**	GMDCN*	NM-FlowGAN	NAFlow	sRGB-Flow	DANet
G4 800	+0.61	−0.02 → +0.01	+0.45 → +0.25	+0.49 → +0.23	−0.34 → −0.35	−0.19 → −0.11	+0.26 → +0.03
GP 6400	+0.03	−0.19 → −0.01	+0.03 → +0.14	–	−0.71 → −0.47	–	−0.06 → +0.04
IP 1600	−0.11	−0.30 → +0.02	−0.25 → +0.04	−0.25 → +0.20	−1.02 → −0.35	−0.38 → −0.07	−0.15 → +0.04
N6 800	+0.53	−0.01 → +0.01	+0.32 → +0.27	+0.14 → +0.23	−0.90 → −0.35	−0.19 → −0.08	−0.05 → +0.03
S6 800	+0.68	+0.04 → 0.00	+0.41 → +0.28	+0.33 → +0.30	−0.72 → −0.37	−0.25 → −0.15	+0.03 → +0.04

Table S12 Every SIDD generator on a clean image from outside SIDD (Kodak *kodim19*, the evaluated fits and checkpoints, one draw per method and class). SIDD values: the same method's first generator run on the class's Validation sources; Real: the real SIDD residual. Detail slope: least-squares slope (through the origin) of the residual on the clean image's detail, the clean image minus its Gaussian blur (σ = 1 pixel), per channel; positive means the output is sharper than the clean image. –: class outside the 19-source block. *Our implementation of the published model.

Radius	0.025	0.05	0.075	0.1	0.2	0.4	0.6	0.8
Newson (codes)	1.70	3.29	4.91	6.53	12.95	25.51	37.07	47.12
PGNM + B	+3.6%	+1.5%	+1.1%	+0.3%	+0.9%	+0.7%	+0.4%	+0.9%
Zhang23 + B	−0.4%	+0.3%	+0.6%	+1.1%	+1.5%	+2.2%	+0.7%	−1.1%
GrainNet	+60.5%	+9.4%	+3.2%	+2.1%	+1.6%	+0.2%	+0.2%	+0.4%
Ind. Newson render	−1.0%	+0.1%	−0.2%	+0.1%	+0.1%	+0.5%	−0.2%	−0.1%

Table S13 Film grain on the gray Kodak image. Residual standard deviation relative to $\mu = B * x$: the Newson reference in 8-bit codes, then each method's difference from it, with the evaluated film fits and settings (one draw).

	Offset (codes)					Strength (codes)				
Radius	Newson	**PGNM** + B	Zhang23 + B	GrainNet	Ind. Newson render	Newson	**PGNM** + B	Zhang23 + B	GrainNet	Ind. Newson render
0.025	0.05	0.09	0.02	3.61	0.05	1.49	1.59	1.50	2.15	1.49
0.05	0.07	0.11	0.05	3.46	0.07	2.89	2.96	2.93	3.10	2.89
0.075	0.09	0.13	0.07	3.34	0.09	4.32	4.38	4.38	4.43	4.31
0.1	0.11	0.14	0.09	3.24	0.11	5.73	5.81	5.82	5.84	5.73
0.2	0.20	0.23	0.18	2.91	0.20	11.37	11.54	11.59	11.47	11.37
0.4	0.37	0.43	0.45	2.34	0.37	22.28	22.67	22.75	22.26	22.30
0.6	0.54	0.58	0.92	2.26	0.54	32.22	32.87	32.29	32.37	32.22
0.8	0.73	0.76	1.51	2.77	0.71	41.02	41.94	40.06	41.28	41.03

Table S14 Offset and strength of film-grain outputs relative to the expected image $\mu = B * x$, inside the 3-pixel border, on the scored outputs (60 test images, five runs). In each merged clean-intensity bin of μ (as for conditional W_1): offset, the pixel-weighted mean of |bin mean residual|; strength, the pixel-weighted mean of the bin standard deviation.

	Largest |bin mean|				GrainNet
Radius	Newson	**PGNM**	Zhang23	Ind. Newson render	range
0.025	0.21	0.25	0.03	0.21	−6.3 to +6.9
0.05	0.22	0.30	0.01	0.22	−6.0 to +6.2
0.075	0.23	0.30	0.01	0.22	−5.8 to +5.5
0.1	0.20	0.27	0.03	0.19	−5.6 to +4.9
0.2	0.28	0.27	0.16	0.29	−5.2 to +3.1
0.4	0.40	0.69	1.32	0.41	−5.3 to +2.4
0.6	0.38	0.83	2.82	0.36	−6.2 to +3.0
0.8	0.39	0.75	4.33	0.41	−6.7 to +3.2

Table S15 Mean film-grain residual by intensity, in 16 equal-width bins of μ, pooled over the 60 test images and five runs (8-bit codes): the largest magnitude over bins for each method and the range of GrainNet's bin means.

E Denoising details

This section gives the setup behind Tab. 2 of the main paper and the complete results. The setup was fixed before any denoiser was trained.

Training data and noise sources. We use NM-FlowGAN's denoiser split [18]: 123 SIDD-Medium scene instances with both captures (246 sRGB pairs), which are all Medium instances of the 15 camera–ISO classes NM-FlowGAN supports (G4 ISO 100/800; GP 100/800; IP 100/400/800/1600; N6 100/800; S6 100/400/800/1600/3200), so every generator is conditioned natively. Each image is cut into 96×96 patches at stride 48, with the last row and column clamped to the image edge, giving 1,473,956 patches per epoch. Every epoch draws a fresh permutation and augmentation (rotation by a multiple of 90° and a horizontal flip, applied to the clean patch and its real noisy patch before renoising), and every arm draws fresh noise for every patch and epoch, conditioned on the image's camera and ISO:

- *Real*: the patch's own real noisy observation, as the upper reference.
- *PGNM*: one preset per class, fitted by Alg. S1 on every 256×256 crop of the same 246 captures, so PGNM sees no other data. Each patch gets an independent latent field of its own size instead of a bank window (Sec. I.4), computed on CPU in the data loader.
- *GMDCN*: the checkpoint of Tab. 1, with its camera–ISO condition map.
- *NAFlow* [23]: class-conditional sampling of the released checkpoint.
- *NAFlow-NAS*: the same checkpoint with its published noise-aware sampling, which encodes the patch's real noisy observation, weights all class Gaussians by their likelihood, and samples from the mixture. It receives information the other synthetic arms do not.
- *NM-FlowGAN*: the released sampler with its native camera and ISO inputs.

All generators use their native settings. Every noisy patch, real or synthetic, is clipped to $[0, 1]$ and rounded half up to 8 bits. Neural generators run in float32 with TF32 disabled.

Denoiser and schedule. Every arm trains the DnCNN of NM-FlowGAN's denoiser configuration: 18 convolution layers of 64 features, batch normalization (momentum 0.9, $\epsilon = 10^{-4}$) on the 16 middle layers, and a residual output. Inputs are 8-bit codes divided by 256, and the loss is the mean squared error in code units. Adam (learning rate 10^{-4}, $\beta = (0.9, 0.999)$) halves its learning rate after every fifth epoch, for 30 epochs at batch 128 (11,516 iterations per epoch), in float32 with TF32 disabled, on one RTX 3090 per run. Three training seeds were planned; only seed 1 was run, for all six arms. For a given seed, all arms share the initial weights, patch order and augmentation. The reported model is the epoch-30 checkpoint; no checkpoint, epoch or setting was chosen from the evaluation curves (Fig. S5).

Test set. SIDD Validation: all 1,280 blocks of the 40 images, the 19 images whose camera–ISO class is among the 15 training classes, and the other 21 images. Scores and their computation follow Sec. J; paired intervals resample the eight physical scenes.

Reference outside the comparison. NAFlow's own released denoiser, trained by its authors with noise-aware sampling on all of SIDD-Medium, scores 37.23 dB on SIDD Validation; it is outside the comparison because its training data and schedule differ.

Results. On SIDD Validation (Tabs. S16 and S17), PGNM's denoiser is 0.72 dB below the real-pair reference on all 40 images (0.77 and 0.68 dB on the 19 and 21 images in and outside the training classes) and above every other synthetic arm on all three sets; the 95% interval of each lead excludes zero except against NAFlow-NAS on the 21 images, which is a tie.

	All 40		19 in training classes		21 in other classes	
Training noise	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
Real noise (reference)	37.94	0.896	38.16	0.910	37.74	0.883
GMDCN	36.15	0.872	36.22	0.884	36.08	0.862
NM-FlowGAN	35.15	0.875	35.31	0.888	35.00	0.862
NAFlow	36.72	0.878	36.80	0.889	36.66	0.868
NAFlow-NAS	36.88	0.883	36.96	0.893	36.81	0.873
PGNM	**37.22**	**0.889**	**37.39**	**0.903**	**37.06**	**0.877**
Reference outside the comparison						
NAFlow's released DnCNN	37.23	0.888	37.37	0.900	37.10	0.878

Table S16 Denoising on SIDD Validation by subset (training seed 1, epoch 30). Training classes: the 15 camera–ISO classes of the training data; the other 21 images belong to other classes. Bold: best synthetic noise. NAFlow-NAS also reads the real noisy training patch. Reference: NAFlow's own released denoiser, trained by its authors with noise-aware sampling on all of SIDD-Medium. The main paper reports the All 40 columns.

PGNM − row (dB)	All 40	19 in training classes	21 in other classes
Real noise (reference)	−0.72 [−0.89, −0.57]	−0.77 [−1.01, −0.52]	−0.68 [−0.81, −0.55]
GMDCN	+1.07 [+0.85, +1.29]	+1.17 [+0.74, +1.56]	+0.98 [+0.77, +1.15]
NM-FlowGAN	+2.06 [+1.62, +2.49]	+2.07 [+1.39, +2.65]	+2.05 [+1.70, +2.56]
NAFlow	+0.49 [+0.27, +0.70]	+0.59 [+0.22, +0.92]	+0.40 [+0.12, +0.66]
NAFlow-NAS	+0.33 [+0.12, +0.54]	+0.43 [+0.08, +0.74]	+0.25 [−0.10, +0.56]

Table S17 Paired PSNR differences on SIDD Validation: PGNM's denoiser minus the row's, mean over blocks, with a 95% interval from resampling the 8 physical scenes (100,000 draws, seed 11). An interval containing zero is a tie.

F Video delivery details

This section completes Tab. 5 of the main paper with every arm and rate and the setup. It illustrates delivery on one clip and is not a codec benchmark. The setup was fixed before any shot was selected or encoded, and every arm and rate is reported.

Source and clip. Netflix's *Nocturne* [32] (2018, CC BY 4.0) provides a clean master (JPEG 2000, 12-bit RGB, PQ, P3-D65, 3840 × 2160, 60 fps) and a grain version with Netflix's synthetic Filmbox grain (Vision3 stock, strength 1.2) on the same master (ProRes 422, 10-bit). The pair is the ground truth: the residual (grain version minus clean master) also contains the grain file's ProRes and 4:2:2 coding error, which no arm can remove; the grain version is also slightly darker than the master (by 0.5–1.8 codes in 8 bits, already present in PQ). Among the eight *Nocturne* shots of the AOM test material, we took the one whose first frame has median BT.709 luma closest to 128 (master frames 2370–2539, median 82; the other shots are darker) and its first 100 frames (1.67 s). The clean frames are read from the AOM test archive of the shot, whose first frame is identical to master frame 2370, and every grain frame was checked to match the clean frame of the same index. Both files are cropped to the center 1920 × 1080, keeping the grain at its native 4K size, and converted to 8-bit SDR by one fixed pipeline: the ST 2084 transfer function, P3-D65 to BT.709, division by the 203 cd/m^2 reference white, a Hable filmic curve applied to $\max(R, G, B)$ with the RGB ratio kept, clipping, and the inverse BT.1886 curve. The codec domain is BT.709 limited-range 8-bit Y′CbCr 4:2:0, with the same converter for every arm.

Arms. *Grain coded* encodes the grain version, with and without libaom's temporal filtering (alt-ref and keyframe filtering), which otherwise removes grain before coding. The grain-synthesis arms encode either the clean master (a perfect denoiser, so any saving comes from modeling the grain) or the grain version denoised with ffmpeg `nlmeans=3.5:5:3:9:7`, the 3GPP setting for these film-grain sequences, and add grain after decoding. PGNM is fitted by the unchanged Fit (Alg. S1) on clip frames 0, 10, …, 90, on the 8-bit RGB pairs (clean or denoised, grain version), and renders frame t with the unchanged renderer, bank seeds 6100–6103 and window seed 9000 + t, identical across rates and arms. AV1's grain table comes from libaom's `noise_model` with its defaults (autoregression lag 3, 32 × 32 flat-block finder) on the same frames, and aomdec applies it. VVC's grain arm uses VVenC's film-grain analysis, which estimates grain from its own temporally filtered input and accepts no clean or denoised input, so it is not matched to the master or nlmeans inputs; VVdeC synthesizes the grain.

Encoders and rates. libaom v3.6.0 runs libaom's own film-grain example settings (`-good -cpu-used=4 -end-usage=q -lag-in-frames=25 -auto-alt-ref=2`, two threads, 8-bit 4:2:0) at the quantizers $Q \in \{15, 23, 31, 39, 47, 55, 63\}$ of Netflix's AOM *Nocturne* encodes, plus a lossless row. VVenC v1.14.0 uses its medium preset with two threads at QP $\in \{17, 22, \ldots, 47\}$, and VVdeC v3.2.0 decodes. The bitrate is the bitstream payload over 1.67 s. PGNM's model is charged as the zlib-compressed float64 arrays the renderer uses: 161,269 bytes, or 774 kbps on this clip (the main paper's rule; 160,469 bytes and 770 kbps for the nlmeans fit) and 1.9 kbps amortized over the 664.5 s film, since one grain setting covers the whole film. AV1's grain parameters are inside its bitstream; the rate difference between the AV1-grain and PGNM encodes of the clean master, about 65 kbps at Q 31, makes PGNM's model the cheaper side information beyond about 20 s of video.

Measurements. Grain fidelity uses the SIDD metric code of Sec. J unchanged, with the clean master as clean image and the grain version as real image, on all 100 frames. DISTS and LPIPS use GrainNet's definitions on the full crop of frames 0, 4, …, 96, against the grain version. The main arms were scored on CPU; the supplementary groups were scored on GPU after each group's first frame matched the CPU values within 10^{-4} for every output.

Main-table rows and later arms. The main paper's rows were fixed by rate before the final scores: each grain-coded encoder at

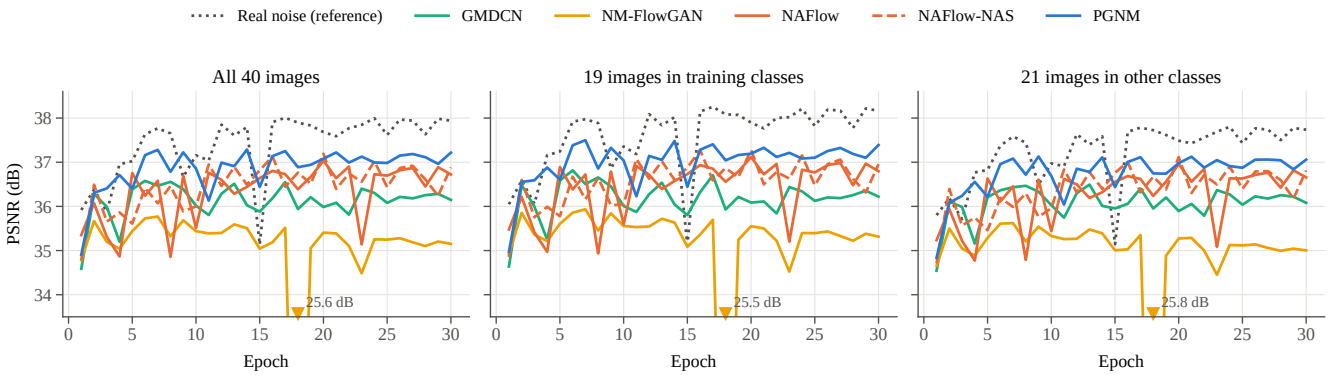

Figure S5 SIDD Validation PSNR after every training epoch (seed 1). Descriptive monitoring only: every reported number uses the epoch-30 checkpoint, and no checkpoint, epoch or setting was chosen from these curves. The learning rate halves after every fifth epoch. A triangle at the lower edge marks a value below the plotted range.

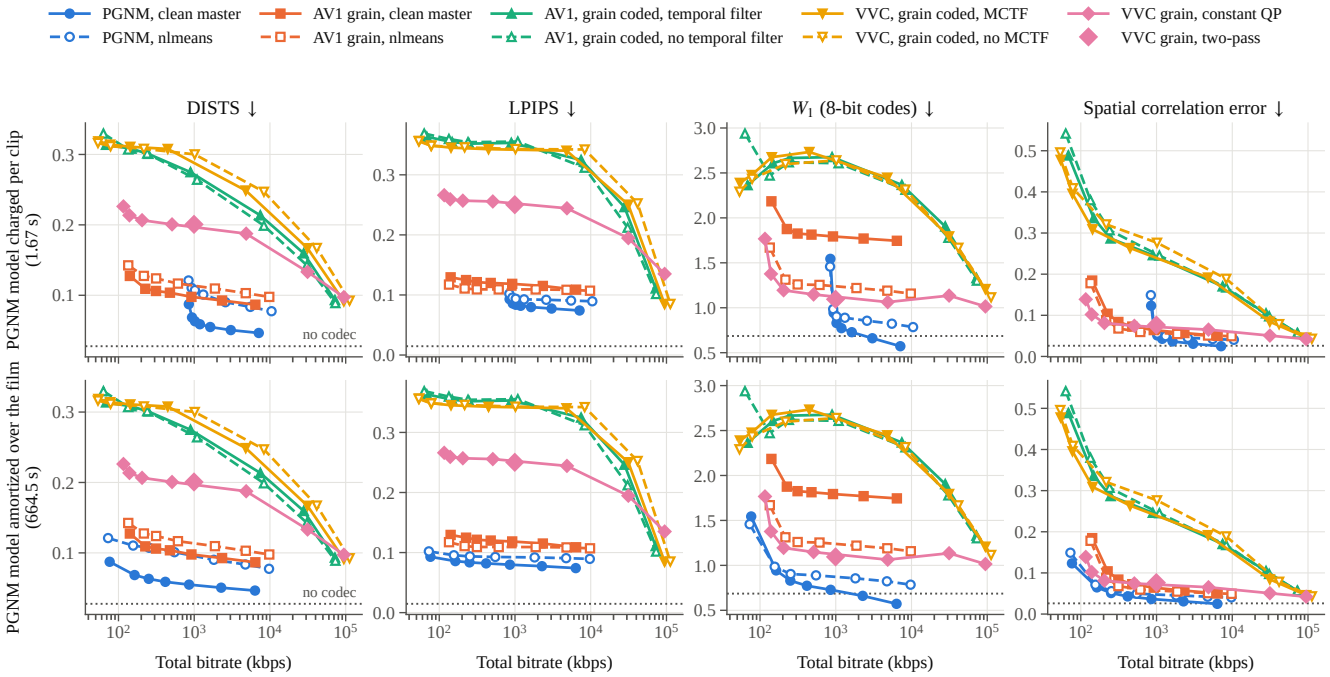

Figure S6 Rate–distortion curves of video delivery (*Nocturne*, 100 frames; lower is better). Each curve runs over its encoder's rate ladder (libaom Q 15–63, VVenC QP 17–47; lossless encodes omitted); the single diamond is VVC grain with two-pass rate control. Total bitrate includes PGNM's 161 KB model, charged in full to this 1.67 s clip (top, conservative) or amortized over the 664.5 s film, which one grain setting covers (bottom); only PGNM's curves move between the rows. Grain coded: the grain version encoded directly, with the encoder's temporal filter on (solid) or off (dashed). Dotted: the grain version after 4:2:0 conversion alone. Values in Tabs. S18 and S19.

		kbps		Appearance↓		Grain fidelity↓				
Arm	Q	clip	film	DISTS	LPIPS	W_1	Spatial	Color	Strength (CV)	PSNR
No codec (4:2:0 only)	–	–	–	0.028	0.015	0.69	0.026	0.100	0.86 (0.005)	39.45
Grain coded	15	72,098	72,098	0.093	0.111	1.33	0.056	0.208	0.77 (0.036)	34.51
	23	27,903	27,903	0.160	0.246	1.90	0.105	0.281	0.72 (0.053)	32.87
	31	7,462	7,462	0.213	0.325	2.37	0.170	0.328	0.68 (0.077)	31.88
	39	894	894	0.275	0.353	2.68	0.246	0.334	0.65 (0.113)	31.47
	47	247	247	0.301	0.352	2.67	0.287	0.345	0.66 (0.135)	31.33
	55	148	148	0.308	0.354	2.61	0.336	0.377	0.70 (0.144)	31.15
	63	69	69	0.314	0.363	2.36	0.488	0.436	0.85 (0.152)	30.39
	lossless	642,047	642,047	0.028	0.015	0.69	0.026	0.100	0.86 (0.005)	39.45
Grain coded, no temporal filter	15	73,038	73,038	0.089	0.102	1.30	0.056	0.209	0.78 (0.026)	34.56
	23	30,735	30,735	0.143	0.212	1.78	0.097	0.282	0.73 (0.041)	33.01
	31	8,328	8,328	0.199	0.312	2.31	0.168	0.315	0.69 (0.070)	31.93
	39	1,090	1,090	0.264	0.355	2.61	0.245	0.339	0.66 (0.104)	31.44
	47	240	240	0.301	0.354	2.62	0.307	0.348	0.68 (0.115)	31.24
	55	134	134	0.307	0.360	2.47	0.380	0.387	0.74 (0.117)	30.97
	63	63	63	0.330	0.369	2.94	0.542	0.454	1.01 (0.098)	29.73
PGNM, clean master	15	7,118	6,346	0.046	0.074	0.57	0.025	0.023	1.10 (0.017)	29.17
	23	3,041	2,269	0.051	0.077	0.66	0.031	0.022	1.12 (0.024)	29.07
	31	1,628	856	0.055	0.080	0.73	0.037	0.024	1.14 (0.032)	29.00
	39	1,188	416	0.059	0.082	0.77	0.043	0.027	1.15 (0.039)	28.94
	47	1,023	251	0.063	0.084	0.83	0.051	0.033	1.16 (0.045)	28.88
	55	935	163	0.069	0.086	0.94	0.065	0.045	1.18 (0.053)	28.79
	63	849	76	0.087	0.093	1.54	0.123	0.097	1.26 (0.070)	28.42
	lossless	327,771	326,999	0.039	0.068	0.23	0.015	0.055	1.02 (0.005)	29.47
AV1 grain, clean master	15	6,412	6,412	0.087	0.109	1.75	0.050	0.163	0.80 (0.049)	30.19
	23	2,332	2,332	0.093	0.115	1.77	0.057	0.170	0.83 (0.057)	30.06
	31	919	919	0.098	0.118	1.79	0.064	0.176	0.85 (0.066)	29.96
	39	478	478	0.104	0.120	1.81	0.073	0.184	0.87 (0.074)	29.88
	47	313	313	0.106	0.121	1.83	0.084	0.194	0.88 (0.081)	29.81
	55	226	226	0.109	0.125	1.88	0.104	0.210	0.91 (0.089)	29.69
	63	141	141	0.127	0.129	2.18	0.184	0.270	1.01 (0.103)	29.23
	lossless	327,065	327,065	0.066	0.091	1.81	0.045	0.144	0.69 (0.042)	30.57
PGNM, nlmeans	15	10,494	9,726	0.077	0.089	0.79	0.041	0.176	0.99 (0.032)	29.94
	23	5,502	4,734	0.084	0.091	0.82	0.042	0.180	1.00 (0.035)	29.82
	31	2,584	1,815	0.090	0.092	0.86	0.045	0.183	1.01 (0.039)	29.72
	39	1,317	549	0.101	0.093	0.89	0.049	0.185	1.01 (0.044)	29.63
	47	1,022	254	0.107	0.094	0.90	0.056	0.191	1.02 (0.051)	29.56
	55	924	156	0.110	0.096	0.98	0.071	0.203	1.05 (0.058)	29.46
	63	841	73	0.121	0.102	1.46	0.149	0.254	1.14 (0.076)	29.00
	lossless	245,548	244,780	0.064	0.085	0.72	0.039	0.164	0.99 (0.027)	30.28
AV1 grain, nlmeans	15	9,789	9,789	0.098	0.107	1.16	0.049	0.197	0.92 (0.041)	30.32
	23	4,795	4,795	0.103	0.109	1.19	0.051	0.202	0.93 (0.043)	30.20
	31	1,876	1,876	0.110	0.109	1.22	0.055	0.206	0.93 (0.047)	30.08
	39	610	610	0.117	0.109	1.25	0.060	0.208	0.94 (0.053)	29.99
	47	314	314	0.124	0.109	1.26	0.068	0.215	0.95 (0.060)	29.92
	55	216	216	0.128	0.111	1.31	0.088	0.228	0.98 (0.067)	29.80
	63	135	135	0.142	0.117	1.67	0.178	0.284	1.07 (0.084)	29.31
	lossless	244,844	244,844	0.082	0.101	1.09	0.047	0.183	0.91 (0.038)	30.70

Table S18 Video delivery with libaom AV1, every arm and quantizer (*Nocturne*, 100 frames, 1920 × 1080, 60 fps). kbps: clip, PGNM's 161 KB model charged to this 1.67 s clip (the main paper's rule); film, the same model amortized over the 664.5 s film. Appearance against the grain version (25 frames); grain fidelity of the residual from the clean master (all 100 frames; W_1 in 8-bit codes). Strength: RMS of the delivered residual over the grain version's, mean over frames (target 1), with its coefficient of variation across frames. PSNR (dB) against the grain version is reported but penalizes an independent grain realization by design. No temporal filter: alt-ref and keyframe filtering off. nlmeans: the grain version denoised with the 3GPP setting.

			Appearance↓		Grain fidelity↓				
Arm	QP	kbps	DISTS	LPIPS	W_1	Spatial	Color	Strength (CV)	PSNR
Grain coded	17	94,536	0.091	0.085	1.21	0.046	0.206	0.78 (0.025)	34.53
	22	31,305	0.167	0.250	1.79	0.084	0.316	0.72 (0.051)	32.64
	27	4,791	0.249	0.340	2.45	0.192	0.329	0.66 (0.095)	31.68
	32	448	0.308	0.343	2.73	0.264	0.323	0.65 (0.140)	31.38
	37	143	0.311	0.345	2.68	0.309	0.348	0.68 (0.159)	31.23
	42	78	0.314	0.349	2.48	0.396	0.409	0.74 (0.173)	30.92
	47	54	0.319	0.357	2.39	0.478	0.473	0.84 (0.191)	30.44
Grain coded, no MCTF	17	112,464	0.092	0.085	1.12	0.043	0.190	0.80 (0.031)	35.01
	22	41,521	0.167	0.252	1.67	0.079	0.301	0.73 (0.058)	32.96
	27	8,325	0.247	0.342	2.32	0.189	0.335	0.68 (0.092)	31.78
	32	1,015	0.300	0.343	2.64	0.276	0.325	0.66 (0.130)	31.37
	37	216	0.308	0.346	2.60	0.321	0.346	0.68 (0.151)	31.20
	42	79	0.312	0.348	2.43	0.409	0.401	0.75 (0.169)	30.90
	47	54	0.316	0.356	2.29	0.496	0.464	0.85 (0.185)	30.41
VVC grain	17	94,598	0.097	0.135	1.01	0.042	0.124	1.13 (0.018)	31.17
	22	31,366	0.133	0.195	1.14	0.051	0.144	1.16 (0.014)	29.78
	27	4,853	0.188	0.244	1.06	0.065	0.154	1.15 (0.025)	29.17
	32	510	0.201	0.256	1.15	0.074	0.158	1.15 (0.034)	28.93
	37	205	0.207	0.257	1.19	0.081	0.165	1.17 (0.049)	28.84
	42	140	0.214	0.259	1.38	0.102	0.183	1.21 (0.063)	28.65
	47	116	0.226	0.266	1.77	0.139	0.210	1.29 (0.082)	28.31
VVC grain, two-pass	935 k target	994	0.201	0.250	1.09	0.076	0.160	1.13 (0.040)	29.05

Table S19 Video delivery with VVC (VVenC 1.14 medium preset, VVdeC 3.2; constant QP except the two-pass row), measured as in Tab. S18. No MCTF: VVenC's motion-compensated temporal filter off. VVC grain: VVenC's film-grain analysis of the grain version, synthesized by VVdeC; the two-pass row targets 935 kbps, the total rate of PGNM at Q 55.

			Saving against each direct encode (%, clip)				Smallest (%)	
Grain synthesis	Q	kbps	AV1	AV1^{nf}	VVC	VVC^{nf}	clip	film
PGNM, master	15	7,118	≥90.1	≥90.3	≥92.5	≥93.7	≥90.1	≥91.2
	23	3,041	≥95.8	≥95.8	≥96.8	≥97.3	≥95.8	≥96.9
	31	1,628	≥97.7	≥97.8	≥98.3	≥98.6	≥97.7	≥98.8
	39	1,188	≥98.4	≥98.4	≥98.7	≥98.9	≥98.4	≥99.4
	47	1,023	≥98.6	≥98.6	≥98.9	≥99.1	≥98.6	≥99.7
	55	935	≥98.7	≥98.7	≥99.0	≥99.2	≥98.7	≥99.8
	63	849	≥98.8	≥98.8	≥99.1	≥99.2	≥98.8	≥99.9
AV1 grain, master	15	6,412	≥91.1	≥91.2	≥93.2	≥94.3	≥91.1	≥91.1
	23	2,332	≥96.8	96.6	97.5	97.9	96.6	96.6
	31	919	98.6	98.5	98.9	99.1	98.5	98.5
	39	478	99.2	99.2	99.4	99.5	99.2	99.2
	47	313	99.5	99.4	99.6	99.7	99.4	99.4
	55	226	99.6	99.6	99.7	99.7	99.6	99.6
	63	141	99.7	99.6	99.7	99.8	99.6	99.6
PGNM, nlmeans	15	10,494	≥85.4	≥85.6	≥88.9	≥90.7	≥85.4	≥86.5
	23	5,502	≥92.4	≥92.5	≥94.2	≥95.1	≥92.4	≥93.4
	31	2,584	≥96.4	96.4	≥97.3	≥97.7	96.4	97.5
	39	1,317	98.0	97.8	98.4	98.7	97.8	99.1
	47	1,022	98.3	98.1	98.6	98.9	98.1	99.5
	55	924	98.4	98.2	98.7	99.0	98.2	99.7
	63	841	98.3	98.1	98.6	98.9	98.1	99.8
AV1 grain, nlmeans	15	9,789	85.5	84.6	88.6	90.7	84.6	84.6
	23	4,795	92.4	91.8	94.0	95.1	91.8	91.8
	31	1,876	96.7	96.4	97.4	97.9	96.4	96.4
	39	610	98.8	98.7	99.1	99.3	98.7	98.7
	47	314	99.3	99.2	99.5	99.6	99.2	99.2
	55	216	99.5	99.4	99.6	99.7	99.4	99.4
	63	135	99.6	99.6	99.7	99.8	99.6	99.6
VVC grain	17	94,598	−39.1	−48.4	−9.3	10.1	−48.4	−48.4
	22	31,366	22.9	12.2	38.8	51.9	12.2	12.2
	27	4,853	65.5	55.2	75.1	82.4	55.2	55.2
	32	510	95.0	93.5	96.5	97.6	93.5	93.5
	37	205	97.7	96.9	98.4	98.9	96.9	96.9
	42	140	98.1	97.3	98.7	99.1	97.3	97.3
	47	116	97.6	96.7	98.6	99.1	96.7	96.7
VVC grain, two-pass	935 k	994	90.3	87.3	93.1	95.3	87.3	87.3

Table S20 DISTS-matched bitrate savings, the headline measure. For each grain-synthesis rate, the direct encode's bitrate reaching the same DISTS is interpolated linearly in log-rate between adjacent ladder points (first crossing from low rate); saving = 1− rate/matched direct rate. ≥: the direct encode never reaches the arm's DISTS, so its highest rate gives a lower bound. AV1/VVC: the grain-coded encodes; ^{nf}: temporal filter off. Smallest: over the four direct encodes, with PGNM's model charged per clip or amortized over the film; the main paper cites the per-clip range.

Grain synthesis	DISTS	LPIPS	W_1	Spatial	Color
PGNM, master	≥90.1 to ≥98.8	≥90.1 to ≥98.8	≥90.1 to ≥98.7	≥90.1 to ≥98.6	≥90.1 to ≥98.8
AV1 grain, master	≥91.1 to 99.6	90.7 to 99.8	80.4 to 99.1	≥91.1 to 99.0	≥91.1 to 99.7
PGNM, nlmeans	≥85.4 to 98.2	≥85.4 to ≥98.8	≥85.4 to ≥98.7	≥85.4 to 98.6	≥85.4 to ≥98.7
AV1 grain, nlmeans	84.6 to 99.6	86.0 to 99.8	≥86.4 to 99.7	≥86.4 to 99.4	≥86.4 to 99.6
VVC grain	−48.4 to 97.3	−68.3 to 99.2	≥−31.2 to 99.8	≥−31.2 to 99.4	≥−31.2 to 99.8
VVC grain, two-pass	87.3	94.7	≥98.6	97.5	≥98.6

Table S21 Matched savings for every measurement (secondary): the smallest per-clip saving over the four direct encodes, as a range over each arm's rates (%). Matching as in Tab. S20, with each measurement in place of DISTS.

its highest lossy rate with the closer temporal-filter setting, AV1 grain and PGNM at Q 31, and PGNM at the Q whose total rate is closest to AV1 grain at Q 31 (Q 55). Three arms were added after results existed and are reported like the others: the grain-coded arm without libaom's temporal filter (after part of the main scores), the VVC arms (after a preview), and VVC grain with two-pass rate control at 935 kbps, the rate of PGNM at Q 55, because VVC grain had no constant-QP point near it (after the final reports; its row then joined the main table).

Results. Fig. S6 plots the rate–distortion curves of every arm, with PGNM's model charged per clip and amortized over the film; Tabs. S18 and S19 list every arm and rate, Tabs. S20 and S21 the matched savings, and Fig. S7 the delivered frames. Charged per clip, the model adds about 770 kbps to every PGNM point, so PGNM's curves start near 850 kbps; amortized, it adds 1.9 kbps and PGNM on the clean master reaches DISTS 0.087 at 76 kbps, below the DISTS of every lossy grain-coded encode (at least 0.089, at 73 Mbps). Coding the grain costs 9–12 times the bits of the clean master at Q 15–31 (up to 14 times without libaom's temporal filter); from Q 47 on, the encoder drops the grain and the two rates meet. Against even the closest of the four grain-coded baselines, PGNM saves 90.1–98.8% of the bitrate at matched DISTS when synthesizing on the clean master and 85.4–98.2% on nlmeans-denoised video. AV1 grain synthesis reaches similar savings, but each is matched at its own DISTS, which is higher than PGNM's at every quantizer (0.087–0.127 against 0.046–0.087 on the clean master); VVC's own grain pipeline stays near DISTS 0.20 at both 0.51 and 0.99 Mbps. On the clean master at the lossy rates, PGNM's delivered grain is 10–26% stronger in RMS than the grain version's, and AV1's ranges from 20% weaker to 1% stronger; the 4:2:0 round trip of the grain version alone keeps 86% of its RMS. PGNM's signed lookup can represent the grain version's darker offset and AV1's zero-mean grain cannot: delivered mean offsets at Q 31 are $(0.16, -0.01, 0.17)$ codes for PGNM and $(0.74, 1.89, 1.71)$ for AV1 grain on the clean master. Strength, PSNR and offsets are reported for information only.

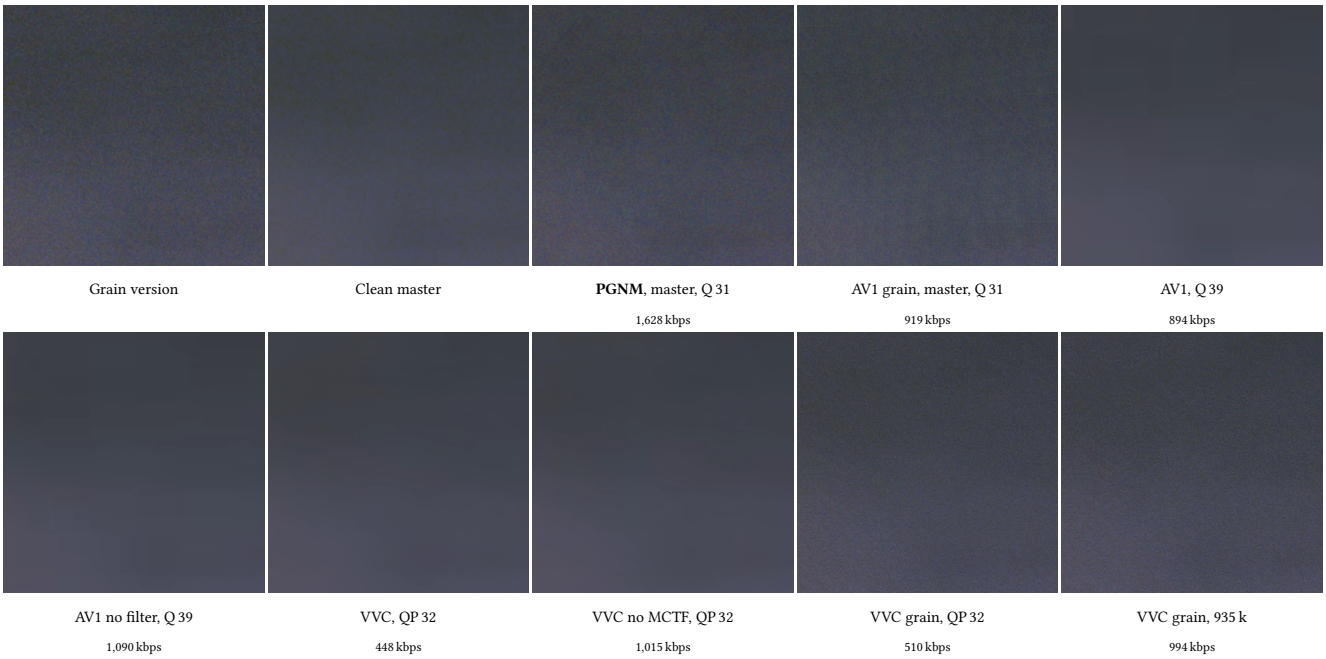

Figure S7 Delivered frames. Center 384 × 384 crop of clip frame 50 (top-left corner at row 348, column 768 of the 1920 × 1080 crop), chosen by rule before viewing: PGNM and AV1 grain at Q 31, each direct encode at the ladder rate closest to PGNM's total rate, and VVC grain at QP 32 and its two-pass run. kbps: total rate, PGNM's model charged per clip. Please zoom in; grain is at native 4K size. *Nocturne* © Netflix, Inc., CC BY 4.0.

G Visual comparisons on SIDD

This section extends Fig. 5 of the main paper to every evaluated camera–ISO class of SIDD [1]. Every crop comes from the evaluation scored in Tab. 1 of the main paper: each method shows the first of its five generator runs, with the fit or checkpoint setting used there. NM-FlowGAN and sRGB-Flow are shown only for the classes their checkpoints cover. Residual rows show the output minus the clean image, amplified by one gain per class that is set from the real residual alone, so every method in a row is shown at the same scale.

Uncurated crops. For every class, one source image, block and 128 × 128 window were drawn with a fixed seed from the clean image only, before any output was viewed (Figs. S8–S11; windows in Tab. S22). The S6 ISO 400 crop of the main paper is one of them.

Highly textured regions. PGNM uses the clean image only through pixel intensity, whereas the neural generators also see the surrounding image structure. For every class, Figs. S12–S15 therefore show the window with the most local contrast in the clean image, selected by a rule fixed from the clean images alone (Tab. S23). The GP ISO 6400 crop of the main paper is one of them.

The blocks shown by NAFlow. NAFlow [23] illustrates its results with two complete 256 × 256 Validation blocks. Matching their real noisy panels against all 1,280 Validation blocks identifies them as image 1, block 0 (GP ISO 10000) and image 6, block 4 (IP ISO 1600). Fig. S16 shows every generator on these two blocks, which were selected by NAFlow's authors rather than by us. On both, PGNM has the lowest small-scale KL, for the displayed draw and for the mean over the five generator runs.

A clean image from outside SIDD. Sec. 5 of the main paper applies every SIDD fit and checkpoint, unchanged, to the Kodak image *kodim19*, whose most textured windows have more than twice the local contrast of the most textured SIDD window. Fig. S17 shows the S6 ISO 800 class, which every method supports, in the image's most textured window.

H Visual comparisons on film grain

This section extends Fig. 6 of the main paper to all eight radii of the Newson renderer [33]. Every crop comes from the evaluation scored in Tab. 3 of the main paper: each method shows its first generator run, the reference is one Newson realization, and an independently seeded Newson run shows the renderer's own sampling variability. PGNM and Zhang23 [48] add grain to the shared blur $B * x$ fitted on training pairs. Residuals are taken relative to $\mu = B * x$ and amplified by one gain per radius, set from the reference residual alone.

Uncurated crops. For every radius, one test image and a 128 × 128 window were drawn with a fixed seed from the clean images only, before any output was viewed (Figs. S18 and S19; windows in Tab. S22). The crops for $r = 0.4$ and $r = 0.8$ in the main paper are among them.

A highly textured region. Figs. S20 and S21 show the window with the most local contrast over all test images at every radius, selected by the rule used for SIDD (Tab. S23).

Class	SIDD source (Validation index)	Block	x	y
G4 ISO 400	0148_007_G4_00400_00400_4400_L (31)	20	34	65
G4 ISO 800	0100_005_G4_00800_00400_3200_N (21)	1	2	9
GP ISO 1600	0067_003_GP_01600_01500_3200_L (13)	26	117	83
GP ISO 3200	0158_007_GP_03200_03200_5500_N (33)	16	125	78
GP ISO 6400	0183_008_GP_06400_06400_5500_N (38)	20	72	70
GP ISO 10000	0109_005_GP_10000_08000_4400_N (23)	8	86	105
IP ISO 1000	0112_005_IP_01000_07500_5500_L (24)	17	98	4
IP ISO 1600	0162_007_IP_01600_01600_3200_L (34)	27	11	22
IP ISO 2000	0071_003_IP_02000_04000_5500_L (14)	27	69	2
N6 ISO 400	0074_004_N6_00400_00040_3200_L (15)	9	54	62
N6 ISO 800	0049_002_N6_00800_00800_5500_L (8)	0	16	0
N6 ISO 3200	0024_001_N6_03200_01500_5500_N (2)	21	83	67
S6 ISO 400★	0176_008_S6_00400_00100_5500_L (37)	8	98	79
S6 ISO 800	0009_001_S6_00800_00350_3200_L (0)	14	103	128
S6 ISO 1600	0103_005_S6_01600_00800_4400_L (22)	31	88	48
S6 ISO 3200	0153_007_S6_03200_03200_5500_L (32)	20	88	108
Radius	Test image (size $h \times w$)		x	y
$r = 0.025$	59 (499 × 333)		173	284
$r = 0.05$	26 (333 × 500)		220	45
$r = 0.075$	45 (334 × 500)		98	164
$r = 0.1$	25 (334 × 500)		311	72
$r = 0.2$	32 (336 × 500)		189	121
$r = 0.4$★	28 (331 × 500)		190	136
$r = 0.6$	17 (375 × 500)		279	240
$r = 0.8$★	29 (333 × 499)		57	13

Table S22 Crop windows of the uncurated comparisons (Figs. S8–S11 and S18–S19): top-left corner (x, y) of each 128 × 128 window; SIDD blocks are 256 × 256, block indices from 0. ★Also shown in the main paper.

Class	SIDD source (Validation index)	Block	x	y	Tex.
G4 ISO 400	0058_003_G4_00400_00500_5500_L (11)	22	128	128	6.6
G4 ISO 800	0174_008_G4_00800_00800_4400_N (36)	5	16	128	14.3
GP ISO 1600	0131_006_GP_01600_01250_4400_N (28)	14	24	80	7.5
GP ISO 3200	0158_007_GP_03200_03200_5500_N (33)	13	8	96	9.8
GP ISO 6400★	0085_004_GP_06400_02000_4400_N (18)	19	120	80	16.9
GP ISO 10000	0109_005_GP_10000_08000_4400_N (23)	3	128	0	14.3
IP ISO 1000	0112_005_IP_01000_07500_5500_L (24)	30	80	24	3.3
IP ISO 1600	0031_001_IP_01600_02000_3200_N (4)	20	40	8	5.0
IP ISO 2000	0071_003_IP_02000_04000_5500_L (14)	22	8	24	3.1
N6 ISO 400	0074_004_N6_00400_00040_3200_L (15)	19	96	120	12.8
N6 ISO 800	0049_002_N6_00800_00800_5500_L (8)	22	104	104	7.8
N6 ISO 3200	0056_003_N6_03200_04000_5500_N (10)	29	48	96	12.3
S6 ISO 400	0176_008_S6_00400_00100_5500_L (37)	26	0	8	15.5
S6 ISO 800	0061_003_S6_00800_00500_4400_L (12)	27	96	56	13.0
S6 ISO 1600	0103_005_S6_01600_00800_4400_L (22)	22	128	0	9.1
S6 ISO 3200	0053_002_S6_03200_02000_5500_N (9)	22	24	96	13.9
Radius	Test image (size $h \times w$)		x	y	Tex.
All radii	38 (333 × 499)		115	19	62.2

Table S23 Crop windows of the highly textured comparisons (Figs. S12–S15 and S20–S21): top-left corner (x, y) of each 128 × 128 window and its texture (Tex.), the median 7 × 7 local standard deviation of the clean window in 8-bit code values. ★Also shown in the main paper.

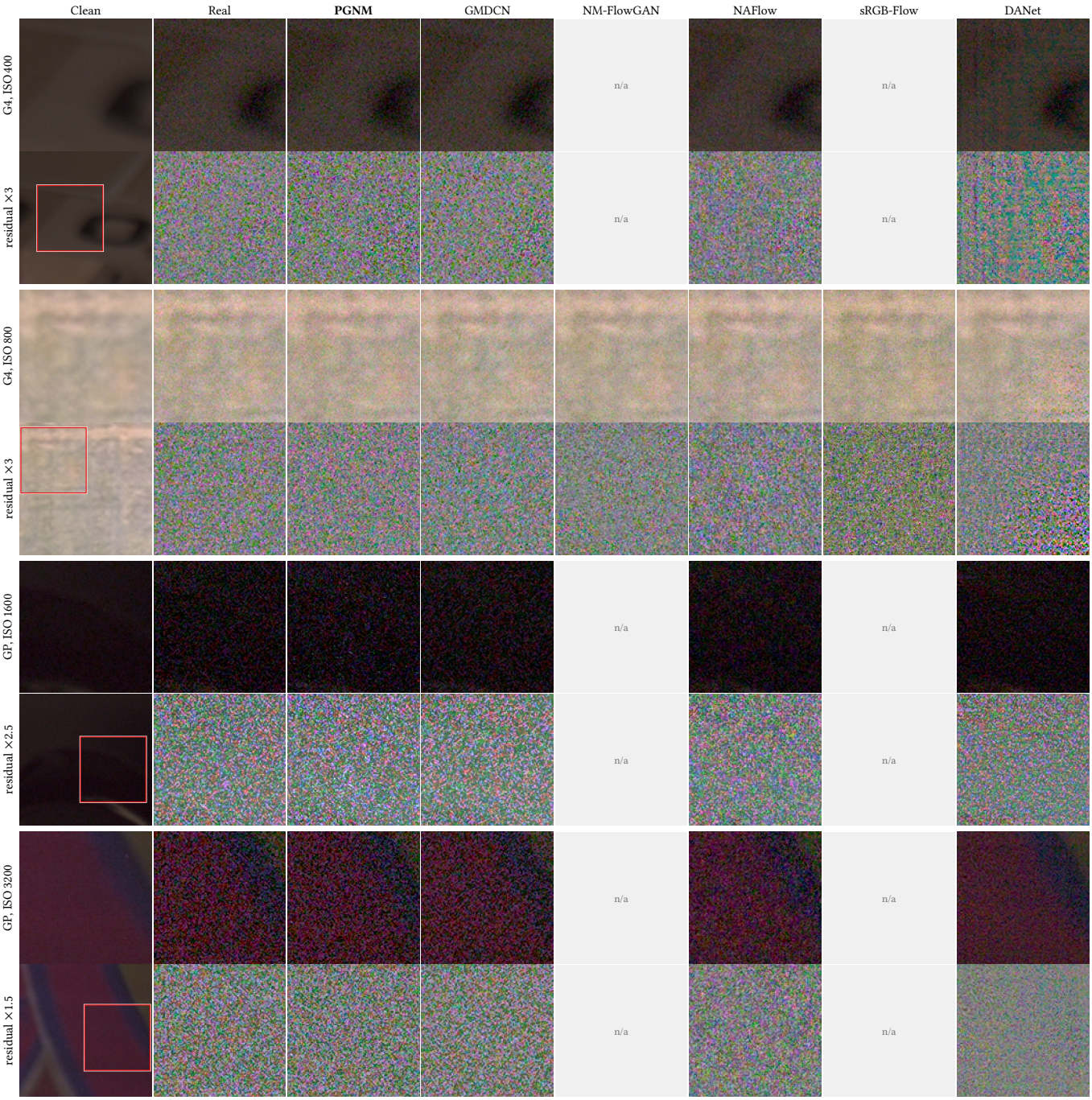

Figure S8 **Uncurated SIDD comparison, every evaluated camera–ISO class (1/4).** Each source, block and 128 × 128 window was drawn with a fixed pseudo-random seed from the clean image only; a window was redrawn only when more than 25% of its clean pixels were near saturation, and no draw was replaced after viewing the outputs. Every method shows its first generator run. sRGB-Flow and NM-FlowGAN are shown only for the 19-source block covered by their checkpoints (n/a otherwise). Second row of each pair: residual relative to the clean image, amplified by the factor at left (set from the real residual alone; gray is zero); under the clean column, the full 256 × 256 block with the crop window marked. Crops are shown at native resolution; zoom in for detail.

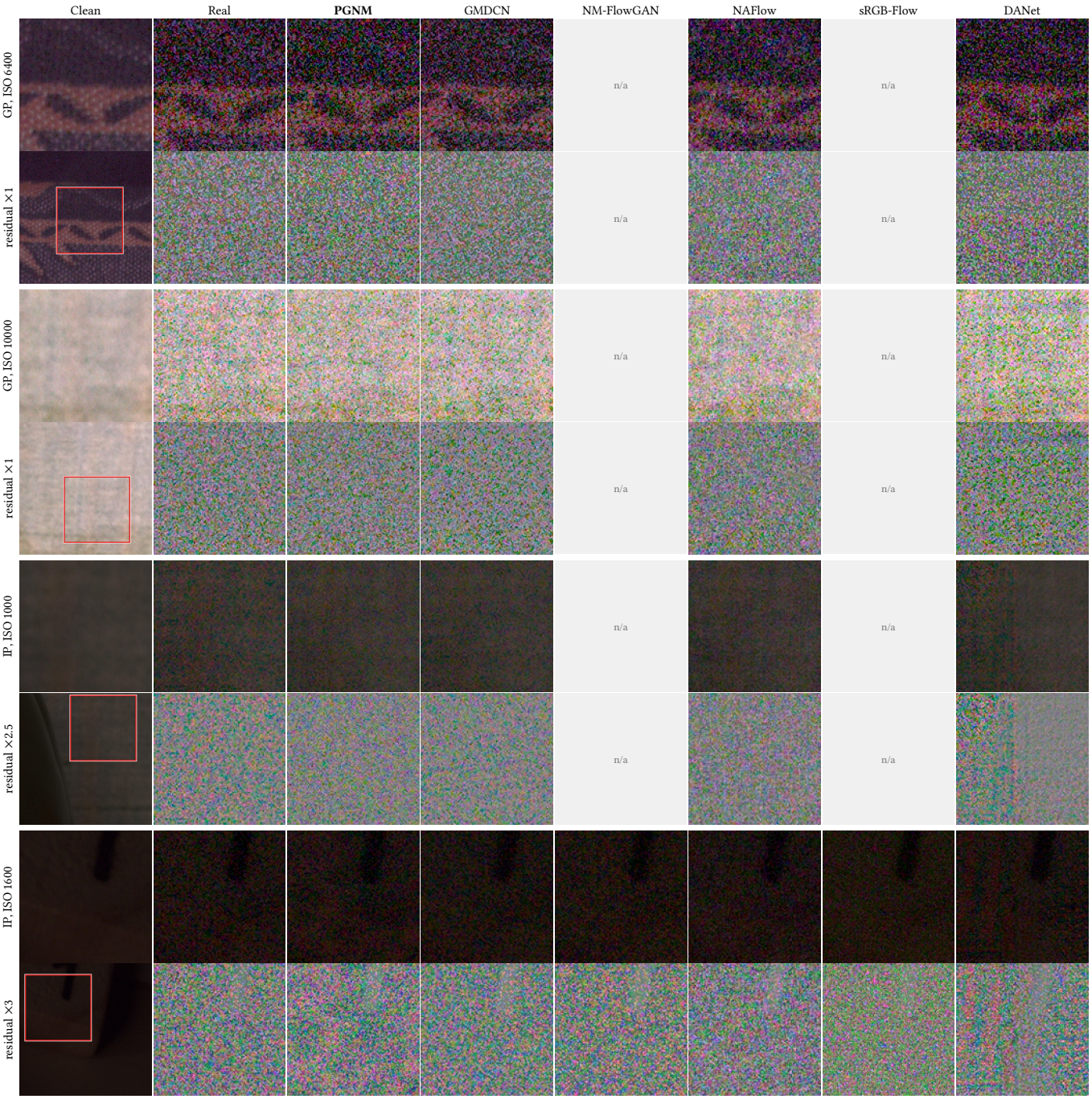

Figure S9 Uncurated SIDD comparison, every evaluated camera–ISO class (2/4). Each source, block and 128 × 128 window was drawn with a fixed pseudo-random seed from the clean image only; a window was redrawn only when more than 25% of its clean pixels were near saturation, and no draw was replaced after viewing the outputs. Every method shows its first generator run. sRGB-Flow and NM-FlowGAN are shown only for the 19-source block covered by their checkpoints (n/a otherwise). Second row of each pair: residual relative to the clean image, amplified by the factor at left (set from the real residual alone; gray is zero); under the clean column, the full 256 × 256 block with the crop window marked. Crops are shown at native resolution; zoom in for detail.

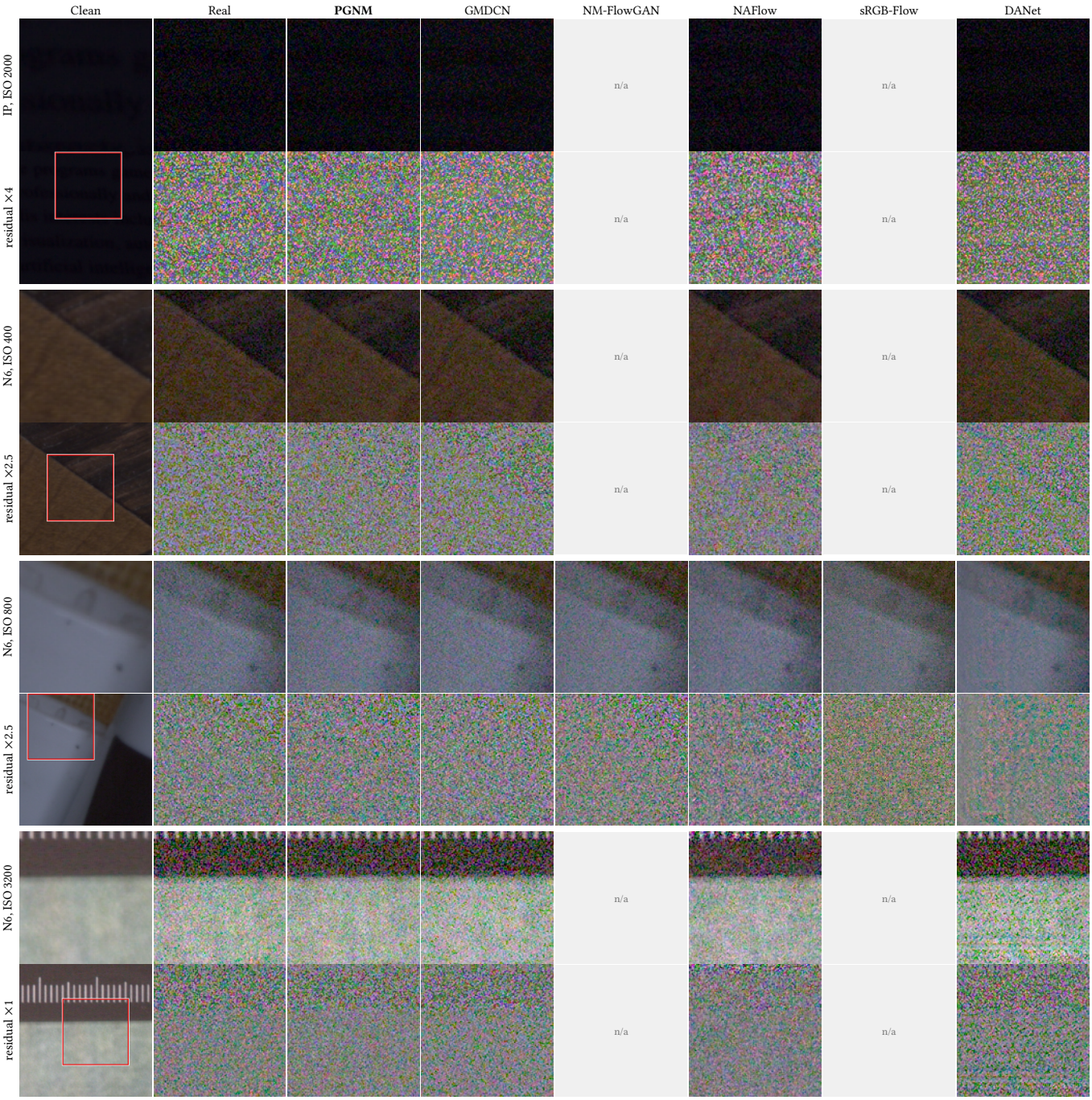

Figure S10 **Uncurated SIDD comparison, every evaluated camera–ISO class (3/4).** Each source, block and 128 × 128 window was drawn with a fixed pseudo-random seed from the clean image only; a window was redrawn only when more than 25% of its clean pixels were near saturation, and no draw was replaced after viewing the outputs. Every method shows its first generator run. sRGB-Flow and NM-FlowGAN are shown only for the 19-source block covered by their checkpoints (n/a otherwise). Second row of each pair: residual relative to the clean image, amplified by the factor at left (set from the real residual alone; gray is zero); under the clean column, the full 256 × 256 block with the crop window marked. Crops are shown at native resolution; zoom in for detail.

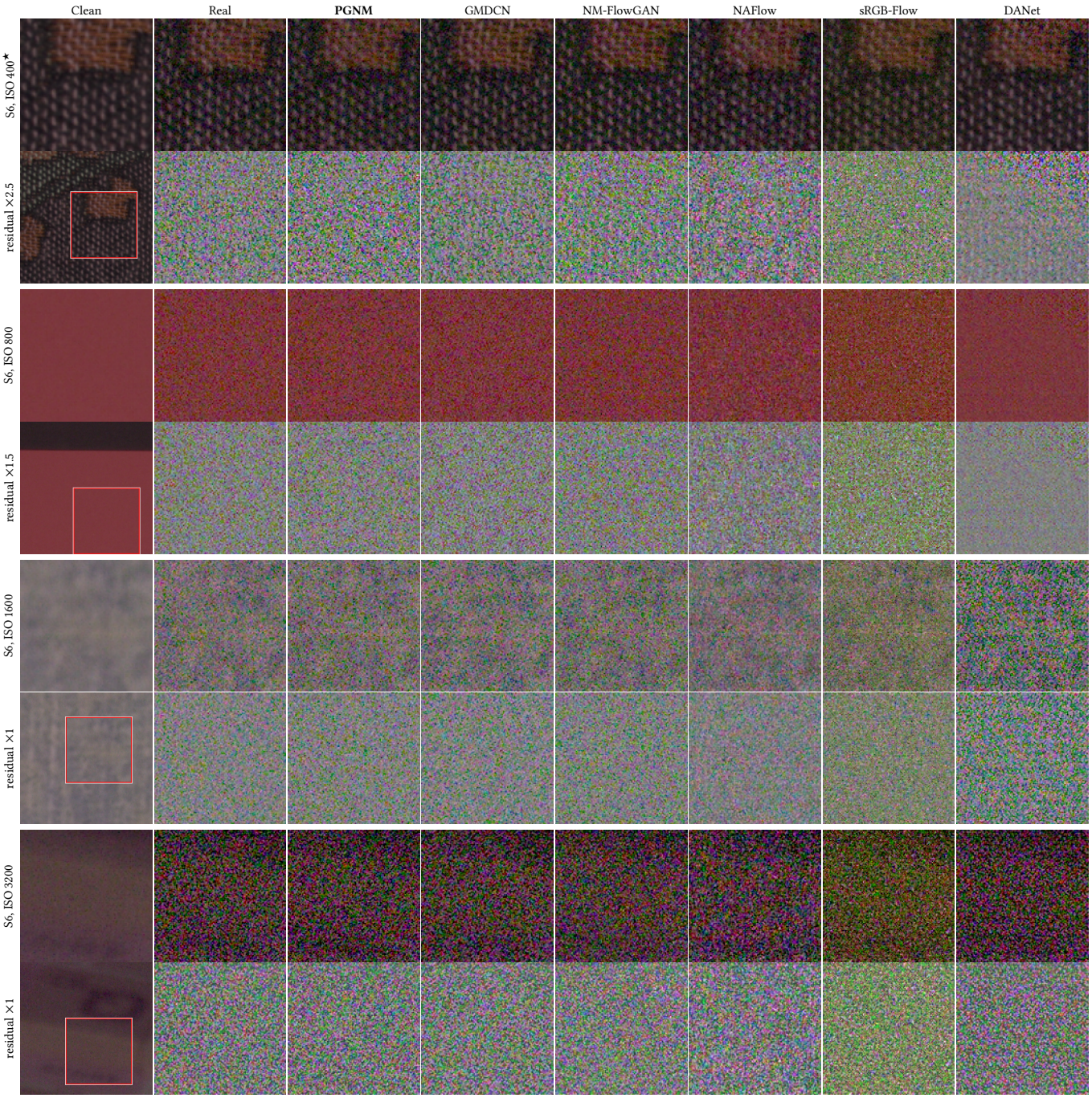

Figure S11 **Uncurated SIDD comparison, every evaluated camera–ISO class (4/4).** Each source, block and 128 × 128 window was drawn with a fixed pseudo-random seed from the clean image only; a window was redrawn only when more than 25% of its clean pixels were near saturation, and no draw was replaced after viewing the outputs. Every method shows its first generator run. sRGB-Flow and NM-FlowGAN are shown only for the 19-source block covered by their checkpoints (n/a otherwise). Second row of each pair: residual relative to the clean image, amplified by the factor at left (set from the real residual alone; gray is zero); under the clean column, the full 256 × 256 block with the crop window marked. Crops are shown at native resolution; zoom in for detail. ★Also shown in the main paper (Fig. 5).

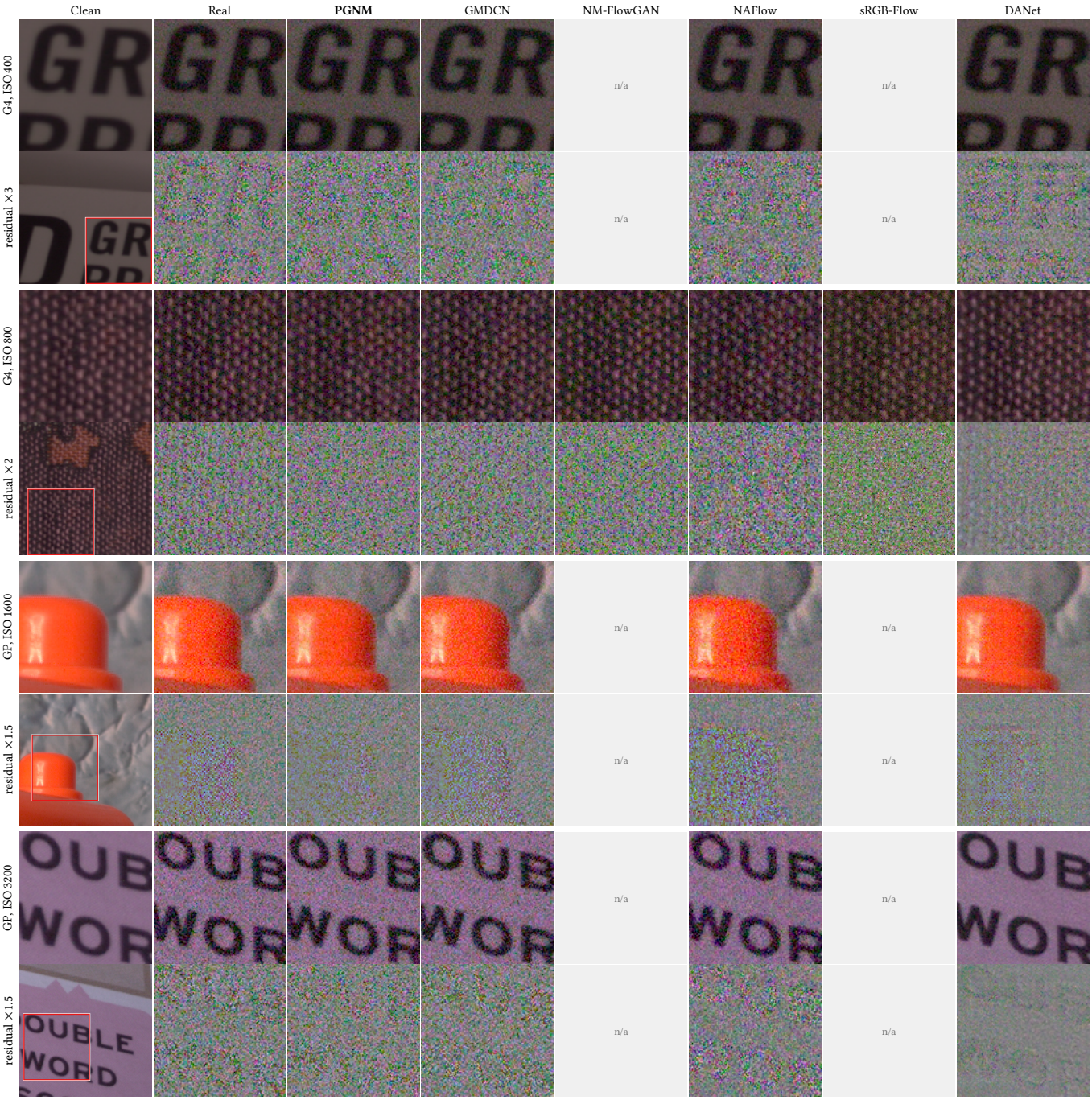

Figure S12 SIDD comparison on highly textured regions, every evaluated camera–ISO class (1/4). In each class, the 128 × 128 window with the highest median 7 × 7 local standard deviation of the clean image (maximum over RGB, the contrast of our flatness mask) was taken over every block of the class's evaluated sources on an 8-pixel grid, excluding windows with more than 25% near-saturated clean pixels; it was fixed from the clean images before any output was viewed. Every method shows its first generator run. sRGB-Flow and NM-FlowGAN are shown only for the 19-source block covered by their checkpoints (n/a otherwise). Second row of each pair: residual relative to the clean image, amplified by the factor at left (set from the real residual alone; gray is zero); under the clean column, the full 256 × 256 block with the crop window marked. Crops are shown at native resolution; zoom in for detail.

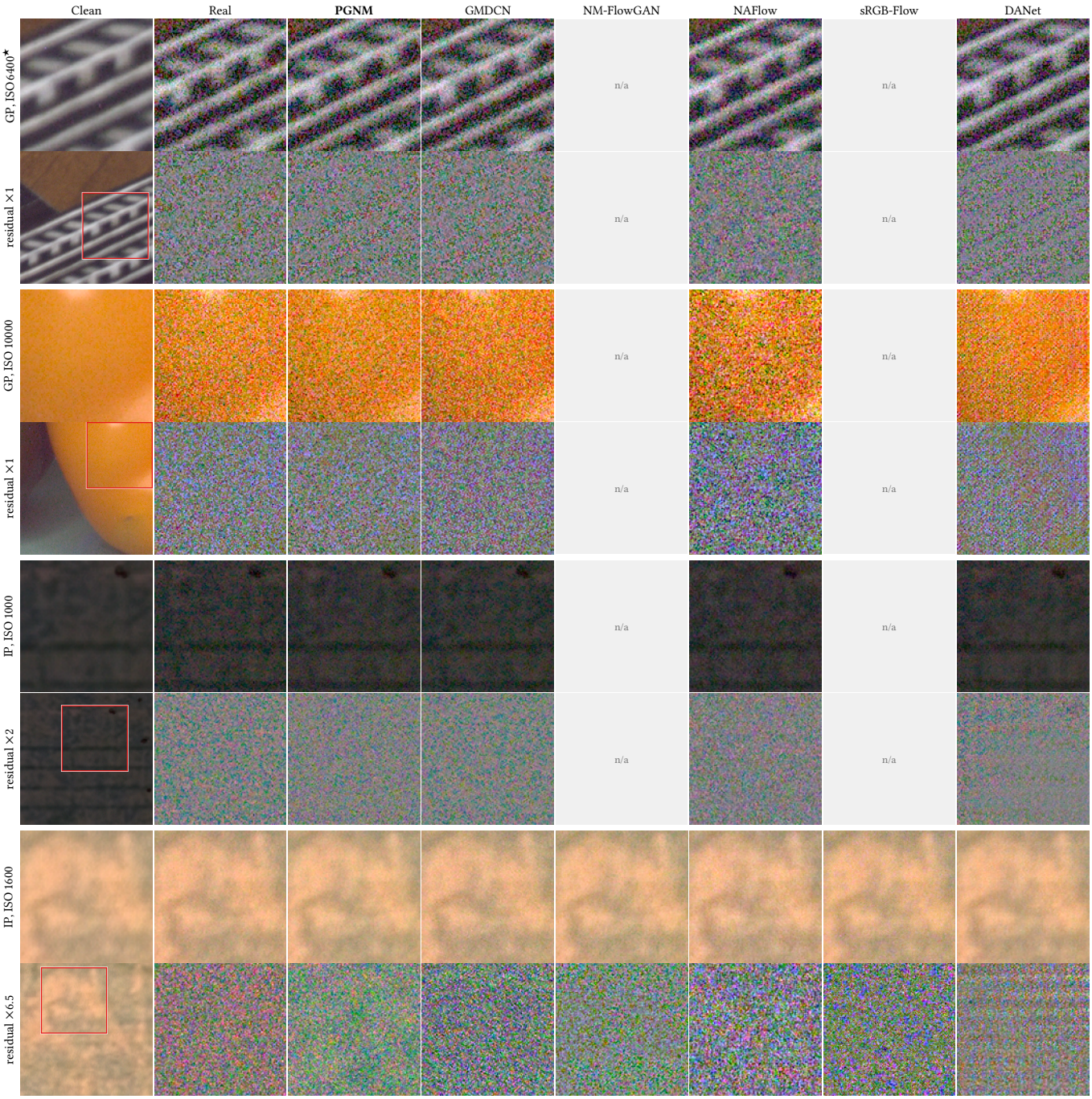

Figure S13 SIDD comparison on highly textured regions, every evaluated camera–ISO class (2/4). In each class, the 128 × 128 window with the highest median 7 × 7 local standard deviation of the clean image (maximum over RGB, the contrast of our flatness mask) was taken over every block of the class's evaluated sources on an 8-pixel grid, excluding windows with more than 25% near-saturated clean pixels; it was fixed from the clean images before any output was viewed. Every method shows its first generator run. sRGB-Flow and NM-FlowGAN are shown only for the 19-source block covered by their checkpoints (n/a otherwise). Second row of each pair: residual relative to the clean image, amplified by the factor at left (set from the real residual alone; gray is zero); under the clean column, the full 256 × 256 block with the crop window marked. Crops are shown at native resolution; zoom in for detail. ★Also shown in the main paper (Fig. 5).

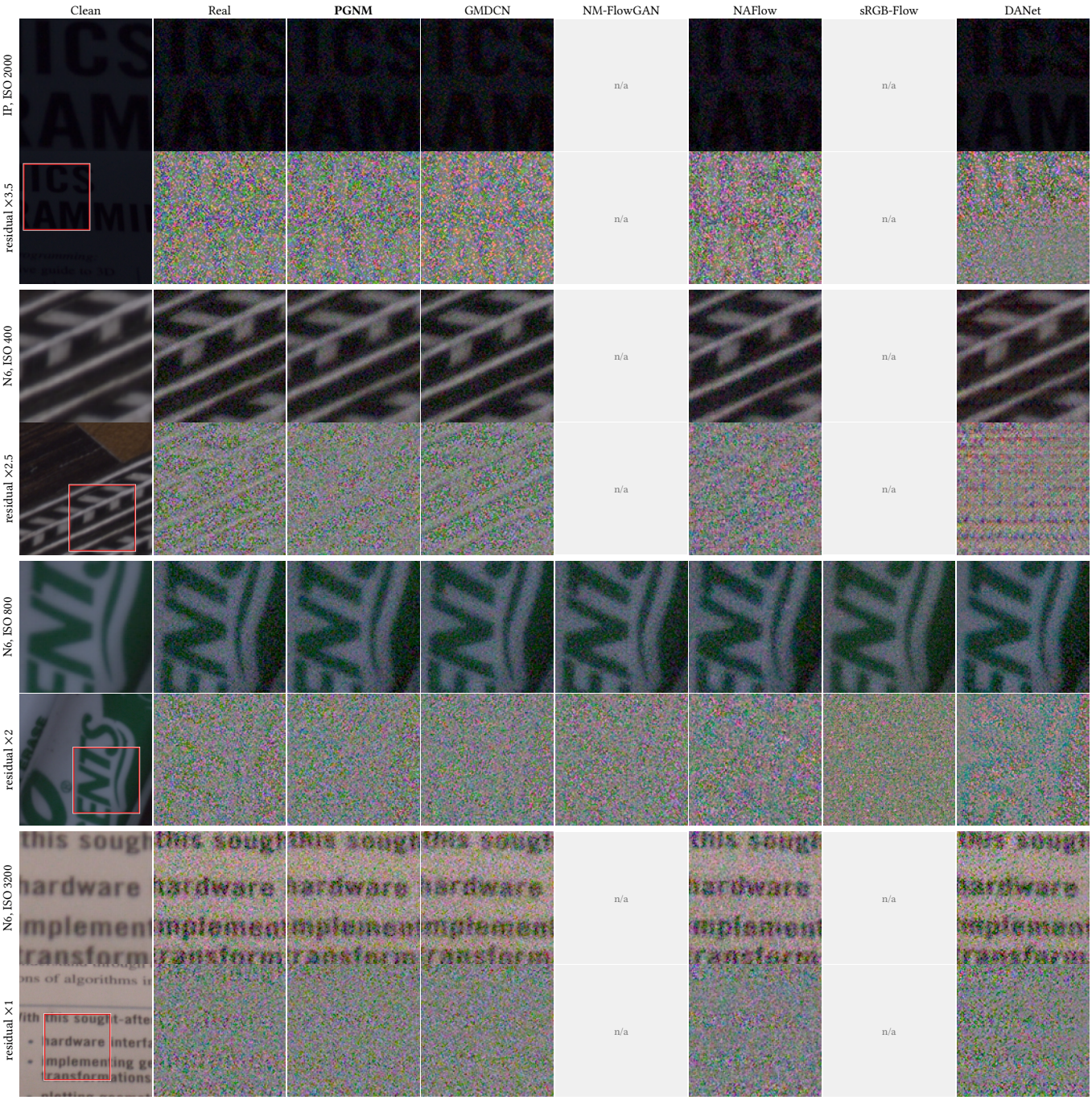

Figure S14 SIDD comparison on highly textured regions, every evaluated camera–ISO class (3/4). In each class, the 128 × 128 window with the highest median 7 × 7 local standard deviation of the clean image (maximum over RGB, the contrast of our flatness mask) was taken over every block of the class's evaluated sources on an 8-pixel grid, excluding windows with more than 25% near-saturated clean pixels; it was fixed from the clean images before any output was viewed. Every method shows its first generator run. sRGB-Flow and NM-FlowGAN are shown only for the 19-source block covered by their checkpoints (n/a otherwise). Second row of each pair: residual relative to the clean image, amplified by the factor at left (set from the real residual alone; gray is zero); under the clean column, the full 256 × 256 block with the crop window marked. Crops are shown at native resolution; zoom in for detail.

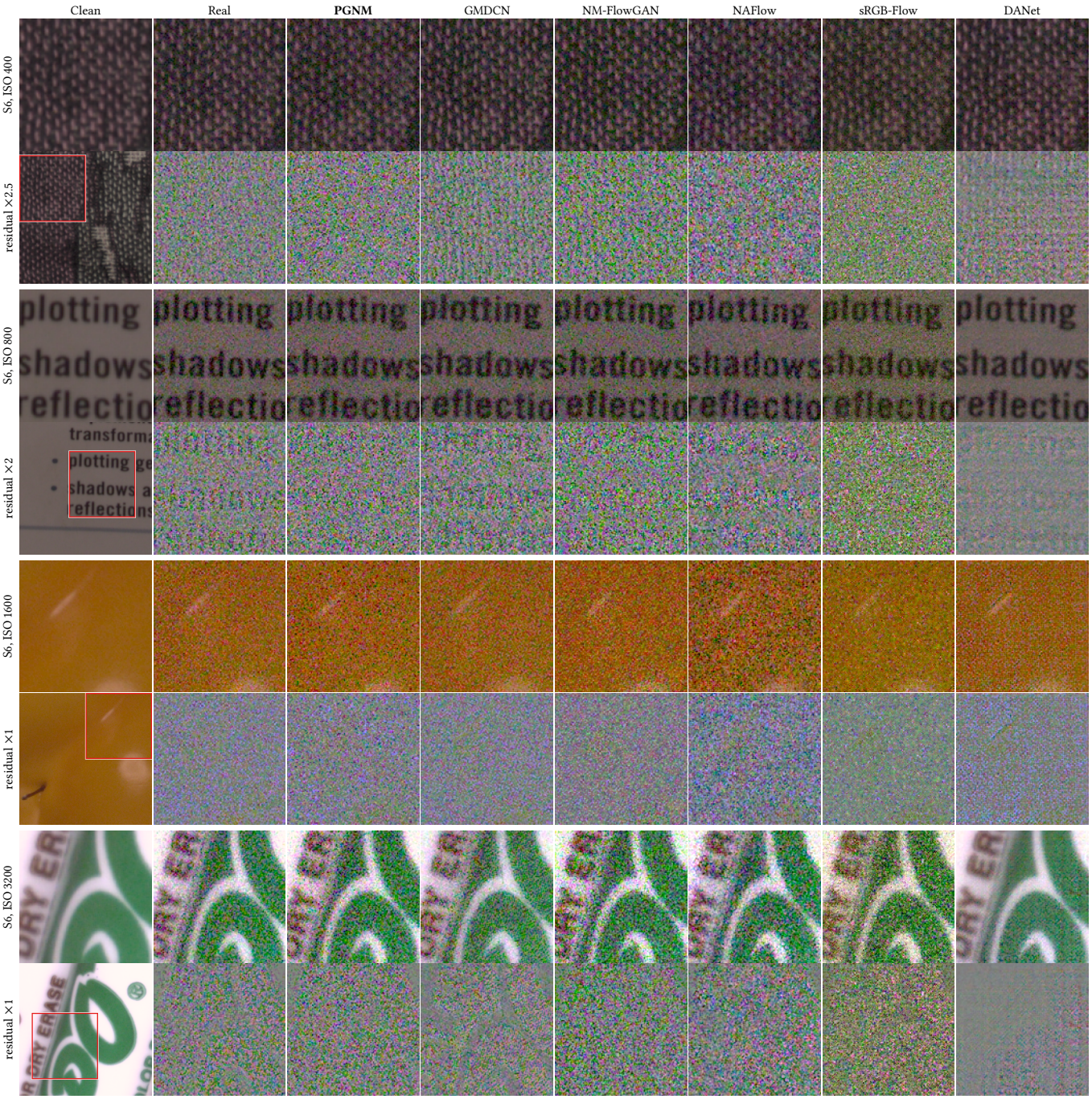

Figure S15 SIDD comparison on highly textured regions, every evaluated camera–ISO class (4/4). In each class, the 128 × 128 window with the highest median 7 × 7 local standard deviation of the clean image (maximum over RGB, the contrast of our flatness mask) was taken over every block of the class's evaluated sources on an 8-pixel grid, excluding windows with more than 25% near-saturated clean pixels; it was fixed from the clean images before any output was viewed. Every method shows its first generator run. sRGB-Flow and NM-FlowGAN are shown only for the 19-source block covered by their checkpoints (n/a otherwise). Second row of each pair: residual relative to the clean image, amplified by the factor at left (set from the real residual alone; gray is zero); under the clean column, the full 256 × 256 block with the crop window marked. Crops are shown at native resolution; zoom in for detail.

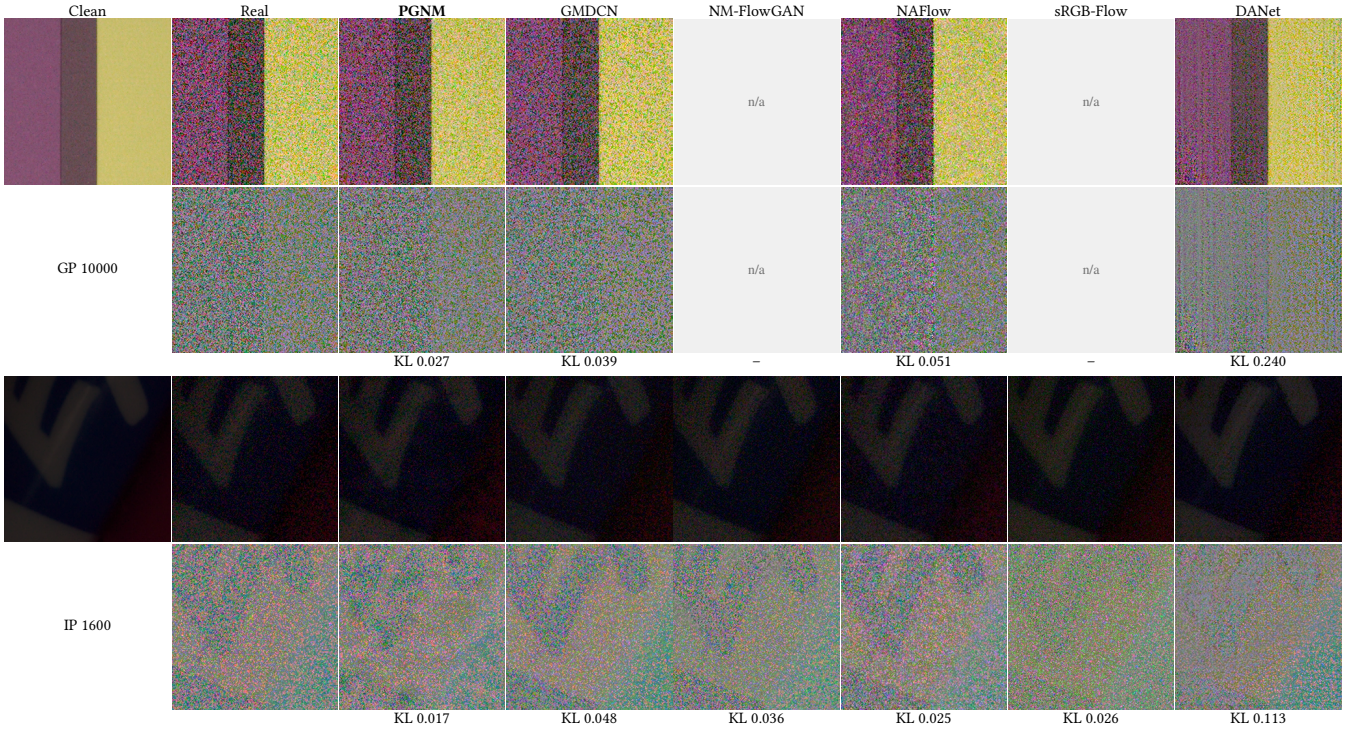

Figure S16 **The two SIDD Validation blocks shown by NAFlow** [23] (its Fig. 4; complete 256 × 256 blocks), rendered by every generator with its fit or checkpoint (first generator run, as scored). Second row of each pair: residual relative to the clean image, amplified by one gain per block set from the real residual (gray is zero). KL: small-scale KL of the displayed draw, averaged over the block's 64 32 × 32 tiles. n/a: class not covered by the checkpoint.

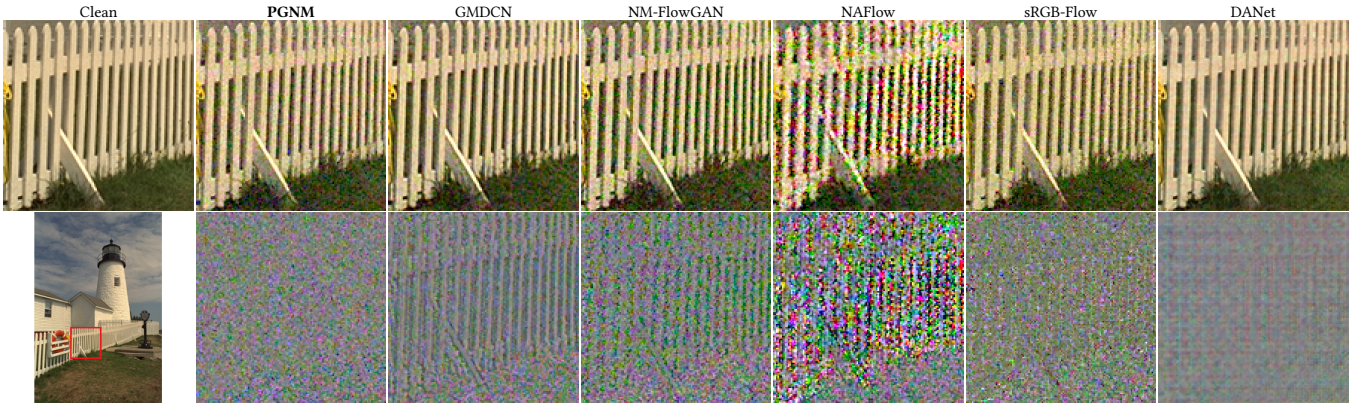

Figure S17 **A clean image from outside SIDD.** Kodak *kodim19* rendered by every SIDD-trained generator with its S6 ISO 800 fit or checkpoint, unchanged (one draw each). Top: output; bottom: residual relative to the clean image, amplified ×1.5 (gray is zero); bottom left: the full image with the 128 × 128 window marked, the image's most textured window by the rule of Figs. S12–S15, fixed before rendering.

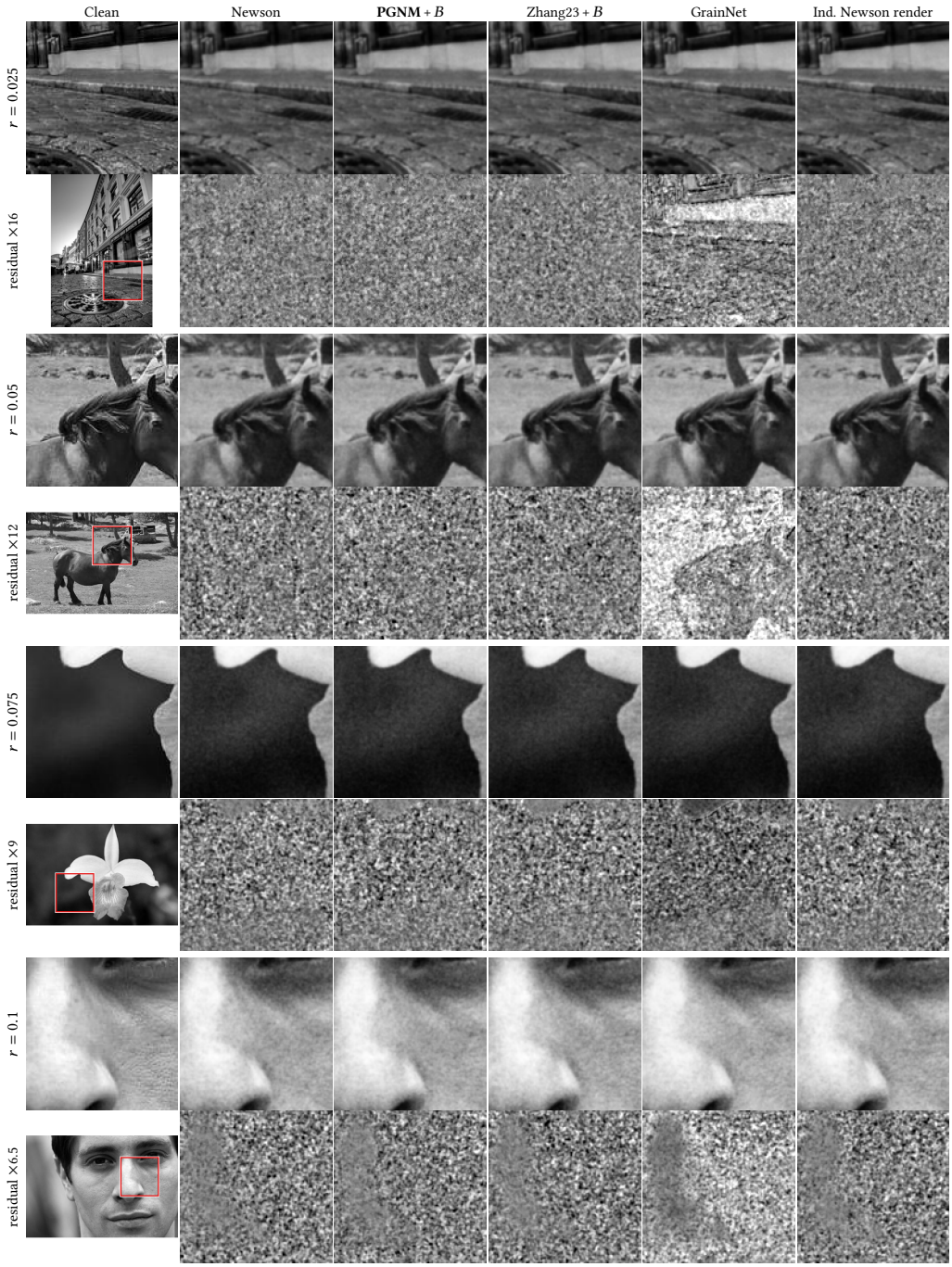

Figure S18 Uncurated film-grain comparison, every radius (1/2). One test image per radius (without replacement) and a 128 × 128 window inside the 3-pixel scoring border were drawn with a fixed pseudo-random seed from the clean image only; a window was redrawn only when more than 25% of its clean pixels were near 0 or 255, and no draw was replaced after viewing the outputs. Every method shows its first generator run; the reference is one Newson realization and the independent Newson render is an independently seeded simulator run, showing sampling variability. PGNM and Zhang23 add grain to the shared training-fitted blur $B * x$. Second row of each pair: residual relative to $\mu = B * x$, amplified by the factor at left (set from the Newson reference residual alone; gray is zero); under the clean column, the full image with the crop window marked. Crops are shown at native resolution; zoom in for detail.

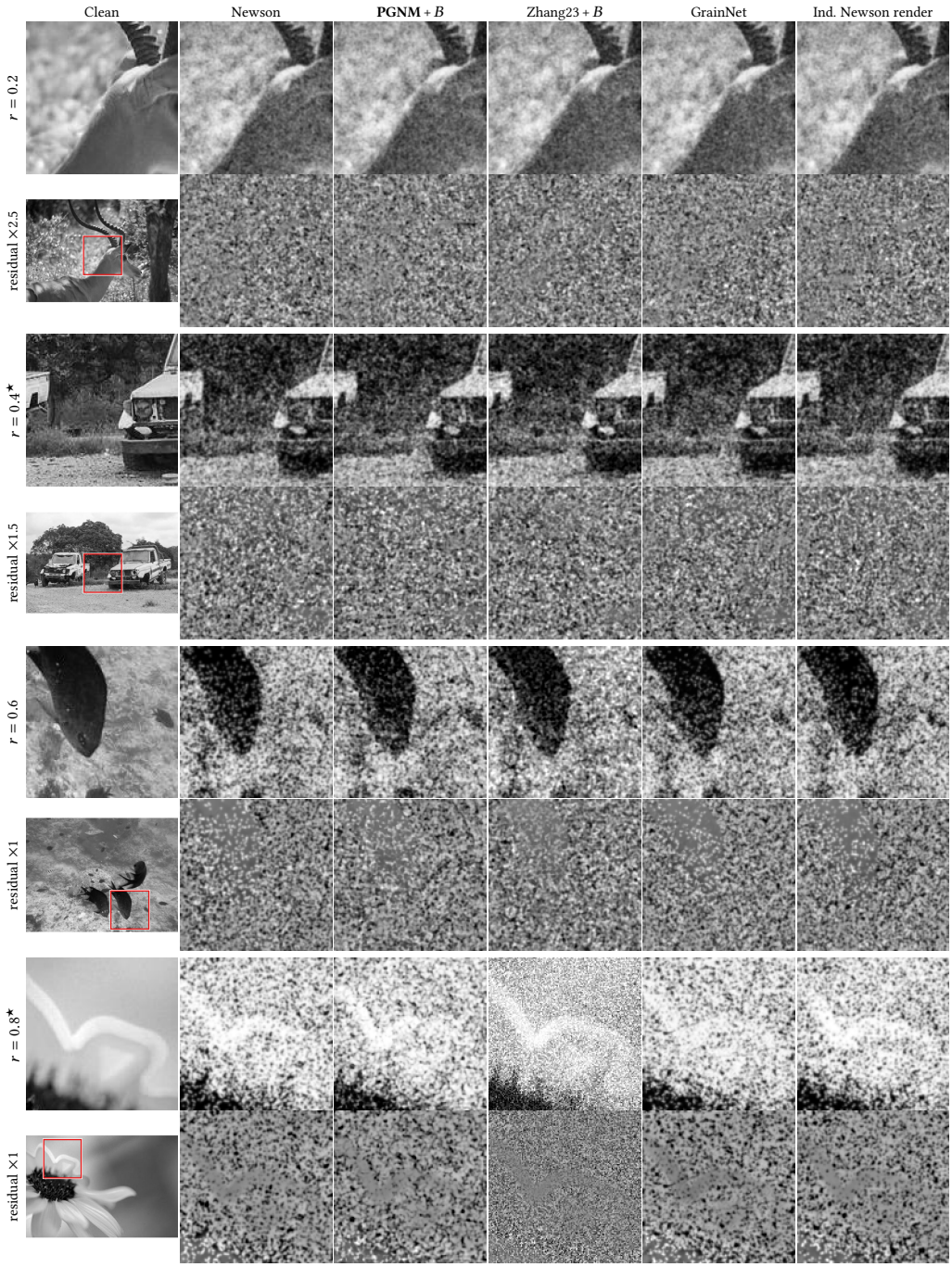

Figure S19 Uncurated film-grain comparison, every radius (2/2). One test image per radius (without replacement) and a 128 × 128 window inside the 3-pixel scoring border were drawn with a fixed pseudo-random seed from the clean image only; a window was redrawn only when more than 25% of its clean pixels were near 0 or 255, and no draw was replaced after viewing the outputs. Every method shows its first generator run; the reference is one Newson realization and the independent Newson render is an independently seeded simulator run, showing sampling variability. PGNM and Zhang23 add grain to the shared training-fitted blur $B * x$. Second row of each pair: residual relative to $\mu = B * x$, amplified by the factor at left (set from the Newson reference residual alone; gray is zero); under the clean column, the full image with the crop window marked. Crops are shown at native resolution; zoom in for detail. ★Also shown in the main paper (Fig. 6).

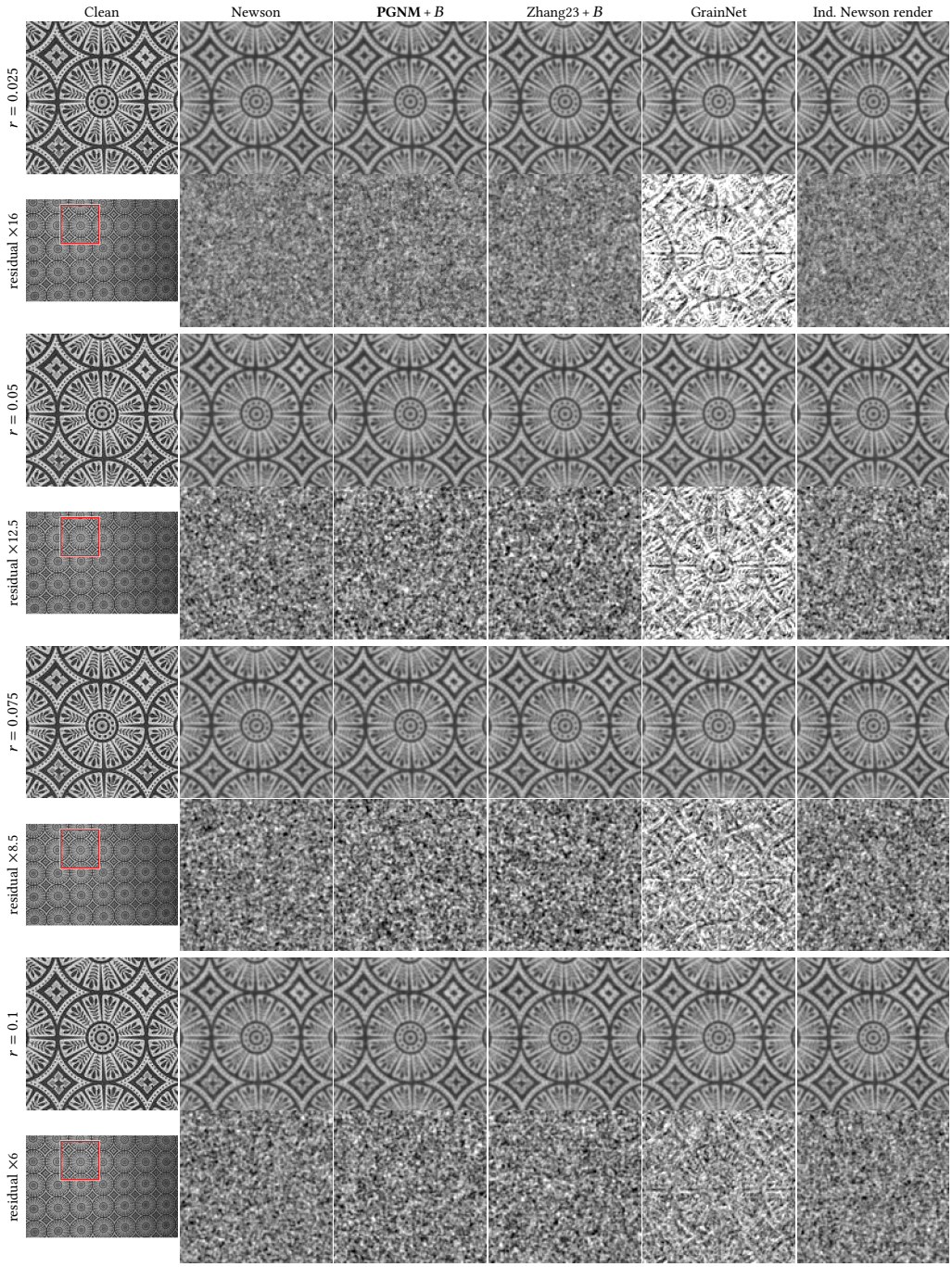

Figure S20 Film-grain comparison on a highly textured region, every radius (1/2). The 128 × 128 window inside the 3-pixel scoring border with the highest median 7 × 7 local standard deviation of the clean image, over all 60 test images on an 8-pixel grid (excluding windows with more than 25% of clean pixels near 0 or 255), is shown at every radius; it was fixed from the clean images before any output was viewed. Every method shows its first generator run; the reference is one Newson realization and the independent Newson render is an independently seeded simulator run, showing sampling variability. PGNM and Zhang23 add grain to the shared training-fitted blur $B * x$. Second row of each pair: residual relative to $\mu = B * x$, amplified by the factor at left (set from the Newson reference residual alone; gray is zero); under the clean column, the full image with the crop window marked. Crops are shown at native resolution; zoom in for detail.

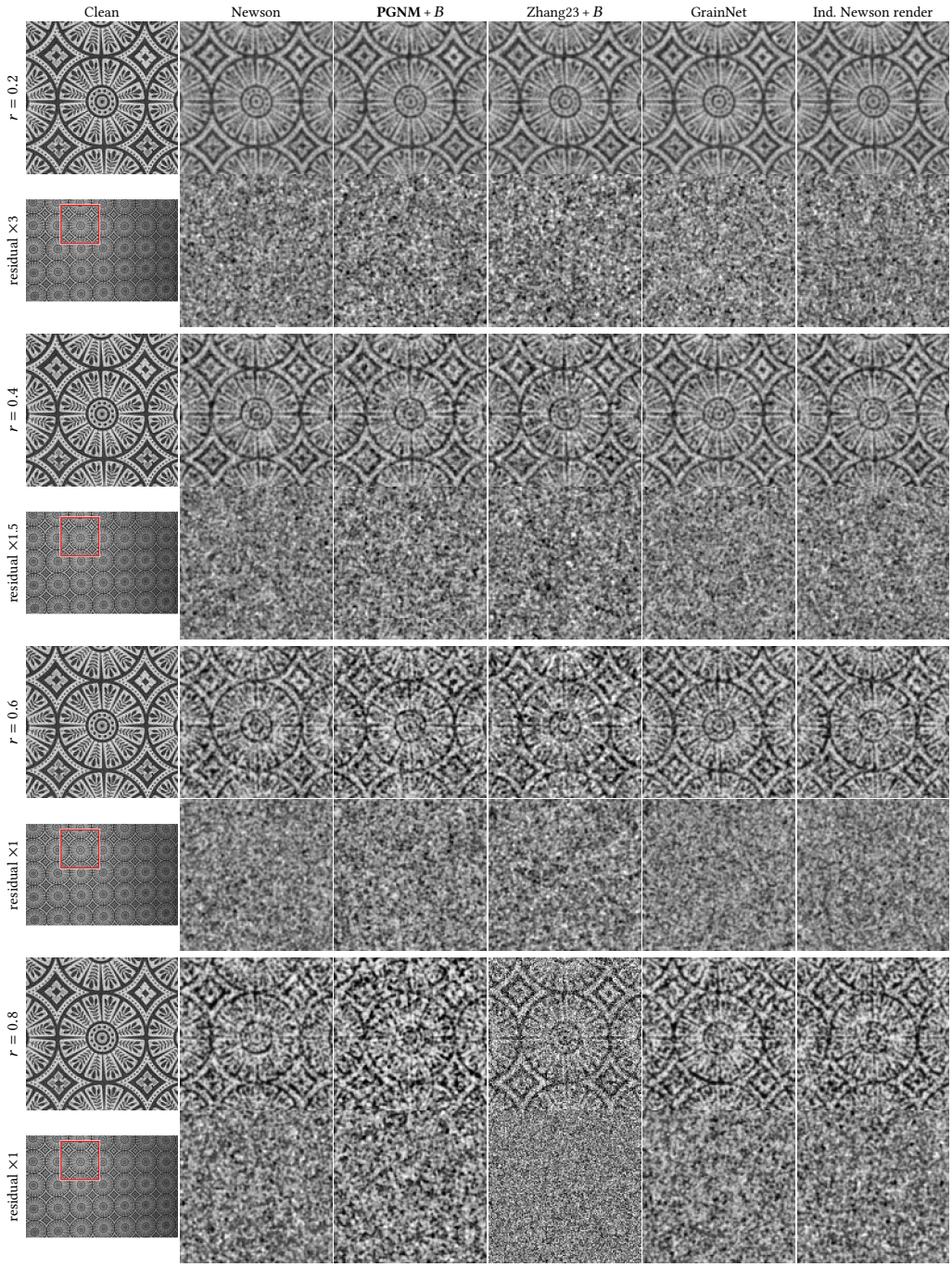

Figure S21 Film-grain comparison on a highly textured region, every radius (2/2). The 128 × 128 window inside the 3-pixel scoring border with the highest median 7 × 7 local standard deviation of the clean image, over all 60 test images on an 8-pixel grid (excluding windows with more than 25% of clean pixels near 0 or 255), is shown at every radius; it was fixed from the clean images before any output was viewed. Every method shows its first generator run; the reference is one Newson realization and the independent Newson render is an independently seeded simulator run, showing sampling variability. PGNM and Zhang23 add grain to the shared training-fitted blur $B * x$. Second row of each pair: residual relative to $\mu = B * x$, amplified by the factor at left (set from the Newson reference residual alone; gray is zero); under the clean column, the full image with the crop window marked. Crops are shown at native resolution; zoom in for detail.

Reproducibility reference

Exact algorithms, numerical conventions, and experimental settings for reproducing the reported results.

I Implementation specification

This section translates the model of Sec. A into an implementation. Fitting produces two arrays: convolution filters that set the latent field's spatial and color dependence, and a lookup that maps each latent value and clean intensity to a residual. Synthesis first constructs Gaussian fields with the fitted filters, then applies the lookup pixel by pixel. Fig. S22 summarizes these stages, Tab. S24 lists their fixed settings, and Algs. S1–S5 specify the numerical operations. Sec. I.4 records the data splits and random seeds used in our experiments; Tab. S26 provides fitted values for checking a reimplementation.

We retain the notation of the main paper (Sec. 3) and Sec. A. In array form, K stores the filters A_{ci}, g stores the values of f_c in Eq. 10, and the bank stores the main paper's latent templates.

I.1 Interfaces and conventions

Images. Each training pair contains aligned clean and noisy RGB arrays of the same shape, with values in $[0, 1]$; different pairs may have different image sizes. An 8-bit image is divided by 255 before use. Fitting uses float64. Synthesis converts the clean image to float32, evaluates the lookup in float64, and returns a float32 image. Delivery to 8 bits uses round-half-up conversion, $\lfloor 255\,\mathrm{clip}(v, 0, 1) + 0.5 \rfloor$.

Preset. The learned part of a preset consists of two arrays. The convolution tensor $K[y, x, i, o]$, of shape $25 \times 25 \times 3 \times 3$, holds the filter from input channel i to output channel o at offset $(y - 12, x - 12)$: $K[y, x, i, o] = A_{oi}(y - 12, x - 12)$ in Eq. 3 of the main paper, with $\sum_{y,x,i} K[y, x, i, o]^2 = 1$ for every o. The lookup $g[c, k, m] = f_c(x_k, t_m)$, of shape $3{\times}256{\times}25$, stores the residual quantiles of Eq. 10 at the clean-intensity grid points $x_k = k/255$ and the Gaussian grid points $t_m = -3{+}0.25m$. Its values are signed residuals in units of the image range ($|g| \le 1$), nondecreasing in m. These are the two fitted arrays; the remaining settings are fixed as listed in Tab. S24. The lookup carries residual amplitude, so it is used directly at synthesis without an additional strength multiplier.

Functions. Fit(pairs) returns a preset from aligned (clean, noisy) training pairs (Alg. S1). Warmup$(H, W, s_0, \ldots, s_3)$ builds a bank of four latent fields from four integer seeds, and Render(x, rng) synthesizes one noisy image (Alg. S5).

Random numbers. *rng* is a NumPy `Generator` created by `numpy.random.default_rng(seed)`, with the PCG64 bit generator. Latent values come from `standard_normal(shape, dtype=float32)`, and *rng*.int(n) denotes `integers(n)`, uniform on $\{0, \ldots, n - 1\}$. Draws are made in the order written.

Operators. U_7 is the 7×7 box mean of each channel with symmetric boundary extension that repeats the edge pixel ($\ldots cba|abc \ldots$; `scipy.ndimage` mode `reflect`). rint rounds to the nearest integer, ties to even. Φ and ϕ are the standard normal CDF and density. Percentiles interpolate linearly between order statistics. The DFT is $\widetilde{G}(\boldsymbol{\omega}) = \sum_{\mathbf{d}} G(\mathbf{d})\, e^{-2\pi \mathrm{i}\langle \boldsymbol{\omega}, \mathbf{d}\rangle/64}$ on a 64×64 grid, applied to every matrix entry: the transform of Sec. A at the frequencies $2\pi\boldsymbol{\omega}/64$.

I.2 Fitting

Fitting follows two branches that use the same aligned training pairs but estimate different properties (Fig. S22). The filter branch measures dependence where residuals are least affected by image structure and clipping. The lookup branch estimates the observed residual distribution at each clean intensity using all training pixels. The pilot and pixel masks belong only to the filter branch. The implementation makes repeated passes over the training pairs, loading one pair at a time.

Selecting and normalizing residuals. Alg. S1 first selects flat pixels using the largest RGB local standard deviation in a 7×7 window. Alg. S2 then estimates how residual strength changes with clean intensity using the clipped-Gaussian model of Eqs. S7–S10. This pilot supplies the normalization scale and clipping guard; the lookup supplies the residual distribution at synthesis. The subbin means and replication counts in Alg. S2 approximate each bin's clean-intensity distribution without retaining its individual pixels.

From correlations to filters. Alg. S3 combines flatness with a guard defined by the clean intensity and pilot in every RGB channel. If this combined mask selects zero pixels over the entire fitting set, correlation fitting uses the flat mask alone for every image, retaining the same pilot normalization. Otherwise the combined mask is retained throughout the fit. Normalized residuals are centered within each image; products and valid-pair counts are then pooled across images for each displacement. Both endpoints must pass the shared mask and remain in the same image. Zero-lag variances normalize the covariances to correlations. The spectral construction of Eqs. S16–S19 converts these measurements into K by tapering, spectral projection, inverse transformation, cropping, and output-energy normalization.

From histograms to a lookup. Alg. S4 counts every residual, including textured and clipped observations. For each clean code, neighboring intensity histograms are pooled until they contain at least 2,000 samples or cover the full intensity range. The lookup stores the residual quantiles at the 25 probabilities $\Phi(t_m)$. Treating mass as uniform within each residual-code bin lets these quantiles vary continuously between integer codes.

Stage	Quantity	Value	Used in
Flatness	local window, contrast	7×7 box (U_7), $\max_c$ of local std	Alg. S1
	threshold τ	clip($2 \times$ 2nd percentile, 0.003, 0.030)	
	percentile sample	every 4th row and column, all training images	
	flat mask	contrast $\leq \tau$ (inclusive)	
Pilot strength	knots	17, at $k/16$; nearest-knot bins	Alg. S2
	subbins per bin; copies of subbin means	32; about 512	
	minimum pixels per knot; predicted clipping limit	500; 0.35	
	bisection interval, iterations; rejection	$[10^{-6}, 2]$, 60; $\hat{\sigma} \geq 1.999$	
	smoothing; floor at use	one pass of $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, ends held; 10^{-6}	
Correlation	clipping guard	$2.5\,\sigma_c(x_c)$, all channels	Alg. S3
	lags	$\{-16, \ldots, 16\}^2$, all 3×3 channel pairs	
	taper	1 for $\|\mathbf{d}\| \leq 8$, $\cos^2$ to 0 at $\|\mathbf{d}\| = 16$	
	spectral grid	64×64	
	kernel support	25×25, energy 1 per output	
Lookup	histogram	3×256 clean codes $\times$ 511 residual codes	Alg. S4
	pooling	$\pm\rho$ codes, $\rho \geq 2$; 2,000 samples or all codes	
	Gaussian grid points	25, $t_m = -3 + 0.25m$	
Synthesis	bank	4 fields of $2H \times 2W \times 3$, float32	Alg. S5
	window	bank index $\neq$ previous; cyclic offsets in $[0, 2H) \times [0, 2W)$	
	lookup evaluation	bilinear; linear extrapolation in t	

Table S24 Constants of PGNM. All values are fixed for every dataset.

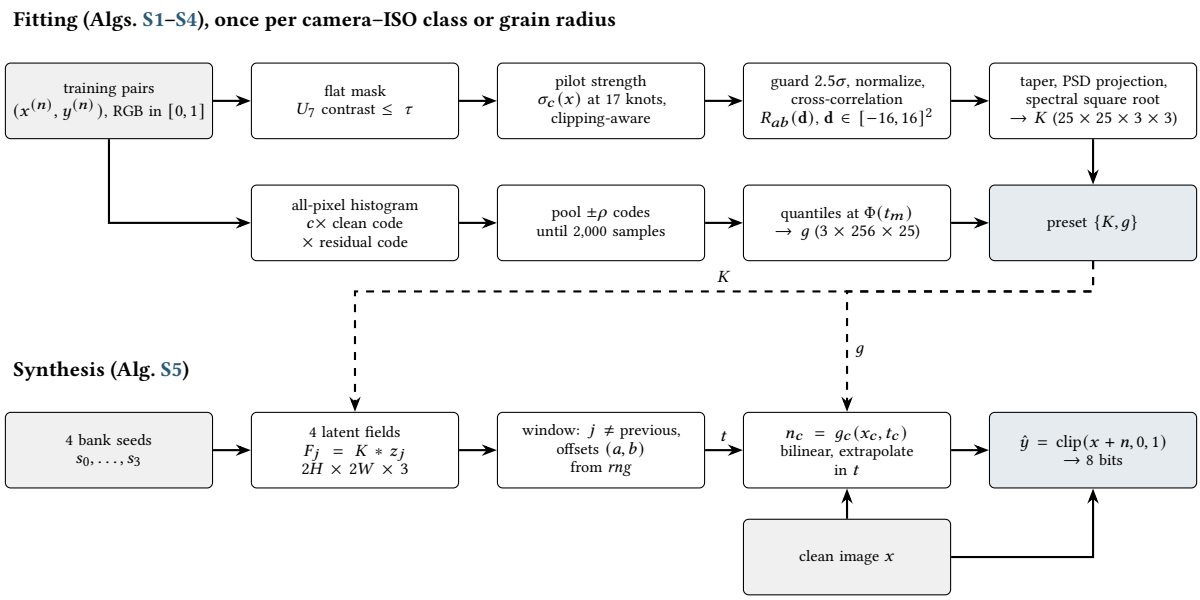

Figure S22 Data flow of PGNM. Fitting produces the preset $\{K, g\}$ from aligned training pairs. The filter branch (top) estimates dependence from selected pixels; the separately fitted lookup branch (bottom) uses all pixels. Synthesis builds the latent bank once per preset and image size and renders each output from one window.

Class	Held-out captures (scene instance / capture)
G4 400	`0123_006_G4_00400_00160_3200_N` /010, /011
G4 800	`0047_002_G4_00800_00640_3200_L` /010
	`0149_007_G4_00800_00800_4400_L` /010
GP 1600	`0157_007_GP_01600_01600_5500_N` /010
	`0200_010_GP_01600_03200_5500_N` /011
GP 3200	`0182_008_GP_03200_03200_5500_N` /011
GP 6400	`0036_002_GP_06400_03200_3200_N` /010
GP 10000	`0065_003_GP_10000_08460_4400_N` /010
IP 1000	`0069_003_IP_01000_02000_3200_N` /011
IP 1600	`0117_005_IP_01600_04160_5500_N` /010
	`0137_006_IP_01600_01600_5500_N` /011
	`0194_009_IP_01600_04000_3200_N` /010
	`0195_009_IP_01600_04000_5500_L` /010
IP 2000	`0070_003_IP_02000_04000_3200_N` /011
N6 400	`0146_007_N6_00400_00400_4400_N` /011
N6 800	`0055_003_N6_00800_01000_5500_N` /011
	`0075_004_N6_00800_00080_3200_L` /010
N6 3200	`0050_002_N6_03200_03200_5500_L` /010
	`0145_007_N6_03200_03200_4400_N` /011
S6 400	`0126_006_S6_00400_00200_4400_L` /010
	`0154_007_S6_00400_00400_5500_L` /011
S6 800	`0010_001_S6_00800_00350_3200_N` /011
	`0081_004_S6_00800_00160_4400_L` /010
	`0177_008_S6_00800_00200_5500_L` /010
S6 1600	`0152_007_S6_01600_01600_5500_L` /010
	`0178_008_S6_01600_00400_5500_L` /010
S6 3200	`0013_001_S6_03200_01250_3200_L` /010
	`0015_001_S6_03200_01600_5500_L` /011
	`0016_001_S6_03200_01600_5500_N` /011

Table S25 SIDD-Medium captures held out from fitting. PGNM is fitted on every crop of all other captures of the class.

A fit requires at least two accepted pilot knots per channel, positive selected residual variances, and nonzero retained filter energy. The numerical checks are given in the algorithms. Insufficient data or a degenerate estimate produces an explicit fitting failure; the mask changes only in the globally empty-guard case described above.

I.3 Synthesis

Synthesis separates field construction from repeated rendering (Alg. S5). Warmup filters independent Gaussian samples with the nine channel-pair kernels using FFT-based convolution. It stores four float32 fields of size $2H \times 2W \times 3$ and reuses them for the same preset and output image size. The filter-energy constraint of Eq. S2 gives each latent value population variance one.

Each call to Render draws a field index, a row offset, and a column offset, in that order. It reads an $H \times W$ cyclic window, applies the lookup, adds the signed residual to the clean intensity, and clips to $[0, 1]$. Eval interpolates bilinearly in clean intensity and Gaussian value; beyond $t = \pm 3$ it extends the first or last line segment.

Cyclic indexing permits any offset without constructing another field. The stored fields themselves are not periodic, so their usual short-range correlation does not continue across a wrap seam. For large H, W, approximately three quarters of random windows cross at least one field boundary. This bank-based procedure is used for every reported SIDD and film score. Denoising training instead constructs a fresh field for each patch, as specified in Sec. I.4.

I.4 Data and random seeds

The same PGNM estimator is used for both datasets: one preset per SIDD camera–ISO class and one per film-grain radius. This subsection specifies the training observations and random streams for each preset; Tab. S26 gives values for checking the resulting fits.

SIDD. One preset is fitted per camera–ISO class on SIDD-Medium sRGB [1]. Each full image pair is cut into 256×256 crops on a 256-pixel grid in raster order, with a final row and column of crops flush with the bottom and right borders (e.g. 252 crops for a 3000×5328 image). A *capture* is one of the two image pairs (`010`, `011`) of a scene instance. Some captures of each class are held out (Tab. S25): one generator, `default_rng(1234)`, visits the classes in sorted order and permutes each class's sorted list of n captures, and the last $\max(1, \lceil 0.2n \rceil)$ captures of the permutation are held out. Every crop from a held-out capture stays outside the PGNM fit. Fitting uses every crop of the remaining captures (Tab. S26).

For evaluation, the class preset renders each 256×256 clean block of the SIDD Validation set once per generator run, with a bank built for 256×256 and the seeds below, and the output is delivered to 8 bits.

Film grain. One preset is fitted per grain radius using GrainNet's released training pairs [27]. Source images identical up to rotation or reflection form one group, so transformed copies remain in the same partition. The groups are sorted by their first file name and permuted by `default_rng(1234)`. Of the n groups, the last $\max(1, \lfloor 0.2n + 0.5 \rfloor)$ are held out and excluded from fitting (Sec. K, Tab. S26).

The grain renderer also blurs the clean image. We therefore first estimate the expected image μ and fit the residual model around it (Sec. 4.2 of the main paper). For each radius, a 7×7 image filter B predicts grainy codes from windows of clean codes. Estimation first removes a 3-pixel border from both images, then uses all complete 7×7 windows within that crop; the regression target centers consequently lie at least six pixels inside the original image. B is fitted without an intercept by float64 least-squares normal equations.

Write the original clean and grainy images in code units as $x_{\text{code}} = 255x$ and $y_{\text{code}} = 255y$. The expected image is $\mu = \text{clip}(B \star x_{\text{code}}, 0, 255)/255$, computed on the whole image by correlation with mode `reflect`; the main paper writes this stage as $B * x$. Before fitting, μ is cast to float32, and y_{code} is converted to float32 and divided by 255. These normalized arrays are cropped by three pixels on each side, replicated to three identical channels, and passed to Alg. S1, which uses float64. This 3-pixel exclusion for residual fitting is distinct from the regression-window requirement above. Rendering uses float32 μ replicated to three channels, and the first output channel is delivered.

Seeds. Setup and window selection use separate random streams. Each experiment uses five generator runs, each with its own four-field bank. On both datasets, the bank of run $run \in \{0, \ldots, 4\}$ is built from the seeds $6100 + 4\,run, \ldots, 6103 + 4\,run$. The window generator *rng* of each output image is seeded from the SHA-256 digest D of a UTF-8 string:

- SIDD: D of `pgnm-protocol3|cvpr-sidd|`⟨class⟩`|`⟨image⟩`|`⟨2026 + *run*⟩`|`⟨block⟩, where ⟨class⟩ is written like `S6_800`, ⟨image⟩ is the Validation image index (0–39) and ⟨block⟩ the block index (0–31); seed = (first 8 bytes of D, big-endian) $\bmod (2^{31} - 1)$.
- Film: D of `cvpr-film-pgnm:`⟨radius⟩`:`⟨image⟩`:`⟨*run*⟩, with the radius written to three decimals (`0.025`) and the test image's

Algorithm S1 FIT: one preset from N aligned training pairs.

Require: pairs $(x^{(n)}, y^{(n)})$, float64 arrays $H_n \times W_n \times 3$ in $[0,1]$, $n = 1, \dots, N$
for $n = 1, \dots, N$ **do**
  $\kappa^{(n)}(\mathbf{p}) \leftarrow \max_c \sqrt{\max\big(U_7(x_c^2)(\mathbf{p}) - U_7(x_c)(\mathbf{p})^2,\ 0\big)}$ ▷ clean-image contrast
end for
$\tau \leftarrow \mathrm{clip}\big(2 \cdot \mathrm{percentile}_2\{\kappa^{(n)}(4i, 4j)\}_{n,i,j},\ 0.003,\ 0.030\big)$
$M_{\mathrm{flat}}^{(n)} \leftarrow [\kappa^{(n)} \le \tau]$ for every n
$\sigma \leftarrow \textsc{Pilot}(\{x^{(n)},\ y^{(n)} - x^{(n)},\ M_{\mathrm{flat}}^{(n)}\})$ ▷ Alg. S2; curves σ_c, $c \in \{R, G, B\}$
if the guarded mask of Alg. S3 selects at least one pixel in the whole set **then**
  $R \leftarrow \textsc{CrossCorrelation}(\mathrm{pairs}, M_{\mathrm{flat}}, \sigma, \mathrm{guard} = \mathrm{true})$
else
  $R \leftarrow \textsc{CrossCorrelation}(\mathrm{pairs}, M_{\mathrm{flat}}, \sigma, \mathrm{guard} = \mathrm{false})$ ▷ same pilot normalization
end if
$K \leftarrow \textsc{Factor}(R)$ ▷ Alg. S3
$g \leftarrow \textsc{Lookup}(\mathrm{pairs})$ ▷ Alg. S4; all pixels, no mask
return $\{K, g\}$

Algorithm S2 PILOT: clipping-aware strength curves from flat pixels.

Require: clean images $x^{(n)}$, residuals $r^{(n)} = y^{(n)} - x^{(n)}$, masks $M_{\mathrm{flat}}^{(n)}$
knots $\xi_k = k/16$, $k = 0, \dots, 16$; arrays $Sq, Num \in \mathbb{R}^{3\times17}$ and $Cnt, Sum \in \mathbb{R}^{3\times17\times32}$, all zero
for each n, channel c and pixel $\mathbf{p}$ with $M_{\mathrm{flat}}^{(n)}(\mathbf{p})$ **do**
  $v \leftarrow x_c^{(n)}(\mathbf{p})$; $k \leftarrow \min(\lfloor 16v + \frac{1}{2} \rfloor, 16)$ ▷ nearest knot
  $Sq[c,k]$ += $r_c^{(n)}(\mathbf{p})^2$; $Num[c,k]$ += 1
  $f \leftarrow \mathrm{clip}\big(16v - k + \frac{1}{2},\ 0,\ 1 - 10^{-9}\big)$; $j \leftarrow \lfloor 32f \rfloor$ ▷ position inside the bin
  $Cnt[c,k,j]$ += 1; $Sum[c,k,j]$ += v
end for
for $c = 1, 2, 3$ **do**
  for $k = 0, \dots, 16$ with $Num[c,k] \ge 500$ **do**
    $\bar{m} \leftarrow Sq[c,k]/Num[c,k]$; **if** $\bar{m} \le 0$ **then** skip this knot
    $Rep[j] \leftarrow \max\big(\mathrm{rint}(512\, Cnt[c,k,j]/\textstyle\sum_{j'} Cnt[c,k,j']),\ 1\big)$ for every occupied subbin j ($Cnt[c,k,j] > 0$)
    $\mathcal{I} \leftarrow$ multiset holding each subbin mean $Sum[c,k,j]/Cnt[c,k,j]$ $Rep[j]$ times ▷ subbin means repeated by weight
    if $\mathrm{mean}_{I\in\mathcal{I}}\, m(2, I) < \bar{m}$ **then**
      $\hat{\sigma} \leftarrow 2$ ▷ m of Eq. S7; rejected below
    else
      $\sigma_- \leftarrow 10^{-6}$, $\sigma_+ \leftarrow 2$
      for 60 bisection steps **do**
        $\sigma_0 \leftarrow (\sigma_- + \sigma_+)/2$
        if $\mathrm{mean}_{I\in\mathcal{I}}\, m(\sigma_0, I) < \bar{m}$ **then**
          $\sigma_- \leftarrow \sigma_0$
        else
          $\sigma_+ \leftarrow \sigma_0$
        end if
      end for
      $\hat{\sigma} \leftarrow (\sigma_- + \sigma_+)/2$
    end if
    $\gamma \leftarrow \mathrm{mean}_{I\in\mathcal{I}}\big[\Phi(-I/\hat{\sigma}) + \Phi(-(1-I)/\hat{\sigma})\big]$ ▷ predicted clipped fraction
    if $\hat{\sigma} < 1.999$ and $\gamma \le 0.35$ **then** accept $\sigma_c[k] \leftarrow \hat{\sigma}$
  end for
  require at least two accepted knots; fill the others by linear interpolation in ξ, constant beyond the outermost accepted knots
  $\sigma_c[k] \leftarrow \frac{1}{4}\sigma_c[k-1] + \frac{1}{2}\sigma_c[k] + \frac{1}{4}\sigma_c[k+1]$ for $k = 1, \dots, 15$ (from the unsmoothed values; ends kept); $\sigma_c \leftarrow \max(\sigma_c, 0)$
end for
return $\sigma_c(v) = \max\big(\mathrm{interp}(\mathrm{clip}(v, 0, 1);\ \xi,\ \sigma_c),\ 10^{-6}\big)$ ▷ piecewise linear in v

Algorithm S3 CrossCorrelation and Factor: from residual correlations to the convolution tensor.

procedure CrossCorrelation(pairs, M_{flat}, σ, guard)
  $P[a,b,\mathbf{d}] \leftarrow 0$, $Q[\mathbf{d}] \leftarrow 0$ for $a,b \in \{1,2,3\}$, $\mathbf{d} \in \{-16,\dots,16\}^2$
  for $n = 1,\dots,N$ **do**
    $s_c(\mathbf{p}) \leftarrow \sigma_c(x_c^{(n)}(\mathbf{p}))$ for every pixel and channel
    $M \leftarrow M_{\text{flat}}^{(n)}$; **if** guard **then** $M(\mathbf{p}) \leftarrow M(\mathbf{p}) \wedge \bigwedge_c \big[2.5\, s_c(\mathbf{p}) < x_c^{(n)}(\mathbf{p}) < 1 - 2.5\, s_c(\mathbf{p})\big]$
    if M has fewer than 2 pixels **then** skip this image
    $u_c \leftarrow (y_c^{(n)} - x_c^{(n)})/s_c$; $w_c \leftarrow \big(u_c - \text{mean}_M\, u_c\big) \cdot M$ ▷ zero outside M
    $P[a,b,\mathbf{d}]$ += $\sum_{\mathbf{p}} w_a(\mathbf{p})\, w_b(\mathbf{p}+\mathbf{d})$; $Q[\mathbf{d}]$ += $\sum_{\mathbf{p}} M(\mathbf{p})\, M(\mathbf{p}+\mathbf{d})$ ▷ pairs inside image n only
  end for
  $C \leftarrow P/\max(Q,1)$; $v_a \leftarrow C[a,a,(0,0)]$; require $v_a > 10^{-12}$
  return $R[a,b,\mathbf{d}] = C[a,b,\mathbf{d}]/\sqrt{v_a v_b}$
end procedure

procedure Factor(R)
  $\eta(\mathbf{d}) \leftarrow \cos^2\!\big(\frac{\pi}{2}\,\text{clip}((\|\mathbf{d}\|_2 - 8)/8,\ 0,\ 1)\big)$
  $R^{\text{s}}[a,b,\mathbf{d}] \leftarrow \frac{1}{2}\big(R[a,b,\mathbf{d}] + R[b,a,-\mathbf{d}]\big)\,\eta(\mathbf{d})$ ▷ symmetrize and taper, Eq. S17
  $G \leftarrow 0 \in \mathbb{R}^{64\times64\times3\times3}$; $G[d_y \bmod 64,\ d_x \bmod 64] \leftarrow R^{\text{s}}[\cdot,\cdot,\mathbf{d}]$ for every $\mathbf{d}$
  $\widetilde{G} \leftarrow \text{DFT}(G)$ ▷ transform each channel-pair lag array
  for each frequency $\boldsymbol{\omega}$ of the 64×64 DFT **do**
    $S_{\boldsymbol{\omega}} \leftarrow \frac{1}{2}\big(\widetilde{G}(\boldsymbol{\omega}) + \widetilde{G}(\boldsymbol{\omega})^{\text{H}}\big) = V\,\text{diag}(\lambda_1,\lambda_2,\lambda_3)V^{\text{H}}$
    $W(\boldsymbol{\omega}) \leftarrow V\,\text{diag}\big(\sqrt{\max(\lambda_l,0)}\big)V^{\text{H}}$ ▷ PSD projection and square root
  end for
  $\Gamma \leftarrow \text{Re}\,\text{IDFT}\big(\overline{W}\big)$ ▷ $\Gamma[\mathbf{d},o,i]$: filter from input i to output o
  $K[y,x,i,o] \leftarrow \Gamma[((y-12) \bmod 64,\ (x-12) \bmod 64),o,i]$ for $y,x \in \{0,\dots,24\}$
  $e_o \leftarrow \sum_{y,x,i} K[y,x,i,o]^2$ for each o; require $e_o > 10^{-12}$
  $K[\cdot,\cdot,\cdot,o] \leftarrow K[\cdot,\cdot,\cdot,o]/\sqrt{e_o}$ for each o
  return K
end procedure

Algorithm S4 Lookup: conditional quantiles from all training pixels.

$\mathcal{H} \leftarrow 0 \in \mathbb{N}^{3\times256\times511}$
for every pair n, pixel $\mathbf{p}$ and channel c **do**
  $k \leftarrow \text{clip}(\text{rint}(255\, x_c(\mathbf{p})), 0, 255)$
  $e \leftarrow \text{clip}(\text{rint}(255\,(y_c(\mathbf{p}) - x_c(\mathbf{p}))), -255, 255) + 255$; $\mathcal{H}[c,k,e]$ += 1 ▷ residual code $e - 255$
end for
for $c = 1,2,3$ and $k = 0,\dots,255$ **do**
  $\rho \leftarrow 2$
  repeat
    $k_- \leftarrow \max(0, k-\rho)$, $k_+ \leftarrow \min(255, k+\rho)$
    $n_{\text{pool}} \leftarrow \sum_{k'=k_-}^{k_+} \sum_e \mathcal{H}[c,k',e]$
    if $n_{\text{pool}} < 2000$ and $(k_-,k_+) \neq (0,255)$ **then** $\rho \leftarrow \rho + 1$
  until $n_{\text{pool}} \geq 2000$ or $(k_-,k_+) = (0,255)$
  $P_e \leftarrow \sum_{k'=k_-}^{k_+} \mathcal{H}[c,k',e]/n_{\text{pool}}$; $F_e \leftarrow \sum_{e' \le e} P_{e'}$, $e = 0,\dots,510$
  for $m = 0,\dots,24$ **do**
    $q \leftarrow \Phi(t_m)$; $i \leftarrow \min(\min\{e : F_e \geq q\}, 510)$; $F_- \leftarrow F_{i-1}$ (0 if $i = 0$)
    $v_m \leftarrow (i - 255 - \frac{1}{2}) + (q - F_-)/\max(P_i, 10^{-300})$ ▷ linear inside the unit bin of code $i - 255$
  end for
  $g[c,k,m] \leftarrow \text{clip}\big(\max_{m' \le m} v_{m'}/255,\ -1,\ 1\big)$ ▷ running maximum keeps g monotone
end for
return g

Algorithm S5 WARMUP and RENDER: the latent bank and one synthesized image.

procedure BUILDFIELD(h, w, rng)
  $z \leftarrow rng$.standard_normal($(h+24,\ w+24,\ 3)$, float32)
  $t_o \leftarrow \sum_i z_i * K[\cdot,\cdot,i,o]$ at the $h \times w$ positions where the kernel lies inside z ▷ true convolution
    (scipy.signal.fftconvolve, mode valid)
  return t ▷ $h \times w \times 3$; unit variance per channel
end procedure

procedure WARMUP($H, W, s_0, \ldots, s_3$)
  $F_j \leftarrow$ float32(BUILDFIELD($2H, 2W$, default_rng(s_j))) for $j = 0, \ldots, 3$; $prev \leftarrow -1$
end procedure

procedure RENDER($x \in [0,1]^{H\times W\times 3}$ float32, rng)
  $\mathcal{J} \leftarrow [\ j \in (0,1,2,3) : j \neq prev\ ]$; $j \leftarrow \mathcal{J}[rng.\text{int}(|\mathcal{J}|)]$; $a \leftarrow rng.\text{int}(2H)$; $b \leftarrow rng.\text{int}(2W)$
  for every pixel $\mathbf{p} = (p_y, p_x)$ and channel c **do**
    $t \leftarrow F_j[(p_y + a) \bmod 2H,\ (p_x + b) \bmod 2W,\ c]$
    $\hat{y}_c(\mathbf{p}) \leftarrow \text{clip}(x_c(\mathbf{p}) + \text{EVAL}(g_c, x_c(\mathbf{p}), t),\ 0,\ 1)$
  end for
  $prev \leftarrow j$; **return** $\hat{y}$
end procedure

function EVAL(g_c, x, t)
  $\alpha \leftarrow 255\,\text{clip}(x, 0, 1)$; $k \leftarrow \min(\lfloor \alpha \rfloor, 254)$; $\theta_x \leftarrow \alpha - k$
  $\beta \leftarrow (t+3)/0.25$; $m \leftarrow \text{clip}(\lfloor \beta \rfloor, 0, 23)$; $\theta_t \leftarrow \beta - m$ ▷ $\theta_t \notin [0,1]$ extrapolates linearly
  $g_- \leftarrow g_c[k,m] + \theta_t\,(g_c[k,m+1] - g_c[k,m])$; $g_+ \leftarrow g_c[k+1,m] + \theta_t\,(g_c[k+1,m+1] - g_c[k+1,m])$
  return $g_- + \theta_x\,(g_+ - g_-)$
end function

Preset	Pairs	Captures	τ	Flat (%)	Corr. (%)	Corr. mask	Energy
SIDD, one preset per camera–ISO class (256 × 256 crops)							
G4 400	1,008	4	0.0087	20.7	16.7	flat + guard	1.0000
G4 800	1,512	6	0.0076	38.4	26.0	flat + guard	1.0000
GP 1600	1,152	6	0.0088	38.1	32.8	flat + guard	1.0000
GP 3200	192	1	0.0280	29.6	3.6	flat + guard	0.9999
GP 6400	576	3	0.0193	50.0	21.7	flat + guard	0.9999
GP 10000	192	1	0.0209	58.8	4.6	flat + guard	0.9999
IP 1000	576	3	0.0061	58.2	12.7	flat + guard	1.0000
IP 1600	3,072	16	0.0070	42.5	15.3	flat + guard	0.9989
IP 2000	192	1	0.0062	60.0	21.9	flat + guard	1.0000
N6 400	663	3	0.0080	18.4	12.7	flat + guard	1.0000
N6 800	1,326	6	0.0069	24.0	20.8	flat + guard	1.0000
N6 3200	1,768	8	0.0230	45.4	5.9	flat + guard	1.0000
S6 400	1,008	4	0.0074	34.6	26.1	flat + guard	1.0000
S6 800	2,268	9	0.0070	33.7	21.3	flat + guard	1.0000
S6 1600	1,512	6	0.0104	10.7	3.3	flat + guard	1.0000
S6 3200	2,772	11	0.0133	23.9	2.4	flat + guard	0.9999
Film grain, one preset per radius (gray, replicated to RGB, 3-pixel border removed)							
$r = 0.025$	106	–	0.0030	37.2	36.9	flat + guard	1.0000
$r = 0.05$	84	–	0.0030	46.5	45.1	flat + guard	1.0000
$r = 0.075$	104	–	0.0030	44.1	43.0	flat + guard	1.0000
$r = 0.1$	104	–	0.0030	41.6	40.5	flat + guard	1.0000
$r = 0.2$	113	–	0.0030	45.1	40.3	flat + guard	1.0000
$r = 0.4$	114	–	0.0030	42.0	25.4	flat + guard	1.0000
$r = 0.6$	101	–	0.0030	41.4	9.4	flat + guard	1.0000
$r = 0.8$	86	–	0.0030	47.2	47.2	flat only	1.0000

Table S26 Checkpoints of the fitted presets. Pairs: training pairs; Captures: SIDD-Medium captures they come from. τ: flatness threshold (Alg. S1). Flat and Corr.: shares of all training pixels in the flat mask and in the mask used for correlation. Corr. mask: flat and clipping guard, or the flat-only fallback. Energy: smallest share, over output channels, of the factor's energy retained by the 25 × 25 crop before normalization.

two-digit identifier (`01`–`60`); seed = first 4 bytes of D, little-endian.

Before each output image, *prev* is reset to -1, so in our experiments the field is drawn uniformly from all four; the rule $j \neq prev$ matters only for consecutive renders without a reset, such as video frames. With these seeds and NumPy's generator, the procedure determines every output up to floating-point rounding in the FFT convolution.

Denoising. The denoising presets are fitted in the same way, on every crop of each class's denoising training images. Training does not use the bank: for every training patch and epoch, a fresh field BuildField$(96, 96, rng)$ is built, and Render's per-pixel loop reads it with offsets $a = b = 0$. Each random stream is seeded with the first 8 bytes, read big-endian and shifted right by one bit, of the SHA-256 digest of a string, for training seed ⟨seed⟩, epoch ⟨epoch⟩ $= 1, \ldots, 30$ and position ⟨position⟩ of the patch in the epoch's order:

- field: `CVPR-DENOISE/1|pgnm-field|`⟨seed⟩`|`⟨epoch⟩`|`⟨position⟩;
- patch order: `CVPR-DENOISE/1|permutation|`⟨seed⟩`|`⟨epoch⟩;
- augmentation: `CVPR-DENOISE/1|augment|`⟨seed⟩`|`⟨epoch⟩`|`⟨position⟩.

J Metric details

The scores in Sec. 4 of the main paper examine complementary properties. Conditional W_1 compares residual distributions at similar clean intensities; spatial and color correlation errors compare relationships across locations and channels; KL compares residual histograms within small tiles. Film-grain evaluation also compares the appearance of delivered images, while denoising evaluates reconstruction of clean images after training on each noise source. Lower values are better for the synthesis errors and appearance distances; higher PSNR and SSIM are better for denoising.

We first specify the common observations and intensity groups, then each estimator and its aggregation. Alg. S6 summarizes one SIDD source image and one generator run. The film-grain adaptations follow below.

Residuals after 8-bit delivery. Synthesis metrics compare the images delivered by each method. A continuous output v in $[0, 1]$ units is converted to 8-bit codes by $\lfloor 255\,\mathrm{clip}(v, 0, 1) + 0.5 \rfloor$. On SIDD, let X_8, Y_8, and $\widehat{Y}_8$ denote the clean, real noisy, and generated code arrays used in Alg. S6. Their normalized values are $x = X_8/255$, $y = Y_8/255$, and $\widehat{y} = \widehat{Y}_8/255$, as in the main paper. The signed residuals are $r = y - x$ and $\widehat{r} = \widehat{y} - x$; thus $255r$ and $255\widehat{r}$ are integer code residuals in $[-255, 255]$.

Pixels included in evaluation. Every SIDD block pixel contributes, including textured and clipped observations. The 32 blocks of one source image form one scoring unit per generator run: intensity statistics pool their pixels, and correlations pool products and pair counts without forming pairs across block boundaries. Evaluation uses neither the flatness mask nor the clipping guard used for fitting. Film scores use the image interior after removing three pixels from each edge.

Grouping similar clean intensities. Conditional statistics compare residuals where the clean intensity is similar. For each channel c, assign pixel p to $j_c(p) = \min(\lfloor 16 x_c(p) \rfloor, 15)$. The groups depend only on the clean image, so real and generated residuals, and all methods, share exactly the same pixels in each group. To avoid estimating a distribution from too few pixels, adjacent bins are merged until each group has at least 512 pixels or only one group remains (Merge in Alg. S6). At each step, the undersized group with the fewest pixels is merged with the adjacent group whose unweighted mean bin index is closest. Ties select the group with the lower first bin. Write b for a merged group, n_{cb} for its pixel count, and $n_c = \sum_b n_{cb}$.

Conditional distribution mismatch: W_1. This error responds to differences in residual bias, strength, and distribution shape at a given clean intensity. In channel c and group b, the real and generated lists contain residuals from the same n_{cb} pixels. Sorting both lists pairs matching empirical percentiles; the mean absolute difference is their exact 1-Wasserstein distance. The score of Eq. 13 weights groups by n_{cb}/n_c, averages channels equally, and multiplies by 255 to report code units. On SIDD's integer residual grid, the same distance before this final factor is $\frac{1}{255}\sum_{v=-256}^{256} |F_{cb}(v/255) - G_{cb}(v/255)|$, where F_{cb} and G_{cb} are the empirical CDFs. This equivalent histogram calculation avoids sorting all block pixels.

Centering and scaling before correlation. To keep conditional bias and strength differences from dominating the correlation comparison, each residual is centered and scaled within its clean-intensity group:

$$z_c(p) = \frac{r_c(p) - m_{cb}}{\max(s_{cb},\, 1/255)}, \qquad j_c(p) \in b. \tag{S20}$$

Here m_{cb} and s_{cb} are the group's mean and population standard deviation. They are computed separately from the real and generated residuals, giving z and $\widehat{z}$ on the shared groups. This is an evaluation normalization; it does not use the pilot strength fitted in Sec. I.

Spatial dependence within each channel. Spatial correlation measures how residuals at two locations vary together as their separation changes. For lag $\delta = (\delta_y, \delta_x) \in \{-8, \ldots, 8\}^2$, let P_δ contain pixel pairs $(p, p+\delta)$ whose endpoints are both in the scoring region and the same block. We compute

$$A_c(\delta) = \frac{1}{|P_\delta|} \sum_{(p,p+\delta)\in P_\delta} z_c(p) z_c(p+\delta), \qquad \rho_c(\delta) = \frac{A_c(\delta)}{A_c(0)}. \tag{S21}$$

The spatial error is the root mean square of $\rho_c(\delta) - \widehat{\rho}_c(\delta)$ over all channels and the 288 nonzero lags. Each lag and its opposite enter separately and have equal values. Zero-padded block FFTs, with at least eight pixels of padding in each dimension, compute these sums without wrap-around.

Color dependence at neighboring locations. Color correlation measures whether a residual in one channel varies with a neighboring residual in another channel. We use the four row–column lags $\delta \in \{(0, 1), (1, 0), (1, 1), (1, -1)\}$ and all six ordered pairs (a, b),

Algorithm S6 SIDD scores of one source image and one generator run. All statistics except KL, which is averaged over tiles, are pooled over the 32 blocks; pixel pairs never cross block boundaries. Zero-variance and support checks follow the text.

Require: delivered code arrays: clean X_8, real Y_8, generated $\widehat{Y}_8 \in \{0,\dots,255\}^{32\times256\times256\times3}$
$r \leftarrow (Y_8 - X_8)/255, \quad \widehat{r} \leftarrow (\widehat{Y}_8 - X_8)/255$
for $c = 1, 2, 3$ **do**
  $j_c(p) \leftarrow \min(\lfloor 16 X_{8,c}(p)/255 \rfloor, 15)$ for every pixel p
  $\{b\}_c \leftarrow \textsc{Merge}(\text{counts of } j_c,\ 512)$ ▷ groups of adjacent bins
end for
$E_{W_1} \leftarrow 255 \cdot \frac{1}{3} \sum_c \sum_b \frac{n_{cb}}{n_c}\ \text{mean}\left|\text{sort}(r_c[b]) - \text{sort}(\widehat{r}_c[b])\right|$
$z \leftarrow \textsc{Standardize}(r), \quad \widehat{z} \leftarrow \textsc{Standardize}(\widehat{r})$
for $c = 1, 2, 3$ and $\delta \in \{-8,\dots,8\}^2$ **do**
  $A_c(\delta) \leftarrow \text{mean}_{(p,p+\delta)\in P_\delta} z_c(p) z_c(p+\delta); \quad \rho_c(\delta) \leftarrow A_c(\delta)/A_c(0)$ ▷ same for $\widehat{z}$
end for
$E_{\text{spatial}} \leftarrow \text{RMS}_{c,\,\delta\neq0}\left(\rho_c(\delta) - \widehat{\rho}_c(\delta)\right)$ ▷ 3×288 values
for $\delta \in \{(0,1),(1,0),(1,1),(1,-1)\}$ and ordered (a,b), $a \neq b$ **do**
  $d_{ab}(\delta) \leftarrow \sqrt{\text{mean}_{P_\delta}[z_a(p)^2]}\,\sqrt{\text{mean}_{P_\delta}[z_b(p+\delta)^2]}$
  $\kappa_{ab}(\delta) \leftarrow \text{mean}_{P_\delta}[z_a(p) z_b(p+\delta)]/\max(d_{ab}(\delta), 10^{-12})$ ▷ same for $\widehat{z}$
end for
$E_{\text{color}} \leftarrow \text{RMS}_{\delta,(a,b)}\left(\kappa_{ab}(\delta) - \widehat{\kappa}_{ab}(\delta)\right)$ ▷ 24 values
for each block k and each of its 64 nonoverlapping 32×32 tiles t **do**
  $h \leftarrow$ counts of $255\,r[k,t]$ (all channels) in bins $[-260,-256),\dots,[256,260]; \quad P \leftarrow (h+1)/\sum(h+1); \quad Q$ likewise from $\widehat{r}$
  $D_{kt} \leftarrow \sum_j P_j \log(P_j/Q_j)$ ▷ p_j, q_j of Eq. 12
end for
$D_{\text{KL}} \leftarrow \text{mean}_k\, \text{mean}_t\, D_{kt}$
return $E_{W_1},\ E_{\text{spatial}},\ E_{\text{color}},\ D_{\text{KL}}$

procedure $\textsc{Standardize}(r)$
  for each channel c and group b of c **do**
    $m \leftarrow \text{mean}(r_c[b]), \quad s \leftarrow \text{std}(r_c[b])$ ▷ population std
    $z_c[b] \leftarrow (r_c[b] - m)/\max(s, 1/255)$
  end for
  return z
end procedure

procedure $\textsc{Merge}(\text{counts } n_0,\dots,n_{15}, \text{minimum } m)$
  $G \leftarrow [\{0\},\{1\},\dots,\{15\}]$ ▷ ordered list of groups of bin indices
  while $|G| > 1$ and some group has fewer than m pixels **do**
    $g \leftarrow$ the undersized group with the fewest pixels (ties: lowest first bin)
    $h \leftarrow$ the neighbor of g in G whose mean bin index is closest to that of g (ties: lower first bin)
    replace g and h by their union
  end while
  return G
end procedure

$a \neq b$. For each, normalize the mean product by the endpoint root mean squares:

$$d_{ab}(\delta) = \sqrt{\frac{1}{|P_\delta|}\sum z_a(p)^2}\,\sqrt{\frac{1}{|P_\delta|}\sum z_b(p+\delta)^2}, \qquad \kappa_{ab}(\delta) = \frac{\frac{1}{|P_\delta|}\sum z_a(p) z_b(p+\delta)}{\max(d_{ab}(\delta), 10^{-12})}, \tag{S22}$$

with all sums over P_δ. The color error is the root mean square of $\kappa_{ab}(\delta) - \widehat{\kappa}_{ab}(\delta)$ over these 24 values. It excludes same-pixel channel correlation (zero lag).

Constant residuals and insufficient support. If a generated channel has $\widehat{A}_c(0) \le 10^{-12}$, its spatial correlation is set to zero; the denominator floor above also defines its color products. If any real channel has $A_c(0) \le 10^{-12}$, both SIDD correlation errors are undefined. They are also undefined when the region has fewer than 4,096 pixels or any lag has fewer than 512 pairs. All complete SIDD scoring units satisfy these size requirements.

Local histogram mismatch: KL divergence. KL compares the frequency of residual values within small regions. It pools channels and clean intensities in each tile; it does not encode the arrangement of residuals inside that tile. Each 256×256 block is divided into 64 nonoverlapping 32×32 tiles. The 3,072 integer code residuals in a tile are counted in 130 width-four bins with edges $-260, -256, \dots, 260$ (the last bin closed). Adding one count per bin gives $p_j = (n_j + 1)/(3072 + 130)$ for the real residuals and q_j for

the generated residuals. Eq. 12 then computes $D_{\mathrm{KL}}(p\|q)$ with natural logarithms. The source score averages tiles within each block, then the 32 blocks. The 256×256 block KL, reported only in this supplement, uses the same histogram estimator on the whole block. NM-FlowGAN's released code applies the estimator before rounding its output; here every method is scored after 8-bit delivery.

Equal camera weighting on SIDD. Scores are first computed for each source image and generator run, then averaged over the five runs. The main table (Tab. 1) gives each camera equal weight, each camera–ISO class equal weight within its camera, and each source image equal weight within its class. This prevents cameras with more sources or ISO settings from dominating the result. The 19-source block applies the same rule to its own sources.

Film residuals around the expected image. The film renderer changes the image mean through blur as well as adding grain. Following Sec. I.4, residuals are therefore measured around the continuous expected image $\mu = \mathrm{clip}(B \star X_8, 0, 255)/255$, where X_8 is the clean gray code image. The fitted 7×7 filter B is applied by whole-image correlation with reflected boundaries, and μ is not rounded. The real and generated residuals are $r = Y_8/255 - \mu$ and $\widehat{r} = \widehat{Y}_8/255 - \mu$.

Film scoring and repeated renders. Conditional W_1 and spatial correlation use one gray channel and the three-pixel interior, with μ replacing x in the intensity groups. Because μ is continuous, W_1 compares sorted residual lists directly. Spatial correlation requires at least one pair at every nonzero lag and nondegenerate reference variance, with the same generated-variance rule as above. Color correlation and KL are not reported for film grain. For each test image and radius, five outputs from a method are compared with five Newson reference renders: every output is paired with every reference, giving 25 scores. We average these scores, then images equally within each radius, then radii equally. The independent Newson render is included as a method with five seeds separate from the reference seeds.

Overall image appearance: DISTS and LPIPS. These distances compare delivered images rather than residuals, complementing the distribution and correlation errors. DISTS [12] and LPIPS [49] (AlexNet) are evaluated for each film output–reference pair after cropping three pixels from each image edge, with the implementations and inputs of GrainNet's released evaluation script [27]. Gray images are represented as three identical channels, explicitly for DISTS and by channel broadcasting for LPIPS, with values in $[0, 1]$. LPIPS inputs are not rescaled to $[-1, 1]$. Aggregation follows the same 25-pair, image, and radius averages as the residual scores.

Clean-image reconstruction: PSNR and SSIM. Denoising scores compare a denoiser's output with the clean block: PSNR measures pixel reconstruction error, while SSIM compares local luminance, contrast, and structure. Under NM-FlowGAN's protocol, noisy 8-bit inputs are divided by 256; outputs are clipped to $[0, 1]$, multiplied by 256, rounded half up, and clipped to 255 before scoring. PSNR uses peak value 255. SSIM follows scikit-image's defaults for 8-bit RGB: a 7×7 uniform window, $K_1 = 0.01$, $K_2 = 0.03$, and sample covariance. Each channel's SSIM map is averaged after discarding its three-pixel border, and the channel scores are averaged. Both PSNR and SSIM are computed per block and averaged equally over the 1,280 Validation blocks.

K Baselines

Common protocol on SIDD. Every baseline uses its released checkpoint and sampling code at its native settings (Tab. S27), except GMDCN, which we implemented. Each 256×256 Validation block is generated separately at native size and batch 1, five times with derived seeds, and every output is clipped to $[0, 1]$ and rounded half up to 8 bits like PGNM's. Baselines receive the camera and ISO of the block's class where their model accepts them; none reads the real noisy block.

Methods.

- **NAFlow** [23] samples from its class-conditional latent distribution for the block's camera–ISO label at the released sampling temperature; its noise-aware mode, which reads a real noisy image, is not used.
- **GMDCN** [37] is our implementation of the published model. It predicts the residual from the clean image and a camera–ISO condition map and adds it to the clean image.
- **DANet** [45] uses the generator of its released model. It receives no camera or ISO input.
- **sRGB-Flow** [26] takes the clean image, camera and ISO. Its checkpoint is trained on 15 camera–ISO classes, 7 of our 16 (G4 ISO 800, IP ISO 1600, N6 ISO 800 and the four S6 classes), which form the 19-source block.
- **NM-FlowGAN** [18] takes the clean image, camera and ISO, and its checkpoint is trained on the same 15 classes. It accepts the ISO values 100, 400, 800, 1600 and 3200 for every camera, but we report it only on the 19-source block of its training classes.
- **NeCA-W** [16] is not included: its released generator requires an encoding of a real noisy image at sampling time, and we found no reference-free inference rule consistent with our protocol.

Film grain. Fitting uses GrainNet's released training pairs [27]. Their 199 source images, with flipped and rotated copies merged, were split with seed 1234 into 159 for fitting and 40 held out and unused (evaluation uses GrainNet's 60 separate test images), which leaves 84–114 fitting pairs per radius (Tab. S26). PGNM and Zhang23 add grain to the same expected image $\mu = B * x$ (Sec. I.4); GrainNet and the independent Newson render receive the clean image in their native pipelines.

- **Zhang23** [48] is our implementation of the published equations. Its grain-size parameter is estimated from the ratio of lag-1 and lag-$\sqrt{2}$ correlations of flat-pixel residuals (clipped to $[0.3, 4]$), and its radius by least squares on Zhang23's variance expression across clean intensities, after subtracting 8-bit quantization noise; both are fitted on the B-filtered fitting pairs inside PGNM's flat masks. Rendering adds Gaussian noise with the predicted intensity-dependent variance, filtered by a separable 33-tap kernel, to μ. At $r = 0.8$ the measured correlations vanish and the grain-size parameter reaches its lower bound, consistent with the approximation no longer holding for resolved grain.

Method	Released code	Commit
NAFlow	`github.com/dongjinkim9/NAFlow`	`533e1eac581a`
DANet	`github.com/zsyOAOA/DANet`	`ea4d71416815`
sRGB-Flow	`github.com/SamsungLabs/Noise2NoiseFlow`	`54cc3c11e4fc`
NM-FlowGAN	`github.com/YoungJooHan/NM-FlowGAN`	`63983e2c2c5e`
GrainNet	`github.com/Gwilherm-LESNE/Neural_Film_Grain_Rendering`	`865c49400405`
Newson	`github.com/alasdairnewson/film_grain_rendering_gpu`	`5695ff2abea7`

Table S27 Released code and checkpoints used for the baselines, at the listed commits.

- **GrainNet** [27] uses the released checkpoint with the clean gray image, radius and a seed as input.
- **Newson** [33]: references and the independent run use the released GPU renderer, changed only to accept an explicit random seed, with grain standard deviation 0, filter standard deviation 0.8 pixels, zoom 1 and 10^5 Monte Carlo samples. Each image and radius has five reference renders and five independent renders from separate seed sets.

L Cost measurement setup

This section describes how the parameter counts and times of Tab. 4 of the main paper were obtained.

Hardware and environment. All rows ran one at a time on one machine with an Intel Core i9-14900K. Every process was pinned to four efficiency cores and limited to two threads (OpenMP, MKL, Numba and PyTorch), at the lowest scheduling priority and with a 16 GB memory cap. GPU rows used one NVIDIA RTX A6000, capped at 12 GB, with the same CPU pinning. Each method ran in its own environment with the PyTorch version its code requires (1.8.1 to 2.5.1, CUDA 11.1 to 12.1). The arithmetic is single precision; cuDNN was allowed to use TF32 convolutions in every environment, and two environments (DANet, sRGB-Flow) also allowed TF32 matrix products.

Input and timed region. The input is one synthetic 1920×1080 8-bit RGB frame of color ramps, batch 1. SIDD rows use the G4 ISO 800 condition and its PGNM preset; film rows use the radius 0.1 setting, applied to each color channel. The timed region starts from the 8-bit frame in host memory and ends with the method's noisy frame, in floating point, in host memory; it includes type conversion and all host–device copies, and CUDA is synchronized before and after each call. Model loading, the first call and the final 8-bit quantization are excluded; the quantization, common to all methods, took 50–100 ms on the two CPU threads. GPU rows report the median of 50 calls after 10 warm-up calls, CPU rows the median of 10 calls after one.

PGNM rows. *New field per frame* draws $(H + 24) \times (W + 24) \times 3$ Gaussian values, filters them with the $25 \times 25 \times 3 \times 3$ tensor (FFT convolution on CPU, direct convolution on GPU), and applies the lookup, addition and clipping; this is comparable to the baselines, which draw new noise for every frame. *Bank playback* is the renderer of all reported results (Alg. S5): it reads one cyclic window of the four $2H \times 2W \times 3$ float32 fields (398 MB at this resolution). *Bank construction* builds these fields once per preset and resolution; on CPU it includes compiling the lookup kernel, and on GPU it is the process's first convolution. The bank pays for itself after $307/(20.4 - 10.2) \approx 30$ frames on GPU and $23.45/(1.04 - 0.051) \approx 24$ on CPU. The GPU rows use a PyTorch port of the renderer. Filtering the same white noise, the port's latent field differs from the CPU field by at most 1.02×10^{-5} (unit variance); given the same field, its noisy image differs by at most 1.8×10^{-7} before rounding and is identical after 8-bit delivery, on a 256×384 crop (height by width) of the same frame. The film PGNM preset has the same array sizes as the SIDD one and was not timed separately.

Baselines. Neural baselines run their generators (released, or ours for GMDCN) through our adapters (Sec. K) in inference mode, including their per-call random seeding; NM-FlowGAN's reflection padding to 1088 rows and cropping back are inside the timed region. GrainNet, a gray model, runs once per color channel. Zhang23 has no GPU implementation, and its time excludes the shared blur B.

Parameter counts. A PGNM preset holds $25 \cdot 25 \cdot 3 \cdot 3 = 5{,}625$ filter values and $3 \cdot 256 \cdot 25 = 19{,}200$ lookup values; the fixed Gaussian knot grid, the residual step and the bank seeds are not counted. Film presets add the 49 values of the shared 7×7 blur, and Zhang23 has two fitted values per radius plus the same blur. Neural baselines count the learned parameters of their noise generators.